From Human Markets to Machine Markets: Transient Amplification, an Exact Depth Condition, and a Maximal Pre-Trade Guard

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Abstract

Eigenvalues and volatility do not bound transient losses before a market settles. We show that cumulative transient risk in a limit order book equals $S = \tilde{\kappa}/(m - q\tilde{\kappa})$, that a book overshoots if and only if $\varphi\varepsilon > (1 - a)\kappa$, and that a pre-trade guard on the summed impulse response is maximal and invariant to unobserved trader cohorts (including under carried inventory and interval-bounded hidden drift). Certifying S requires $O(\sigma^2\epsilon^{-2}\log(1/\epsilon))$ experiments regardless of cohort spacing, yielding a minimax-optimal guard. Across 21 pre-registered primary tests, 12 pass: in thin Binance perpetuals, six-hour estimates forecast next-hour adverse excursions out of sample across five tests ($\Delta R^2 = +0.85$ to $+4.20$ percentage points; $+2.79$ $[+1.43, +4.94]$ beyond HAR/EWMA volatility and $+0.56$ $[+0.27, +1.16]$ beyond VPIN), replicate in

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thin US equities (+0.50 [+0.27, +0.78]), and lower tail loss by 8.1% ([−16.1%, −1.7%]) and 22.6% ([−39.2%, −2.0%]) against static limits.

Keywords: transient amplification, non-normal dynamics, limit order book, market resilience, liquidity externality, pre-trade risk control, pre-registration, replication.

JEL: G01, G10, G14, G18, C32, C58.

Reproducibility. Every number in the paper is produced by a script or notebook cell that writes a result file from public data or from a closed form; Appendix D maps each table, figure and number to its source. Pre-registered specifications, their SHA-256 hashes and test-set unlock counters are in the replication package. Every simulation number was re-run for this version.

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1 Introduction

On October 10, 2025, crypto derivatives markets liquidated an estimated $19 billion of leveraged positions in a single day, the largest such event on record (21Shares, 2025); on May 6, 2010, US equity prices fell and recovered by several percent within minutes (Kirilenko et al., 2017). In both episodes the losses that mattered were transient: positions were closed out on the path, before prices settled. This paper is about that path risk in markets where algorithms trade with each other.

Algorithmic markets are closed-loop systems: trading moves prices, prices move algorithms, and algorithms trade. A common shortcut equates the stability of such a system with its eigenvalues lying inside the unit circle. That is the wrong object for a leveraged intermediary, whose survival depends on the path of losses before the system settles. A market can be spectrally stable and still push a dealer through its maintenance margin within seconds.

This paper asks three questions. When does a limit order book amplify a flow shock, so that prices keep moving after the shock has passed? How can a dealer or a venue control transient loss without identifying who is trading, which the companion paper (Niroula and Niroula, 2026) shows is expensive or impossible from public data? And when the traders cannot be identified, what information certifies a safe trading decision, what does that information cost, and how much safe trading must be given up when it is missing? We give exact answers in a linear book model and then confront them with data, reporting the results as they came out.

1. Spectral stability does not bound transient loss. The exact safe shock radius of a stable linear market is the loss budget divided by its peak transient gain, and that gain can be arbitrarily large at any fixed spectral radius (Theorem 3.1). Markets with identical eigenvalues, one-step volatility and stationary price variance can have arbitrarily large worst-case transient gain, as the companion paper shows (Niroula and Niroula, 2026). Supervision

through eigenvalues or volatility alone is therefore insufficient in principle.

2. When a book amplifies. We distinguish peak overshoot, a response that rises above its direct impact, from cumulative amplification, a summed response far above it. When order flow depletes one side of the book and the book refills at rate r, the price response overshoots its direct impact if and only if $\varphi\varepsilon > (1-a)\kappa$ (Theorem 4.2(i)). With momentum traders the one-period margin $C_3 = \varphi\varepsilon + \kappa^2\eta q - (1-a)\kappa$ is sufficient (Theorem 4.2(iii)) but no one-period condition is exact (Remark 4.3); momentum alone cannot make a stable book overshoot (Theorem 4.2(iv)), and an explicit condition rules overshoot out (Theorem 4.2(v)). Cumulative risk has a closed form, $S = \tilde{\kappa}/(m - q\tilde{\kappa})$: long-run impact divided by reversion net of momentum (Theorem 4.4). It needs no depletion, and it is the form our evidence supports: crypto books show no peak overshoot, but thin books revert slowly, so their summed response is large.

3. The maximal safe guard. For bounded shocks, worst-case loss over a horizon is inventory times the shock bound times the summed absolute impulse response, the classical ℓ_1 gain of control theory. Under a declared risk contract (fixed inventory, per-period shock bound, mark-to-market loss), a pre-trade guard on this summed gain is necessary and sufficient for staying within the loss budget, so it admits the largest safe set of actions (Lemma 6.1). A guard on the peak response, the quantity closest to the eigenvalue intuition, admits unsafe actions. A Grönwall bound carries the guard to an estimated operator (Theorem 6.3), a contraction-block certificate makes it computable on a finite horizon (Lemma 6.2), and its inputs are identified from prices without identifying the algorithms (Proposition 2.2 and Theorem 5.5). We embed the guard in a dealer's balance sheet with margin, hedging and fill accounting (Theorem 6.8) and show that strategic re-optimization by filtering algorithms reinforces it (Proposition 6.11).

4. Information and safety. The information that certifies a safe decision is not the information that identifies the traders. Two markets can agree exactly on their early price responses and differ threefold in risk (Example 5.1), and any supervisor who must rarely admit an unsafe level then gives up a computable amount of safe capacity (Proposition 5.2). A finite window of responses certifies the summed response to an interval computed exactly from the Markov–Krein extremal measures, which shrinks geometrically with the window under a bound on memory length and is unbounded without one (Theorem 5.3). With noisy reset experiments the summed response is certified at the parametric rate, $N = O(\sigma^2 \epsilon^{-2} \log(1/\epsilon))$, whatever the number and spacing of the traders' memories, while identifying them costs $(\Delta\eta)^{-(4K-2)}$ or more (Theorem 5.5); the companion paper's hardest pairs have identical summed responses. A guard built on the certificate has a closed form, admits an interval of levels, flags infeasible states, and is minimax-optimal in lost capacity up to a logarithm (Theorem 6.5). When admission also changes the book, the safe levels need not form an interval, so certifying the largest one no longer certifies those below it; we certify the whole safe set by interval branch and bound (Proposition 6.6, Theorem 6.7). The window bounds are proved rather than computed (Lemma 5.4), and certificates that allow correlated noise keep their level where the simple one fails (Proposition 5.7).

5. The value of disclosure. Disclosure that rules out the extremal compatible economy raises safe capacity, and its value grows as memory lengthens, approaching the full oracle capacity; disclosure of a noisy proxy can be worth less than nothing (Theorem 7.1, Table 5).

6. Who controls the risk. Reversion is a public good. In equilibrium dealers underprovide it, momentum traders over-provide feedback, fewer dealers mean more transient risk, and the rebate that restores the first best is $(1 - \theta/N)\Lambda S^2/\tilde{\kappa}$, a function of the guard's own statistic (Proposition 8.1). When dealers also choose depth, fewer dealers raise risk through larger impact while the unit-free multiple $S/\tilde{\kappa}$ stays fixed, the pattern we see across Binance perpetuals.

7. Evidence. On Binance BTCUSDT futures the amplification statistic is negative on 93.5% of days, depth does not carry an overshoot, and the statistic fails a pre-registered forecasting test (Section 9); unit-free spot amplification is indistinguishable from one (Section 10). Across 30 perpetuals, Binance's depth bands, which are re-centred on the current mid, show taker flow replenishing the side it hits; on a fixed price grid in Bybit's full order books, the same flow depletes it, so the replenishment finding is an artefact of the band construction (Section 11.4). The amplifying regime is more frequent in thin books (Section 11). There, six-hour estimates of the amplification margin forecast the size of the next hour's adverse excursion out of sample in four pre-registered tests: on the original ten symbols (+1.2 points of R^2), on ten new symbols (+0.85), on fresh days with price-impact controls (+1.6), and on further fresh days beyond a full volatility-forecasting baseline (+2.8, with better tail classification). The summed response S, the guard's own statistic, carries the same information (+2.1 points beyond the volatility baseline), the gain survives adding VPIN to the baseline (+2.0 points post hoc, +0.56 in a pre-registered test on new symbols), unit-free versions work, and the content comes mostly from impact-weighted reversion, as Theorem 4.4 implies. Within a volatility quintile, hours in the top quintile of the statistic have next-hour excursions about 26 basis points larger than hours in the bottom quintile, enough to roughly double the hourly liquidation probability of a 20× position. The effect is much weaker in mid-depth books and absent in BTCUSDT. On Bybit it fails with trade data alone; with order-book depth it meets a pre-registered criterion, but the gain is about one eightieth of Binance's (Table 23). In a regulated market, 23 thin US stocks with consolidated trades, the depth-free statistic passes a fifth pre-registered forecasting test (+0.50 points of R^2 beyond the same volatility baseline), again through impact-weighted reversion; unlike on Binance, mid-size stocks show a gain too (Table 24). As a position limit, a six-hour estimate of the response adds nothing to volatility, but once the operator is certified from the previous three days, scaling a volatility limit by the gain multiple lowers tail loss per unit of position by 2.4% in a fifth pre-registered test, a narrow pass; limits built on the estimated response alone

remain worse than volatility (Section 11.14).

Simulations with a known truth confirm the coverage, attainment and optimality claims and show where the certificate fails: when memory is longer than declared, and when feedback strengthens after certification (Section 12.1). On days that no earlier part used, two pre-registered primaries pass: the forecasting model, frozen at its 2023–2024 estimates, adds 4.2 points of R^2 beyond the volatility baseline, and the uncertainty-aware guard lowers tail loss per unit of position by 8% against a static limit at 91% of its activity, doing much better than depth or spectral-radius throttles; it does not beat a volatility cap, and its bootstrap upper bound does no better than the point estimate (Section 12.2). On thirty perpetuals that no earlier part used, of which a seeded rule kept five thin ones, three pre-registered primaries pass: $\hat{C}_3$ adds 0.73 points of R^2 beyond the volatility baseline and 0.56 beyond the baseline plus VPIN, turning the post-hoc VPIN comparison into a confirmatory one, and the guard lowers tail loss by 23% against a static limit, with a wide interval (Section 12.3). A pre-registered test on a venue no earlier part used, Coinbase spot, fails: in the six thin pairs that trade often enough, the statistic adds nothing (Section 12.4). Post hoc, a placebo that keeps the flow's persistence but breaks its alignment with prices adds nothing on Binance, where the real statistic adds 2.7 points, so the content is the alignment rather than order splitting; and the guard has lower tail loss than certified peak-response, margin-style and participation rules on the Part 15 sample, but not than a six-hour peak guard (Section 12.5). Three further analyses were specified in an earlier draft before their data were downloaded but were not hash-registered (Section 12.6): on fresh days for 19 thin coins, the predicted interaction of $\hat{C}_3$ with open-interest-to-depth leverage fails (-0.18 bp, CI $[-3.02, +3.73]$) while the spot gain is 15.2% of the perpetual gain ($+0.99$ vs. $+6.50$ points, ratio CI $[-0.015, +0.308]$; Section 12.7); in Binance's BTCUSDT spot zero-fee experiment, the primary reversion and gain-multiple hypotheses are not supported while impact falls 18% in the two spot events ($t = -3.86$, exploratory, single treated pair; Section 12.8); and in dollars, the guard alone lowers hourly expected shortfall on \$1M of average position by

$3,045 and $10,518 against a static limit on the untouched days and symbols, while its gain when combined with a volatility cap ($398 and $2,557) is not statistically significant (Section 12.9). Finally, two pre-registered tests on untouched 22nd-of-month days (Sections 12.10–12.11) test the liquidation channel and multi-hour notional capacity directly: $\hat{C}_3$ again strongly forecasts next-hour adverse excursions ($\Delta R^2 = +6.52$ points, $t = 12.21$), but neither forecasts five-minute open-interest contractions ($\Delta R^2 = -0.80$ points) nor is mediated by them (mediation share -6.8%, both primaries fail), showing that the thin-book effect reflects slow book reversion rather than five-minute open-interest liquidations; and in a non-resetting 18-hour leveraged position test, adding the certified guard to a volatility cap raises mean notional capacity at a matched 39.7% liquidation rate by +$332,848 (+36.9%), but the 95% confidence interval ([−$20,390, +$1,180,808]) includes zero (primary fails).

What we do not claim. The information-safety theorems concern declared observation and risk contracts (controlled reset experiments, exact windows, tagged or audited data); they are not finite-sample guarantees for observational order-book records, for which we report a bootstrap diagnostic and its out-of-sample performance. Our cohorts are any traders whose demand follows a linear filter of past prices, human or automated; nothing in the evidence attributes an effect to algorithms rather than to people. We do not claim that crypto books amplify shocks through depth: flow depletes the side it hits on a fixed price grid, but we observe no peak overshoot, and the depth term carries no forecasting content. The thin-book forecasting result is reduced-form, and its size depends on the venue: largest in thin Binance books, about a fifth as large in thin US stocks, negligible on Bybit and absent on Coinbase; it improves tail classification only against the full volatility baseline. We do not claim that the guard replaces volatility in practical position sizing: its theorems are statements about worst-case loss under a certified operator, an estimated six-hour response adds nothing to volatility, and a three-day certified estimate improves a volatility limit only modestly (2.4%, a narrow pass) while the guard alone does worse. The October 10,

2025 episode is reported descriptively and supports no inference (Section 11.15). One test cell of the first thin-book test was executed twice with unchanged code and specification (Section 11.5). The equilibrium is a stylized model; thinner books do carry more cumulative risk per dollar of flow, as it predicts, but through larger impact rather than through the weaker reversion it names.

1.1 Related literature

Closest work. Three strands are closest. Models of transient impact and resilience (Obizhaeva and Wang, 2013; Bouchaud et al., 2004, 2018) take the decay of impact as given; we derive when depletion, refill and momentum make a book overshoot, give cumulative impact in closed form, and ask who supplies the reversion that bounds it. Models of liquidity spirals (Brunnermeier and Pedersen, 2009; Shleifer and Vishny, 1992) explain how funding constraints amplify shocks; our amplification is a property of book dynamics and does not need funding constraints, and our guard is the pre-trade control that bounds the resulting loss. Sornette et al. (2023) show that non-normal interactions create transient bubbles; we give a necessary and sufficient pre-trade control for such transients and test the conditions for them in order-book data.

Market microstructure and resilience. Transient impact and resilience follow Obizhaeva and Wang (2013), Bouchaud et al. (2004) and Bouchaud et al. (2018); optimal execution with impact is in Almgren and Chriss (2001) and Cartea et al. (2015). Depth imbalance as a short-horizon price predictor is documented by Cont et al. (2014). Biais et al. (2005) survey the field; price impact as a measure of information and illiquidity goes back to Kyle (1985) and Amihud (2002). Inventory-based dealer models go back to Ho and Stoll (1981), and Avellaneda and Stoikov (2008) give the standard high-frequency market-making model. Empirically, resiliency, the speed at which a book refills after a liquidity shock, is measured by Biais et al. (1995), Degryse et al. (2005) and Large (2007),

Table 1: Map of results

Question	Answer	Where
Does $\rho(J) < 1$ bound transient loss?	No; safe radius B/M_C can vanish at fixed ρ	Thm 3.1, Companion paper
Can volatility monitors help?	No; fixed variance, unbounded gain	Companion paper
When does a book amplify?	Iff $\varphi\varepsilon > (1-a)\kappa$; with momentum $C_3 > 0$ suffices, no exact one-period rule	Thm 4.2(i), Thm 4.2(iii), Rem 4.3
Can momentum alone amplify?	Not in a stable book	Thm 4.2(iv)
How large is cumulative risk?	$S = \tilde{\kappa}/(m - q\tilde{\kappa})$	Thm 4.4
Which guard is safe and largest?	Summed absolute gain; peak gain is unsafe	Lem 6.1
Robust to estimation error?	Yes, $M/(1 - \epsilon S)$	Thm 6.3
Needs identification of algorithms?	No; inputs are invariant to cohort splits	Prop 2.2, Thm 5.5
Do learners undo the guard?	No; gains fall when admission falls, and every finite-horizon summed response rises with the gains in the cohort economy	Prop 6.11
Is reversion efficiently supplied?	No; under-provided, and the fix is priced by S	Prop 8.1
Is BTCUSDT on the amplifying side?	No: $\hat{C}_3 < 0$ on 93.5% of days	Table 7
Does depth carry an overshoot?	No; all overshoots negative	Figure 4
Does $\hat{C}_3$ forecast tail risk in BTCUSDT?	No (pre-registered)	Table 9
Is spot amplification above one?	No; unit-free gain ≈ 1.0000	Section 10
Does flow deplete thin books?	Yes on a fixed price grid; the apparent replenishment is a band-recentring artefact	Tables 15, 16
Does $\hat{C}_3$ forecast tail size in thin books?	Yes, in four pre-registered tests on Binance	Tables 17, 19, 21, 22
Is it just volatility?	No: +2.8 pts beyond HAR, EWMA and hour effects; S itself +2.1	Table 22
Is it just VPIN?	No: +2.0 pts beyond the baseline plus VPIN (post hoc); +0.56 on new symbols (pre-registered)	Section 11.10, Table 30
Is it tick size, impact or scale?	No; survives controls, unit-free versions work	Tables 18, 20, 21, 22
Does it hold on a second venue?	Bybit: absent with trades; with depth, significant but negligible	Section 11.11, Table 23
In a regulated market?	Thin US stocks: yes, +0.50 pts (pre-registered); mid-size stocks too	Table 24
Does the estimated response size positions better?	Six-hour: no; certified three-day: −2.4% tail loss, narrow pass; alone: worse	Tables 25, 26
How large is the effect?	top vs bottom quintile $\approx$ 26 bp at fixed volatility	Section 11.17
Can equal early data hide different risk?	Yes, threefold; a computable capacity loss follows	Ex. 5.1, Prop. 5.2
What does a finite window certify?	An exact Markov–Krein interval; nothing without a memory bound	Thm 5.3
Certify risk vs identify traders?	$\sigma^2\epsilon^{-2}\log(1/\epsilon)$ vs $(\Delta\eta)^{-(4K-2)}$	Thm 5.5, Table 3
Optimal guard under uncertainty?	Closed form, interval, minimax capacity loss	Thm 6.5
If admission changes the book?	Safe set can be two intervals; certified by branch and bound	Prop 6.6, Thm 6.7
If tap noise is correlated?	Simple certificate undercovers (75%); t and Hotelling certificates exact	Prop 5.7
What is disclosure worth?	Most when memory is long; noisy proxies	Thm 7.1, Table 5

who estimates it with a self-exciting point process for order events; Foucault et al. (2005) show that it is set by the mix of patient and impatient traders and the tick size. Our reversion term m is the reduced-form counterpart of resiliency, and $S = \tilde{\kappa}/(m - q\tilde{\kappa})$ shows how it combines with impact and momentum: a book that is resilient in this sense can still carry a large cumulative response when momentum feedback absorbs most of its reversion. Our book model is a minimal two-state version of resilience dynamics, chosen so that the amplification condition is exact.

Feedback, fire sales and automated markets. Liquidity spirals and fire sales are in Shleifer and Vishny (1992) and Brunnermeier and Pedersen (2009); automated-market crashes in Kirilenko et al. (2017); and the market-design case for discrete auctions in Budish et al. (2015). Menkveld (2013) documents high-frequency market makers as the new intermediaries, and algorithmic trading has been found to improve liquidity and price discovery in normal times (Hendershott et al., 2011; Brogaard et al., 2014); our question is what happens on the path after a shock. Sornette et al. (2023) show that non-normal interactions can generate transient bubbles in unleveraged systems. We add an exact condition for when a book amplifies, a control that does not need the identity of the traders, and evidence on whether the condition holds in a large market.

Early warning and liquidity crashes. The best-known early-warning metric for flow-driven crashes is VPIN (Easley et al., 2012), whose forecasting content beyond volume and volatility is disputed (Andersen and Bondarenko, 2014). Equilibrium models explain crashes through endogenous liquidity withdrawal (Huang and Wang, 2009) and illiquidity contagion across assets (Cespa and Foucault, 2014). Our statistic differs in kind: it is a structural quantity, the summed response of the book to a unit of flow, with a closed form and a control-theoretic meaning, and we evaluate it out of sample against a full volatility-forecasting baseline. In a post-hoc comparison on the same test hours, $\hat{C}_3$ keeps about 73% of its gain once VPIN is added to the baseline (Section 11.10). Crypto markets are studied as a

laboratory for microstructure and arbitrage by Makarov and Schoar (2020).

Non-normal dynamics and robust control. Transient growth of stable non-normal systems and the Kreiss matrix theorem are classical (Trefethen and Embree, 2005; Kreiss, 1962); the sharp en constant is due to Spijker (1991). Standard H_∞ control bounds frequency-domain gain under known coordinates. The time-domain worst case under bounded disturbances, the ℓ_1 norm of the impulse response, is the object of ℓ_1-optimal control (Vidyasagar, 1986; Dahleh and Pearson, 1987; Dahleh and Diaz-Bobillo, 1995), and our guard's worst-case identity is that classical fact. What we add is its use as an admission rule on a dealer's balance sheet, its invariance to the unobservable cohort coordinates and identification from prices, and its link to who supplies reversion in equilibrium. Cross-impact under no dynamic arbitrage is studied by Schneider and Lillo (2019).

Regulation. SEC Rule 15c3-5 requires broker-dealers with market access to enforce pre-trade controls on credit and capital (SEC, 2010). Our guard is a candidate pre-trade control with a provable maximality property.

Forecast evaluation. Our predictive tests use Clark and West (2007) and Diebold and Mariano (1995) with Newey and West (1987) standard errors and block resampling in the spirit of Künsch (1989), the volatility baseline follows Corsi (2009) and J.P. Morgan/Reuters (1996), local projections follow Jordà (2005), and we correct for multiple testing as in Holm (1979). Following the case for pre-registration in science (Nosek et al., 2018) and the concerns about multiple testing in financial economics (Harvey et al., 2016; Harvey, 2017), we pre-register specifications, count every evaluation of a held-out period, and report the failures with the successes.

1.2 Outline

Section 2 sets out the closed-loop market. Section 3 shows why spectral stability is not solvency. Section 4 derives the amplification condition. Section 6 constructs and analyzes the guard, and Section 8 asks who supplies reversion in equilibrium. Sections 9 and 10 report the BTCUSDT evidence, and Section 11 the thin-book program, its replications and its failures. Section 13 discusses policy, and Section 14 concludes.

2 The closed-loop market

2.1 Price formation

Time is discrete with bars of length Δ. Let p_t be the price, $p_t^* = p_{t-1}^* + \xi_t$ the unobserved fundamental and $x_t = p_t - p_t^*$ the pricing error. Competitive dealers post aggregate depth $D > 0$, so price impact is $\kappa = 1/D$, and fundamental arbitrage removes a fraction $\kappa\varphi_f$ of the error each period. With algorithmic demand u_t and liquidity noise, the pricing error follows

$$x_{t+1} = a x_t + \kappa u_t + w_{t+1}, \qquad a = 1 - \kappa\varphi_f \in [0, 1). \tag{1}$$

Algorithmic cohorts trade $u_t = \sum_i q_i y_{i,t}$ with exponential memory $y_{i,t+1} = (1-\eta_i)y_{i,t} + \eta_i x_{t+1}$, a reduced form that arises when an adaptive desk updates its belief by steady-state Kalman filtering or regularized online gradient descent and maximizes expected continuation return $\beta_i y_{i,t} u_{i,t}$ minus quadratic execution cost $\frac{1}{2}\lambda_{\mathrm{tc},i} u_{i,t}^2$ and inventory penalty $\frac{1}{2}\gamma_{\mathrm{inv},i} I_{i,t}^2$ ($I_{i,t} = (1-\delta_{\mathrm{inv}})I_{i,t-1} + u_{i,t}$, $|I_{i,t}| \le \bar{I}_i$), yielding $u_{i,t}^* = q_i y_{i,t} - \psi_{\mathrm{inv},i} I_{i,t-1}$ with $q_i = \beta_i/(\lambda_{\mathrm{tc},i} + \gamma_{\mathrm{inv},i})$ and $\psi_{\mathrm{inv},i} = \gamma_{\mathrm{inv},i}(1-\delta_{\mathrm{inv}})/(\lambda_{\mathrm{tc},i} + \gamma_{\mathrm{inv},i})$ (Niroula and Niroula, 2026; this single-agent filtering-and-control rule is distinct from the joint dealer–trader equilibrium of Section 8). The baseline equations set $\psi_{\mathrm{inv},i} = 0$, the limit in which inventory holding penalties are small relative to execution costs ($\gamma_{\mathrm{inv},i} \ll \lambda_{\mathrm{tc},i}$) or inventory is reset at the bar horizon; Proposition 2.3 below relaxes this and gives the closed-form summed response when cohorts carry inventory

across bars with $\psi_{\text{inv},i} > 0$. Stacking $z_t = (x_t, y_{1,t}, \ldots, y_{K,t})^\top$ gives $z_{t+1} = Jz_t + bw_{t+1}$, where $b = (1, \eta_1, \ldots, \eta_K)^\top$ because cohorts update on the same period's price (a unit of flow enters through κb), and

$$J = \begin{pmatrix} a & \kappa q_1 & \cdots & \kappa q_K \\ \eta_1 a & 1 - \eta_1 + \eta_1 \kappa q_1 & \cdots & \eta_1 \kappa q_K \\ \vdots & & \ddots & \vdots \\ \eta_K a & \eta_K \kappa q_1 & \cdots & 1 - \eta_K + \eta_K \kappa q_K \end{pmatrix}. \tag{2}$$

2.2 Stability and the role of depth

Proposition 2.1 (Stability with one cohort)**.** *With one cohort (q, η) and $a = 1 - \kappa\varphi_f$, the market is Schur stable if and only if*

$$q < \varphi_f, \qquad \kappa < \frac{4 - 2\eta}{(2-\eta)\varphi_f - \eta q} \ \text{ when } (2-\eta)\varphi_f > \eta q, \qquad |1 - \kappa\varphi_f|(1-\eta) < 1. \tag{3}$$

Higher impact κ (thinner depth) lowers price persistence a; feedback is destabilizing only when cohort aggressiveness q reaches the arbitrage strength φ_f.

The condition $q < \varphi_f$ is the familiar statement that momentum demand must be weaker than value arbitrage. The upper bound on κ rules out oscillatory instability when the book is very thin.

2.3 What an observer can use

The companion paper shows that prices identify only the merged rate–weight measure $M = \sum_i q_i \eta_i \delta_{1-\eta_i}$ of the cohorts. A pre-trade control must therefore depend on the market only through objects that are the same for every cohort representation of M.

Proposition 2.2 (Input–output invariance)**.** *Two populations with the same merged measure generate the same transfer function $H(z) = \sum_k g_k z^{-k} = \kappa z/[z - a - \kappa G(z)]$, $G(z) =$*

$\sum_i q_i\eta_i z/(z-1+\eta_i)$, *and the same impulse coefficients* $g_k = \kappa\, c_{\text{price}}^\top J^k b$ *for every* $k \geq 0$. *Under any invertible change of state coordinates* $z \mapsto Tz$, *the Euclidean gain* $\|CT^{-1}(TJT^{-1})^k\|_2$ *depends on* T, *whereas the price path*

$$\Delta p_{t+k} = \sum_{j\geq 1} g_{k-1+j} u_{t-j} + \sum_{m=0}^{k-1} g_{k-1-m} u_{t+m}$$

and its worst case over $\|u\|_\infty \leq \bar{u}$ *depend only on* $\{g_k\}$ *and are invariant to both* T *and cohort splitting.*

Numerically, splitting a cohort of 1,000 random economies into two with the same memory rate leaves every impulse coefficient unchanged up to rounding ($\max_k |g_k(\mathcal{E}^A) - g_k(\mathcal{E}^B)| \leq 2.5 \times 10^{-12}$ over 200 lags; `recompute_oct20.py`). Proposition 2.2 is the reason we build the guard on impulse coefficients and avoid state norms in latent coordinates. When a state norm is needed, we fix units by observable calm-market innovation standard deviations, which is what makes the spot statistic of Section 10 unit-free.

Proposition 2.3 (Summed response with carried cohort inventory)**.** *Suppose each cohort* i *carries inventory* $I_{i,t} = (1-\delta_{\text{inv}})I_{i,t-1} + u_{i,t}$ *with* $\delta_{\text{inv}} \in (0,1]$ *and submits order flow* $u_{i,t} = q_i y_{i,t} - \psi_i I_{i,t-1}$ *with* $q_i \geq 0$ *and* $\psi_i \geq 0$. *Then:*

(i) The closed-loop transfer function from a flow impulse to price is $H_{\text{inv}}(z) = \kappa z/\big[z - a - \kappa G_{\text{inv}}(z)\big]$ *with*

$$G_{\text{inv}}(z) = \sum_{i=1}^{K} \frac{q_i\eta_i z(z-1+\delta_{\text{inv}})}{(z-1+\eta_i)(z-1+\delta_{\text{inv}}+\psi_i)},$$

which is invariant to splitting any cohort that shares (η_i, ψ_i). *Whenever the augmented state matrix is Schur stable, the signed cumulative price response is*

$$S(\psi) = \sum_{t=0}^{\infty} h_t(\psi) = H_{\text{inv}}(1) = \frac{\kappa}{1 - a - \kappa\Phi_{\text{eff}}(\psi)}, \qquad \Phi_{\text{eff}}(\psi) \equiv \sum_{i=1}^{K} \frac{\delta_{\text{inv}}}{\delta_{\text{inv}} + \psi_i}\, q_i \leq \Phi. \tag{4}$$

(ii) If $0 \leq \psi_i \leq 1 - \delta_{\text{inv}}$ *and* $\delta_{\text{inv}} \geq \eta_i$ *for each* i *and* $a + \kappa\Phi_{\text{eff}}(\psi) < 1$, *every open-loop*

feedback tap $k_m(\psi)$ and hence every closed-loop impulse coefficient $h_t(\psi)$ is nonnegative, so the worst-case ℓ_1 gain equals the signed sum, $\|h(\psi)\|_1 = \sum_{t=0}^{\infty} |h_t(\psi)| = S(\psi)$. When ψ_i is large enough that inventory payback reverses the sign of late taps, $\|h(\psi)\|_1 > S(\psi)$.

Inventory mean-reversion $\psi_i > 0$ acts as an automatic stabilizer that attenuates each cohort's long-run feedback weight from q_i to $\frac{\delta_{\text{inv}}}{\delta_{\text{inv}}+\psi_i} q_i$ (across 500 random stable economies (4) matches the simulated state-space impulse sum to 1.78×10^{-15}, and in the monotone regime $\|h(\psi)\|_1 = S(\psi)$ to 1.78×10^{-15}, reducing S from 2.7778 at $\psi = 0$ to 2.5000 at the calibrated worked example; in the sign-reversal regime $\|h(\psi)\|_1 = 2.0960 > S(\psi) = 1.7241$; `recompute_p2_step2_theory.py`).

Proposition 2.4 (Quote cost of ambiguity)**.** *For a one-unit ticket with adverse mark $Z_\theta \sim N(\mu_\theta, \sigma_\theta^2)$ and a CARA dealer with risk aversion γ, let $c_\theta = \mu_\theta + \gamma\sigma_\theta^2/2$. Under competition with zero robust certainty equivalent, the informed quote is c_{θ_0} and the ambiguous quote $\sup_\theta c_\theta$; with inverse demand $(V_0 - V)/b$ and marginal cost c, raising an interior quote from s_0 by Δ costs $b(s_0 - c)\Delta + \frac{b}{2}\Delta^2$ in surplus.*

Ambiguity about the algorithms is costly only when it changes the worst relevant cost. A guard that bounds worst-case transient exposure can therefore narrow the ambiguity premium; whether it does in a real market is an empirical question we do not answer here.

2.4 Discussion of the assumptions

Linearity and small shocks. All dynamics are linear. This is the right approximation for the transient response to flow shocks of ordinary size, which is what a pre-trade control must bound one step ahead; it is not a model of regime changes such as exchange outages, where nonlinear liquidity withdrawal dominates.

One book, one asset. Sections 4–6 treat a single instrument; Proposition 3.2 shows how directed cross-impact enlarges transient gains in a network. Cross-margined portfolios need the multi-asset operator, which we leave to future work.

Bounded shocks. The guard bounds losses for shocks in a declared envelope. Choosing the envelope is a risk-management decision; a larger envelope trades admitted flow for safety. The theorems hold for any envelope, and the empirical work does not depend on it.

Observables. Every input to the guard (impulse coefficients, an operator estimate and its confidence radius, inventory and margin) is observable to a dealer or venue. Nothing requires the identities or memory parameters of the algorithms.

3 Spectral stability is not solvency

Theorem 3.1 (Safe radius, Kreiss bounds and the failure of the spectral radius)**.** *Let* $z_t = J^t z_0$ *with* $\rho(J) < 1$*, loss budget* $B > 0$ *and loss output* C*.*

(i) The largest initial-state radius with $\sup_t \|CJ^t z_0\|_2 \le B$ *for every* $\|z_0\|_2 \le r$ *is*

$$r^*_{\text{exact}} = \frac{B}{M_C}, \qquad M_C \equiv \sup_{t\ge 0} \|CJ^t\|_2. \tag{5}$$

(ii) With $C = I$ *and the Kreiss constant* $\mathcal{K}(J) = \sup_{|z|>1}(|z|-1)\|(zI-J)^{-1}\|_2$*, the Kreiss matrix theorem gives* $\mathcal{K}(J) \le \sup_t \|J^t\|_2 \le e\,n\,\mathcal{K}(J)$ *for* $J \in \mathbb{R}^{n\times n}$*, hence*

$$\frac{B}{e\,n\,\mathcal{K}(J)} \;\le\; r^*_{\text{exact}} \;\le\; \frac{B}{\mathcal{K}(J)}. \tag{6}$$

(iii) For any $\rho \in (0,1)$ *and any* $B > 0$ *there is a family* J_θ *with* $\rho(J_\theta) = \rho$ *for all* θ *and an output* $C = (1,0)$ *such that* $r^*_{\text{exact}}(J_\theta) \to 0$ *as* $\theta \to \infty$*.*

Table 2 illustrates the theorem for $J_\theta = \left(\begin{smallmatrix} 0.8 & \theta \\ 0 & 0.4 \end{smallmatrix}\right)$: the spectral radius is 0.8 throughout, while the price-row gain rises from 1 to 120 and the safe radius falls by the same factor. The Kreiss constant tracks the gain within the factor en.

The theorem concerns a specified initial-state ball and output norm. Changing state

Table 2: Safe radius and Kreiss bounds at a fixed spectral radius

θ	$\rho(J_\theta)$	M_C (price row)	$\sup_t \|J_\theta^t\|$	$\mathcal{K}(J_\theta)$	r^*_{exact} $(B = 1)$
0	0.8	1.00	1.00	1.00	1.000
1	0.8	1.36	1.37	1.06	0.735
5	0.8	6.03	6.04	3.43	0.166
20	0.8	24.01	24.01	13.42	0.042
100	0.8	120.00	120.00	66.99	0.008

$J_\theta = \left(\begin{smallmatrix} 0.8 & \theta \\ 0 & 0.4 \end{smallmatrix}\right)$; Kreiss constant by grid search over $|z| > 1$.

coordinates changes the disturbance set, so latent-state units cannot be read as monetary losses; this is why Section 6 works with input–output gains instead.

Spectral blindness. The companion paper shows more: for any eigenvalues $0 < \lambda_2 < \lambda_1 < 1$, one-step price variance and sufficiently large stationary price variance, there are two-state markets sharing all three quantities whose worst-case transient gain is arbitrarily large (Niroula and Niroula, 2026, spectral-blindness theorem), so neither eigenvalue nor volatility monitors bound transient loss.

At $\lambda = (0.8, 0.7)$ with one-step variance 1 and stationary variance 5, the gain is 1.77, 5.10, 16.95, 50.85 and 169.50 as the off-diagonal coupling rises from 1 to 100. Figure 1 shows the geometry for a constructed matrix with spectral radius 0.5: its pseudospectra cross the unit circle, and the price-row gain grows with the coupling before decaying.

No amplification without depth. In the cohort economy (2) with nonnegative weights and $\rho(J) < 1$, the price impulse response is a nonnegative mixture of decaying exponentials with total weight κ; it is completely monotone and peaks at the direct impact (Niroula and Niroula, 2026, positive-geometry lemma).

This is a negative result with a positive use. The cohort matrix (2) is non-normal, yet its price output never amplifies. Non-normality is necessary for price amplification but not sufficient, and adaptive momentum feedback alone does not supply it. Section 4 shows that depth depletion does.

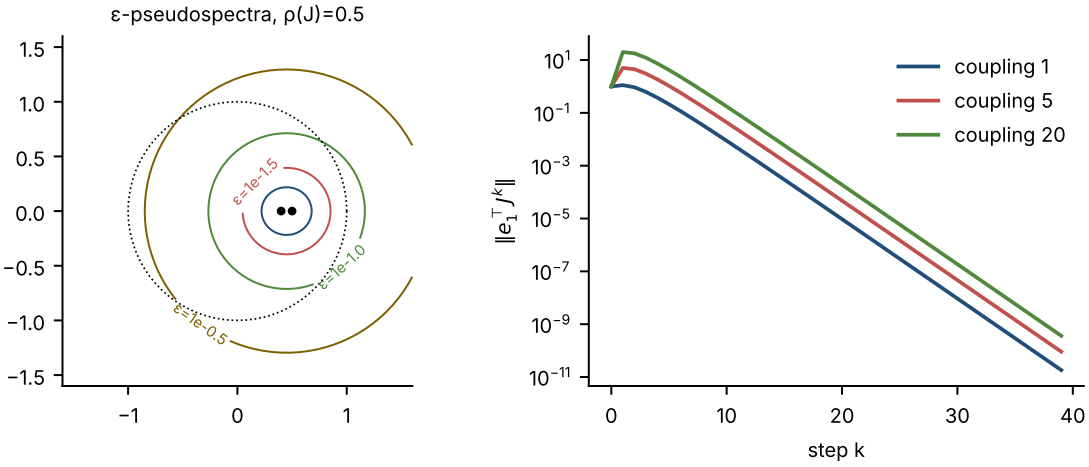


Figure 1: Left: ε-pseudospectra of $J = \left(\begin{smallmatrix} 0.5 & 5 \\ 0 & 0.4 \end{smallmatrix}\right)$; the eigenvalues (dots) are well inside the unit circle (dotted), but the $\varepsilon = 10^{-1}$ contour reaches beyond it. Right: price-row gain $\|e_1^\top J^k\|$ for couplings 1, 5 and 20.

3.1 Multi-asset coupling

Market makers trade many correlated instruments. Let $x_t \in \mathbb{R}^d$ be pricing errors and $u_t \in \mathbb{R}^d$ flows, with $x_{t+1} = Ax_t + Ku_t + w_{t+1}$, $A = \mathrm{diag}(a_1, \ldots, a_d)$, a cross-impact matrix K, and momentum flow $y_{t+1} = (I - H)y_t + Hx_{t+1}$, $u_t = Qy_t$. The stacked Jacobian is

$$J_{\text{multi}} = \begin{pmatrix} A & KQ \\ HA & I - H + HKQ \end{pmatrix} \in \mathbb{R}^{2d \times 2d}. \tag{7}$$

Proposition 3.2 (Directed cross-impact shrinks the safe radius)**.** *Suppose a leader–satellite structure ($K_{12} = 0$, $K_{21} \geq 0$) with $A, Q, K \geq 0$ entrywise, $H \in (0,1)^d$, stable diagonal blocks and a fixed output $C \geq 0$. Let J_{diag} be the decoupled Jacobian with $K_{21} = 0$. Then every power J_{multi}^t is entrywise non-decreasing in K_{21}, so $M_C(J_{\text{multi}}) \geq M_C(J_{\text{diag}})$ and $r^*_{\text{exact}}(J_{\text{multi}}) \leq r^*_{\text{exact}}(J_{\text{diag}})$; likewise $\mathcal{K}(J_{\text{multi}}) \geq \mathcal{K}(J_{\text{diag}})$.*

The proposition concerns nonnegative downstream coupling under a fixed output and shock set. Hedged or signed monetary outputs need separate treatment, and the size of the effect requires an empirical calibration we do not provide. Persistence and transient

amplification are nevertheless different things: in a network of identical cohort economies coupled by symmetric nonnegative cross-impact, which stays symmetric in suitable coordinates, cross-impact of 0.05 raises the spectral radius from 0.822 ($d = 1$) to 0.973 ($d = 10$) while the Kreiss constant stays at one (`recompute_oct20.py`). Directed coupling, as in Proposition 3.2, is what turns persistence into amplification.

4 When does a book amplify? An exact condition

Transient path risk takes two forms, and the distinction organizes both the theory and the evidence. The first is *peak overshoot*, a response that rises above its direct impact; Theorem 4.2 shows that it requires depth depletion to outweigh impact-weighted reversion. The second is *cumulative amplification*, a summed response S much larger than direct impact; Theorem 4.4 shows that it arises without any depletion whenever reversion is slow relative to impact and momentum. The guard of Section 6 bounds the second. The data of Sections 9–11 show no peak overshoot in crypto books, even though flow depletes the side it hits on a fixed price grid, and they support the second form in thin books.

4.1 A book with depletion and refill

Let x_t be the price displacement, d_t the signed depth imbalance (bid minus ask depth near the mid, relative to total) and u_t signed order flow. Flow moves the price directly and depletes one side of the book, which refills at rate r; depth imbalance moves the price:

$$x_t = a\, x_{t-1} + \varphi\, d_{t-1} + \kappa\, u_t, \tag{8}$$

$$d_t = (1 - r)\, d_{t-1} + \varepsilon\, u_t, \tag{9}$$

with $\kappa > 0$, $a \in [0, 1)$ and $r \in (0, 1]$. Write $b = 1 - r$. With the sign convention $d = (B - A)/(B + A)$, a buy that consumes ask depth raises d, so $\varepsilon > 0$ means *depletion* of the

side that is hit, and $\varphi > 0$ means that a bid-heavy book pushes the price up. The impulse response to a unit flow shock at $t = 0$ is $h_0 = \kappa$ and

$$h_k = a\, h_{k-1} + \varphi\varepsilon\, b^{k-1}, \qquad k \geq 1. \tag{10}$$

Definition 4.1. The book *amplifies* if $\max_k h_k > h_0$: the price keeps moving in the direction of the shock after the shock has passed.

4.2 Peak overshoot

Add momentum traders whose flow $f_t = (1-\eta)f_{t-1} + \eta q\, x_{t-1}$ enters both (8) and (9) alongside u_t; the two-state model is the case $q = 0$. Write $C_2 \equiv \varphi\varepsilon - (1-a)\kappa$ and $C_3 \equiv \varphi\varepsilon + \kappa^2\eta q - (1-a)\kappa$.

Theorem 4.2 (Peak overshoot: depletion versus momentum)**.** *(i)* Two-state condition. *Without momentum ($q = 0$) the book amplifies if and only if $C_2 > 0$; when $C_2 \leq 0$ the response peaks at impact,* $\max_k h_k = \kappa$.

(ii) Shape. *Let $q = 0$, $0 < a < 1$ and $0 < b < 1$. If $a \neq b$, $h_k = Aa^k + Bb^k$ with $B = \varphi\varepsilon/(b-a)$ and $A = \kappa - B$; if $a = b$, $h_k = a^{k-1}(a\kappa + \varphi\varepsilon k)$ for $k \geq 1$. When the book amplifies, h viewed as a function of real $k \geq 0$ has a single maximum, at*

$$k^* = \frac{\log\!\big(-A\log a/(B\log b)\big)}{\log(b/a)} \quad (a \neq b), \qquad k^* = \frac{1}{\log(1/a)} - \frac{a\kappa}{\varphi\varepsilon} \quad (a = b),$$

the second being the limit of the first as $b \to a$; the peak lag is $\lfloor k^* \rfloor$ or $\lceil k^* \rceil$ (and is 1 when $k^* < 1$, because $h_1 > h_0$). The cumulative response is $\sum_k h_k = \kappa/(1-a) + \varphi\varepsilon/[(1-a)r]$ in all cases, and faster refill lowers both the peak and the cumulative response. When $a = 0$ ($h_k = \varphi\varepsilon b^{k-1}$) or $r = 1$ ($h_k = a^{k-1}(a\kappa + \varphi\varepsilon)$) the response peaks at lag 1.

(iii) One-period margin with momentum. $h_1 - h_0 = C_3$, *so* $C_3 > 0$ *implies that the book amplifies.*

(iv) Momentum alone cannot amplify. *If* $\varphi\varepsilon = 0$, $q \geq 0$ *and the system is stable, then* $h_k = c_1\lambda_1^k + c_2\lambda_2^k$ *with residues* $c_1, c_2 \geq 0$, $c_1 + c_2 = \kappa$ *and* $|\lambda_2| \leq \lambda_1 = \rho < 1$, *so* $h_k \leq \kappa\rho^k < \kappa$ *for every* $k \geq 1$*; in particular* $C_3 < 0$.

(v) A sufficient condition for no overshoot. *For any signs of* φ, ε *and* q, *if*

$$\bar{C}_3 \equiv |\varphi\varepsilon|\Big(1 + \frac{\kappa|q|}{r}\Big) + \kappa^2|q| - (1-a)\kappa \;\leq\; 0,$$

then $\sup_k |h_k| = \kappa$. *When* $\varphi\varepsilon \geq 0$ *and* $q \geq 0$, $C_3 > 0 \Rightarrow$ *amplification* $\Rightarrow \bar{C}_3 > 0$.

Proof. (i) *If.* $h_1 - h_0 = a\kappa + \varphi\varepsilon - \kappa = C_2 > 0$. *Only if.* Suppose $C_2 \leq 0$. If $\varphi\varepsilon < 0$, then by (10) $h_k = a^k\kappa + \varphi\varepsilon\sum_{j<k} a^{k-1-j}b^j \leq a^k\kappa \leq \kappa$. If $\varphi\varepsilon \geq 0$, let $\Delta_k = h_k - h_{k-1}$. Then $\Delta_1 = C_2 \leq 0$ and, from (10), $\Delta_{k+1} = a\Delta_k + \varphi\varepsilon(b^k - b^{k-1}) = a\Delta_k - \varphi\varepsilon\, r\, b^{k-1} \leq a\Delta_k$, so by induction $\Delta_k \leq 0$ for all k and h peaks at $h_0 = \kappa$.

(ii) Solving (10) gives the two forms. For $a \neq b$, $\frac{d}{dk}h = Aa^k \log a + Bb^k \log b$ has at most one zero because $(b/a)^k$ is monotone, which gives k^*; for $a = b$, $\frac{d}{dk}a^{k-1}(a\kappa + \varphi\varepsilon k) = a^{k-1}[(a\kappa + \varphi\varepsilon k)\log a + \varphi\varepsilon]$ vanishes once, at the stated k^*, and expanding the first formula in $b - a$ gives the same limit. A unimodal function of real k attains its integer maximum at $\lfloor k^* \rfloor$ or $\lceil k^* \rceil$. Summing the geometric series gives the cumulative response, which falls in r; the peak falls in r because h_k is decreasing in r term by term for $\varphi\varepsilon > 0$.

(iii) With a unit shock at $t = 0$, $x_0 = \kappa$, $d_0 = \varepsilon$ and $f_1 = \eta q\kappa$, so $h_1 = a\kappa + \varphi\varepsilon + \kappa^2\eta q$.

(iv) Without the depth channel the state (x_t, f_t) evolves by $M = \left(\begin{smallmatrix} a+\kappa\eta q & \kappa(1-\eta) \\ \eta q & 1-\eta \end{smallmatrix}\right)$, which is entrywise nonnegative, and $h_k = \kappa(M^k)_{11}$. The off-diagonal product is nonnegative, so the eigenvalues are real, and by the Perron–Frobenius theorem $\lambda_2 \leq m_{11} \leq \lambda_1 = \rho$. Writing $(M^k)_{11} = c_1'\lambda_1^k + c_2'\lambda_2^k$ with $c_1' = (m_{11} - \lambda_2)/(\lambda_1 - \lambda_2) \in [0,1]$ and $c_2' = 1 - c_1'$ gives the residues $c_j = \kappa c_j' \geq 0$ and $(M^k)_{11} \leq \rho^k$. Since $m_{11} \leq \rho < 1$, $C_3 = \kappa(m_{11} - 1) < 0$.

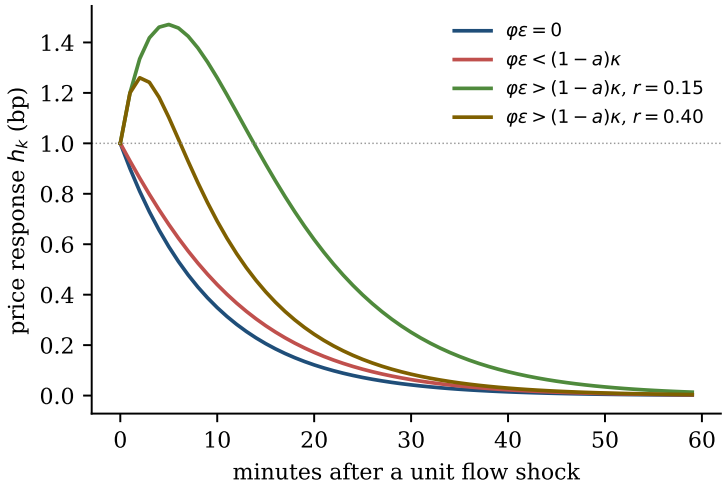


Figure 2: Illustrative impulse responses of model (8)–(9) ($a = 0.9$, $\kappa = 1$, $\varepsilon = 0.1$). The response overshoots only when $\varphi\varepsilon > (1-a)\kappa$; faster refill r lowers the overshoot. Parameters are illustrative, not estimates.

(v) Suppose $|x_j| \le \kappa$ for $j < k$. Then $|f_j| \le |q|\kappa$ for $j \le k$ and $|d_{k-1}| \le |\varepsilon|\big(b^{k-1} + \sum_{j\ge1} b^{k-1-j}|q|\kappa\big) \le |\varepsilon|(1 + |q|\kappa/r)$, so $|x_k| \le a\kappa + |\varphi\varepsilon|(1 + \kappa|q|/r) + \kappa^2|q| \le \kappa$. Induction from $x_0 = \kappa$ completes the proof. □

Interpretation. The left side $\varphi\varepsilon$ of C_2 is the extra price pressure that a unit of flow creates one period later through the book it leaves behind. The right side $(1-a)\kappa$ is the part of the direct impact that reverts in one period. The book amplifies exactly when the depth channel adds more than mean reversion removes. Refill r does not enter the condition; it governs how long an overshoot lasts.

We verify part (i) in 1,995 random parameter draws and the refill comparative static in 565 amplifying draws, with no exceptions. The peak lag lies in the stated bracket in all of 150,039 amplifying draws with $a \neq b$ and all of 20,000 confluent draws; it is not always the integer nearest to k^*, which an earlier version claimed and which fails in 15% of draws (`recompute_audit.py`, `hardening/p2_sec4_checks.py`). Figure 2 illustrates the four cases.

With two slow modes the converse of part (iii) fails, and no condition on the one-period margin is exact.

Remark 4.3 (Counterexamples to an exact one-period condition). Between C_3 and $\bar{C}_3$ the sign of C_3 does not decide. With $\kappa = 1$, $a = 0.98526$, $\varphi = 0.7409$, $\varepsilon = 0.01183$, $r = 0.03436$, $\eta = 0.5092$ and $q = 0.01127$ (all channels reinforcing, $\rho = 0.9995$), $C_3 = -0.00024$ but the response peaks at lag 60 at 1.092κ. With replenishment ($\varphi = 2.614$, $\varepsilon = -0.405$, $a = 0.9426$, $r = 0.033$, $\eta = 0.0016$, $q = 0.0785$, $\rho = 0.998$), $C_3 = -1.11$ yet the response oscillates to 9.2κ at lag 90. In a search over 400,000 parameter draws such cases occur only near the stability boundary or with slow refill and slow momentum.

The spectral reason is visible in the first example. Without depth, part (iv) makes every residue nonnegative, so the response is a nonnegative mixture of decaying modes and cannot rise above κ; we found no negative residue in 10,694 stable momentum-only draws. The depth channel breaks this: the three-state response is $h_k = 1.145\,(0.9995)^k - 0.155\,(0.9626)^k + 0.011\,(0.4853)^k$, with a negative residue on the intermediate mode. That mode subtracts from the response early and dies out faster than the slowest one, so the response first dips ($h_1 < h_0$, hence $C_3 < 0$) and then rises toward the slow mode until lag 60.

4.3 Cumulative transient risk

Theorem 4.4 (Summed response in closed form)**.** *Let $\tilde{\kappa} = \kappa + \varphi\varepsilon/r$ (long-run impact: direct plus depth) and $m = 1 - a$ (reversion). In the stable three-state model,*

$$S \equiv \sum_{k \geq 0} h_k = \frac{\tilde{\kappa}}{m - q\tilde{\kappa}},$$

and when $\varphi\varepsilon \geq 0$ and $q \geq 0$ the system is stable only if $m > q\tilde{\kappa}$. When all $h_k \geq 0$, S is the summed absolute response that enters the guard of Section 6.

Proof. $S = e_1^\top (I - M_3)^{-1} (\kappa, \varepsilon, 0)^\top$ with M_3 the transition matrix of (x, d, f). Direct computation gives $\det(I - M_3) = \eta\,[\,r(m - \kappa q) - \varphi\varepsilon q\,]$ and the stated ratio; the formula matches the simulated sums to a relative error of 7×10^{-14} in 3,000 stable draws. If $\varphi\varepsilon \geq 0$ and $q \geq 0$, M_3 is entrywise nonnegative and stability requires $\det(I - M_3) > 0$, i.e. $m > q\tilde{\kappa}$. □

Cumulative transient risk is long-run impact divided by reversion net of momentum. Two practical consequences follow. First, the statistic that matters for losses is the summed response, which grows without bound as reversion falls to $q\tilde{\kappa}$, even when no single lag overshoots (Theorem 4.2(iv)). Second, C_3 is best read as a one-period amplification margin: it moves with m, q and $\tilde{\kappa}$ in the same direction as S and is what we estimate in Sections 9 and 11. Both statistics are invariant in sign to rescaling flow.

5 Identification and financial safety

The companion paper shows that identifying the cohorts of algorithms from prices is expensive: separating K cohorts whose memories differ by $\Delta\eta$ takes a number of experiments of order $(\Delta\eta)^{-(4K-2)}$, and with general memory accuracy improves only as $1/\log N$ (Niroula and Niroula, 2026). A pre-trade control does not need the cohorts; it needs a bound on the risk they create. This section asks what information certifies a safe trading decision, what that information costs, and how much safe trading must be given up when it is missing.

5.1 Observation and risk contracts

We work in the positive cohort economy (2) with $q_i \geq 0$ and known a and κ. Its merged measure $M = \sum_i q_i \eta_i \delta_{r_i}$, $r_i = 1 - \eta_i$, has kernel $k_m = \int r^m \, dM$, and the price response to a unit of flow obeys $h_0 = \kappa$, $h_{t+1} = a h_t + \kappa \sum_{m \leq t} k_m h_{t-m}$. All $h_t \geq 0$ (Niroula and Niroula, 2026, positive-geometry lemma), and the summed response is

$$S = \sum_t h_t = \frac{\kappa}{1 - a - \kappa\Phi}, \qquad \Phi = \int \frac{dM(r)}{1 - r} = \sum_i q_i, \tag{11}$$

with stability exactly when $a + \kappa\Phi < 1$. A *memory bound* $\bar{r} < 1$ states that every memory rate satisfies $r_i \leq \bar{r}$.

Risk contract. A position I is held fixed, each period's flow shock is bounded by $\bar{u}$, losses are marked to market at every date, and the horizon is unbounded. By Lemma 6.1 the

worst-case loss is $|I|\bar{u}S$, so the safe capacity, the largest shock bound with loss at most the budget $\mathcal{B}$, is $\mathcal{B}/(|I|S)$. Section 6 adds admission, margin and fills. All statements below are deterministic worst cases under this contract; none is a probability-of-ruin statement.

Observation contracts. We consider four: (O1) N independent reset experiments, each returning $h_0, \ldots, h_{T-1}$ with independent Gaussian errors of standard deviation σ per tap, the controlled-probe contract of the companion paper; (O2) an exact window $h_0, \ldots, h_n$; (O3) class-tagged flows; and (O4) audited exposure, which reveals Φ. The theorems concern these contracts. Observational public records of a live book are a different contract, for which we offer evidence (Section 12.2) but no finite-sample guarantee.

Given data D, let $\Theta(D)$ be the set of economies compatible with it. A decision is *certifiably safe* if its worst-case loss is within budget for every economy in $\Theta(D)$, *certifiably unsafe* if it exceeds the budget for every one, and *undecidable* otherwise. This classification is a definition, not a result; it is the finance version of the data-informativity view of control without identification (van Waarde et al., 2020) and of set-membership estimation (Milanese and Vicino, 1991). The results below make it explicit for this economy.

5.2 Two markets that look the same

Example 5.1 (An exact ambiguity witness). Let $a = 0.4$ and $\kappa = 1$. Market A has one cohort ($q = 0.08$, $\eta = 0.5$); market B has two ($q = 0.02/0.95$, $\eta = 0.95$; $q = 0.4$, $\eta = 0.05$). Their merged measures have the same mass and first moment, so in rational arithmetic their price responses coincide at lags 0, 1 and 2 (1, 11/25, 267/1250) and first differ at lag 3. Their summed responses are 1.92 and 5.59, so for a unit position and budget the safe shock bound is 0.52 in A and 0.18 in B.

Proposition 5.2 (Unavoidable loss of safe capacity)**.** *Let two economies have safe capacities $v_0 < v_1$ and data laws P_0, P_1. If a data-driven rule $A(D) \in [0, v_1]$ admits an unsafe level in*

economy 0 with probability at most β, $P_0(A > v_0) \le \beta$, then

$$\mathbb{E}_1[v_1 - A(D)] \;\ge\; (v_1 - v_0)\,\big[1 - \beta - \mathrm{TV}(P_0, P_1)\big]_+.$$

The bound is attained when $P_0 = P_1$ by the rule that admits v_1 with probability β and v_0 otherwise.

Proof. Let $E = \{A \le v_0\}$. Then $P_0(E) \ge 1 - \beta$ and $P_1(E) \ge P_0(E) - \mathrm{TV}$. On E the capacity given up in economy 1 is at least $v_1 - v_0$, and it is nonnegative elsewhere. □

The argument is the two-point method of statistical decision theory (Le Cam, 1986); its use here is to price missing information in units of safe trading. In Example 5.1, a supervisor who sees the first three coefficients and must admit an unsafe level in B with probability at most 5% gives up at least $0.95 \times 0.34 = 0.32$ units of safe shock capacity in A.

5.3 What a finite window certifies

Lags 1 to n of the response determine the first n moments of M exactly, through $k_t = (h_{t+1} - ah_t - \kappa \sum_{m<t} k_m h_{t-m})/\kappa^2$. The summed response depends on M only through $\Phi = \int f\, dM$ with $f(r) = (1-r)^{-1}$. Certifying it from a window is therefore an extremal moment problem.

Theorem 5.3 (Finite-window certification)**.** *Observe $h_0, \dots, h_n$ exactly under a memory bound $\bar{r} < 1$, and suppose the implied moment vector $(k_0, \dots, k_{n-1})$ lies in the interior of the moment space $\mathcal{M}_n([0, \bar{r}])$ (equivalently, the Hankel matrices of the moment sequence and of its localization by $r(\bar{r} - r)$ are positive definite). Let μ_- and μ_+ be its lower and upper principal representations on $[0, \bar{r}]$ (the extremal measures of the Markov–Krein theorem, unique in the interior; μ_+ has an atom at $\bar{r}$), and $\Phi_\pm = \int f\, d\mu_\pm$. On the boundary of the moment space the compatible measure is unique and $\Phi_- = \Phi_+$.*

(i) Sharp interval. *The summed responses of stable economies compatible with the data form the interval $[S(\Phi_-), S(\Phi_+)]$ if $a + \kappa\Phi_+ < 1$, and $[S(\Phi_-), \infty)$ otherwise. Both*

endpoints are attained by economies with at most $\lfloor n/2 \rfloor + 1$ *cohorts.*

(ii) Geometric convergence. $\Phi_+ - \Phi_- \le 2k_0 E_{n-1}$, *where* $E_{n-1} = (2/\bar{r})\,\varrho^{n-1}/(x_0^2 - 1)$ *is the error of the best uniform approximation of* f *on* $[0, \bar{r}]$ *by polynomials of degree* $n-1$, $x_0 = 2/\bar{r} - 1$ *and* $\varrho = x_0 - \sqrt{x_0^2 - 1} < 1$. *A window of length* $n = O\big(\log(1/\epsilon)/\log(1/\varrho)\big)$ *certifies* Φ *to precision* ϵ, *and* $\log(1/\varrho) \sim 2\sqrt{(1-\bar{r})/\bar{r}}$ *as* $\bar{r} \to 1$.

(iii) No certificate without a memory bound. *If memory rates may lie anywhere in* $[0, 1)$, *then for every* n *and every window whose moment vector is interior to the moment space, compatible stable economies have arbitrarily large summed responses: no finite window certifies any finite risk bound.*

Part (i) is an application of the Markov–Krein theorem (Krein and Nudel'man, 1977; Karlin and Studden, 1966): since $f^{(n)}(r) = n!/(1-r)^{n+1} > 0$, the extremes of $\int f\, d\mu$ over measures with given moments are attained at the principal representations. Part (ii) uses Chebyshev's formula for the best approximation of $1/(x_0 - x)$ (Achieser, 1956). The proofs are in Appendix A. Theorem 5.3 turns the three-way classification into a computation: a decision is certifiably safe from the window when $|I|\bar{u}S(\Phi_+) \le \mathcal{B}$, certifiably unsafe when $|I|\bar{u}S(\Phi_-) > \mathcal{B}$, and undecidable in between. In an eight-cohort economy with memory bound 0.95 and true summed response 5.56, the window of lags 0–3 certifies nothing ($a + \kappa\Phi_+ \ge 1$), lags 0–4 certify $S \le 92.8$, lags 0–6 give $[5.01, 7.20]$ and lags 0–8 give $[5.30, 6.00]$; the widths stay inside the bound of part (ii) at every window. For fixed data the upper bound grows without limit as the memory bound approaches one ($\Phi_+ = 2.42$, 9.45, 79.7 and 782 for $\bar{r} = 0.9$, 0.99, 0.999 and 0.9999, from the first four moments of a three-cohort economy), as part (iii) says (`research_extension/information_safety`).

The endpoints in these examples are computed, not proved, in floating point. The next result makes them proved.

Lemma 5.4 (Certified endpoints)**.** *Let* p_+ *and* p_- *be polynomials of degree at most* $n-1$, $p_\pm(r) = \sum_j c_j^\pm r^j$, *with* $p_- \le f \le p_+$ *on* $[0, \bar{r}]$. *Then every economy compatible with the*

window has $\sum_j c_j^- k_j \le \Phi \le \sum_j c_j^+ k_j$. *The Hermite interpolants of* f *at the atoms of* μ_+ *and* μ_-, *with derivative conditions at interior atoms, satisfy the sign conditions and give* Φ_+ *and* Φ_- *exactly.*

Lemma 5.4 turns any computed pair of polynomials into a proof, whatever the rounding errors that produced them: if the sign conditions are verified, the bounds hold. We compute the interpolants in 60-digit arithmetic, round them to double precision, add a slack of 10^{-11} of Φ, verify the sign conditions on $[0, \bar{r}]$ in interval arithmetic with outward rounding (a mean-value form on an adaptive subdivision; Rump, 2010), and evaluate the bounds with outward rounding. In the eight-cohort example the certified intervals for S are $[5.01, 7.20]$ and $[5.30, 6.00]$ for lags 0–6 and 0–8, and every certified endpoint lies outside the 60-digit one by at most 1.5×10^{-8} of its value, using about 62,000 subintervals (test T10; the exact count depends on the platform's floating-point library, the certified bounds do not). The interval endpoints quoted in this paper are therefore results, not approximations.

5.4 What certification costs, and what identification costs

Theorem 5.5 (Certification at the parametric rate)**.** *Consider the class* $\mathcal{K}(\bar{r}, \bar{s})$ *of positive economies with memory bound* $\bar{r}$ *and* $a + \kappa\Phi \le \bar{s} < 1$, *observed under contract (O1). Let* $\bar{\lambda} = 1 - (1 - \bar{r})(1 - \bar{s})$ *and* $\bar{h}_t$ *the average of the* N *responses.*

(i) A valid certificate. *For every economy in the class,* $h_t \le \kappa \bar{\lambda}^t$, *and*

$$\hat{S}_T^+ = \sum_{t<T} \bar{h}_t + z_{1-\gamma}\, \sigma \sqrt{T/N} + \frac{\kappa \bar{\lambda}^T}{1 - \bar{\lambda}}$$

satisfies $P(S \le \hat{S}_T^+) \ge 1 - \gamma$. *With* $T_\epsilon = \lceil \log(2\kappa/((1-\bar{\lambda})\epsilon)) / \log(1/\bar{\lambda}) \rceil$ *and* $N \ge 16 z_{1-\gamma}^2 \sigma^2 T_\epsilon / \epsilon^2$, *the excess* $\hat{S}^+ - S$ *is at most* ϵ *with probability at least* $1 - 2\gamma$. *Certification to precision* ϵ *therefore costs* $N = O(\sigma^2 \epsilon^{-2} \log(1/\epsilon))$.

(ii) A matching lower bound. *Fix an interior single-cohort economy* θ_0 *and let* $C_0 =$

Table 3: Identifying the cohorts versus certifying the risk they create

Rate spacing $\Delta\eta$	0.30	0.20	0.14	0.10	0.08	0.06	0.05	0.04
Identify weights (± 0.01), N	7	88	775	5,928	22,741	128,333	383,859	1,466,362
Certify S (1%), N	18	14	13	12	11	11	11	11

The two-cohort moment-matched pairs of the companion paper ($q = 0.2$, mean rate 0.4, $\tau = 0.01$, $a = 0.5$, $\kappa = 0.8$, per-tap noise 1.6×10^{-3}). Identification: Le Cam requirement at 95% (companion paper, Table on two-point sample complexity). Certification: sufficient N from Theorem 5.5(i) with memory bound equal to the largest rate in the pair and $\bar{s} = a + \kappa q + 0.02$. The two economies of each pair have the same summed response, 2.353. Kind: recomputed (`research_extension/information_safety`, test T03).

$\|\partial h/\partial q\|_2/(\partial S/\partial q)$ *at* θ_0. *Any procedure that is valid at level* $1-\gamma$ *on the class and overshoots* S *by less than* 2ϵ *with probability at least* $1/2$ *at* θ_0 *needs* $N \geq \big((1-2\gamma)\sigma/(2C_0\epsilon)\big)^2(1+o(1))$ *as* $\epsilon \to 0$.

(iii) Independence of the cohort structure. *Neither bound depends on the number of cohorts or on the spacing of their memories, whereas identifying them costs* $N \asymp \sigma^2(\Delta\eta)^{-(4K-2)}$ *with exponential memory and* $\exp(\Theta(1/w))$ *experiments for accuracy* w *with general memory (Niroula and Niroula, 2026).*

The certificate in part (i) is elementary: it is a sum of averaged taps, a Gaussian margin and a tail that the memory bound controls. The content is in the comparison. In contrast to the $\mathcal{H}_\infty$ norm, which Tu et al. (2018) show is no easier to estimate than the model itself, the summed response of a positive economy is a linear functional of the response, so certifying it is a parametric problem, while the cohort parameters are not. Table 3 makes this concrete on the companion paper's hardest two-cohort pairs. Those pairs have *identical* summed responses, because they share the total weight $\sum_i q_i$: the directions in which prices are least informative are directions in which the risk a guard must bound does not move. Identifying their weights to ± 0.01 needs from 7 to 1.5 million reset experiments as the spacing falls from 0.30 to 0.04 (log–log slope -6.06), while certifying the summed response to 1% needs 11 to 18 at every spacing (slope $+0.23$).

Remark 5.6 (Shorter windows cost more experiments). Under exact windows the polynomial bounds of Lemma 5.4 certify Φ to precision ϵ with $n = O\big((1-\bar{r})^{-1/2}\log(1/\epsilon)\big)$ taps

(Theorem 5.3(ii)), whereas the tail bound of Theorem 5.5(i) needs $T_\epsilon = \Theta\big(((1-\bar{r})(1-\bar{s}))^{-1}\log(1/\epsilon)\big)$. With noisy taps the shorter window is not a better certificate. The bound $\sum_j c_j^+ \hat{k}_j$ is linear in the estimated kernel values, so its standard error is $\sigma N^{-1/2}\|\mathcal{J}^\top c^+\|_2$, where $\mathcal{J}$ is the Jacobian of the map from taps to kernel values; the monomial coefficients of an approximating polynomial on $[0,\bar{r}]$ grow geometrically in its degree. In a three-cohort economy with memory bound 0.99 ($S = 3.28$), certifying S to 1% takes 45 taps with the window bound against 4,455 with the tail bound, but $3.5 \times 10^{65}\sigma^2$ experiments against $1.8 \times 10^{8}\sigma^2$; at 0.1%, 56 against 5,482 taps and $3 \times 10^{82}\sigma^2$ against $2.2 \times 10^{10}\sigma^2$ experiments (`hardening/p2_cheb_tradeoff.py`). Power-moment problems are exponentially ill-conditioned, so the parametric rate of Theorem 5.5 is bought with a long, well-conditioned horizon; the window bound is the right tool for exact or near-exact data.

Unknown noise covariance. Theorem 5.5(i) assumes independent taps with a known noise level. Reset experiments need not deliver that: the noise in the taps of one experiment can be correlated. Let contract (O1′) be (O1) with Gaussian noise of unknown covariance Σ across the taps of an experiment, experiments still independent, and let $X_e = \sum_{t<T} h_t^{(e)}$ for experiment e.

Proposition 5.7 (Certification with unknown covariance)**.** *Under (O1′), let $\bar{X}$ and s_X be the mean and standard deviation of $X_1, \dots, X_N$ and $\bar{h}$, $\hat{\Sigma}$ the mean and covariance of the N windows.*

(i) $\hat{S}_t^+ = \bar{X} + t_{N-1,1-\gamma}\, s_X/\sqrt{N} + \kappa\bar{\lambda}^T/(1-\bar{\lambda})$ *satisfies* $P(S \le \hat{S}_t^+) \ge 1-\gamma$ *for every* Σ.

(ii) If $N > T$, the Hotelling region $\{m : N(\bar{h}-m)^\top\hat{\Sigma}^{-1}(\bar{h}-m) \le T_\gamma^2\}$, $T_\gamma^2 = \frac{T(N-1)}{N-T}F_{T,N-T}(1-\gamma)$, contains the mean window with probability exactly $1-\gamma$, so the bounds $c^\top\bar{h} + \sqrt{T_\gamma^2\, c^\top\hat{\Sigma}c/N}$ hold simultaneously for every linear functional c: every horizon, every exposure profile and every guard computed from the same experiments.

(iii) The certificate of Theorem 5.5(i) can cover with probability well below $1-\gamma$ when the taps are positively correlated.

Part (i) is Student's t applied to the sums, and (ii) is Hotelling's theorem (Hotelling, 1931); the content is which assumption each certificate needs. In simulations with twelve taps and nominal 95% coverage (test T11), positively correlated taps (autocorrelation 0.8) cut the coverage of the (O1) certificate to 74–76%, while the t-certificate covers 94–95% and the Hotelling bound 100%, at the cost of a margin three to six times wider. Heavy-tailed noise (t_3) leaves the t-certificate at 95%. Dependence *across* experiments (autocorrelation 0.5) cuts it to 82–85%: independence of the experiments is what a controlled contract buys and what observational data lack, which is why Sections 12.2 and 12.3 report bootstrap bounds as diagnostics rather than certificates. Experiments can also be run one at a time and stopped once the data decide; an anytime-valid version of the certificate keeps its error level under such stopping and needs far fewer experiments when the economy is far from the threshold (Internet Appendix, Proposition C.1).

6 An identification-free pre-trade guard

6.1 The quantity gate

Observable dynamics are $z_{j+1} = Jz_j + Hv_j$ with loss output $l_t = Cz_t$, where v_j is the executed quantity. Let $b_t = CJ^t z_0$ be the drift from past flow and $g_m = c_{\text{price}}^{\top} J^m H$ the impulse coefficients. Which gain bounds the loss depends on the quantity contract. If quantities share a total budget over the window, $\sum_{j<t} |v_j| \le V$, the worst case of $|\sum_{m<t} g_m v_{t-1-m}|$ is $V \max_{m<t} |g_m| \equiv V g_t^*$, attained by placing the whole budget at the lag of the peak, and execution costs at most $c_{\text{exec}} V$. Under a reserve L_{other} the sharp allowance is then

$$V^* = \min_{t:\, g_t^* + c_{\text{exec}} > 0} \frac{B - L_{\text{other}} - |b_t|}{g_t^* + c_{\text{exec}}}, \qquad \bar{R} = V^*/\tau \tag{12}$$

over a window of τ seconds. If instead each period's quantity is bounded, $|v_j| \le V$, the worst case is $V \sum_{m<t} |g_m|$ (attained by $v_{t-1-m} = V \operatorname{sign}(g_m)$) and execution costs at most $c_{\text{exec}} V t$, so the peak gain in (12) must be replaced by the summed gain $\sum_{m<t} |g_m|$ and c_{exec} by $c_{\text{exec}} t$. Pairing a per-period bound with the peak gain understates the worst case, which is the failure Lemma 6.1 formalizes. Either gate needs only impulse coefficients and the current drift.

6.2 Maximality

A dealer with inventory I chooses an admission level $\alpha \in [0, 1]$ (the fraction of incoming algorithmic flow it accepts, or the depth it quotes). The closed-loop operator is $J(\alpha) = J_0 + \alpha B_F$ with price impulse response $g_k(\alpha)$. Shocks satisfy $|u_t| \le \bar{u}$. Over horizon H under a reset initial state $z_0 = 0$ (or after subtracting a known or directly observed initial drift $b_t = C J(\alpha)^t z_0$ from the remaining budget), the shock-driven price change is $\sum_{k<H} g_k(\alpha) u_{t-k}$, and the dealer has a loss budget $\mathcal{B}$. The contract is: inventory $I(\alpha)$ is held fixed over the horizon, the initial state is reset ($z_0 = 0$) or its drift is observed, each period's shock is bounded by $\bar{u}$, and loss is the mark-to-market loss at any date in the horizon; because the partial sums $\sum_{k<h} |g_k|$ increase in h, the worst date is the last. Margin, hedging, fills and non-zero initial states are analysed in Theorem 6.8.

Lemma 6.1 (Summed-gain guard is maximal)**.** *The worst-case loss over horizon H is*

$$L_{\max}(\alpha) = |I(\alpha)|\, \bar{u} \sum_{k<H} |g_k(\alpha)|.$$

Hence the guard "admit α iff $L_{\max}(\alpha) \le \mathcal{B}$" is necessary and sufficient for bounded-shock solvency: it admits every safe α and no unsafe one. Replacing $\sum_k |g_k|$ by $\max_k |g_k|$ can admit unsafe levels; replacing it by $H \max_k |g_k|$ is safe but conservative.

Proof. By Hölder's inequality $|\sum_k g_k u_{t-k}| \le \bar{u} \sum_k |g_k|$, with equality for $u_{t-k} = \bar{u} \operatorname{sign}(g_k)$, which is admissible. Multiplying by $|I(\alpha)|$ gives $L_{\max}$, so the admission set equals the safe

set. Since $\max_k |g_k| \le \sum_k |g_k| \le H \max_k |g_k|$, the peak guard under-states loss and the H-scaled guard over-states it. □

The worst-case identity in Lemma 6.1 is not new as mathematics: the induced ℓ_∞ gain of a linear convolution operator is the ℓ_1 norm of its kernel, the fact on which ℓ_1-optimal control of persistent bounded disturbances is built (Vidyasagar, 1986; Dahleh and Pearson, 1987; Dahleh and Diaz-Bobillo, 1995). We use it rather than claim it. The finance content lies elsewhere: the dealer's admission level changes the operator itself, so the guard is a rule over an action set rather than a controller design; the bound is carried through margin, hedging and fill accounting (Theorem 6.8); its single input is invariant to how the unobserved traders are split into cohorts and is identified from prices (Proposition 2.2 and Theorem 5.5); the peak-response and eigenvalue rules it is compared with are the ones closest to current supervisory practice; and the same statistic prices the corrective rebate in equilibrium (Proposition 8.1).

Numerically, the guard coincides with a brute-force adversary's maximal safe α in 200 of 200 random states. The peak-gain guard admitted an unsafe level in 2,000 of 2,000 states, the H-scaled guard was conservative in all 2,000, and $L_{\max}$ was non-decreasing in α in all 2,000, so the admission set was an interval in these examples.

6.3 Computability: a finite-horizon certificate

Lemma 6.2 (Contraction-block tail bound)**.** *Let $\rho(J(\alpha)) < 1$. Suppose that at horizon H there is a block length $H_0 \le H$ with $q(\alpha) \equiv \|J(\alpha)^{H_0}\|_2 < 1$, and let $\bar{M}_{H_0}(\alpha) = \max_{0\le r<H_0} \|c_{\text{price}}^\top J(\alpha)^r\|_2$ and $M_{C,H}(\alpha) = \max_{k\le H} \|c_{\text{price}}^\top J(\alpha)^k\|_2$. Then*

$$\sup_{k\ge 0} \|c_{\text{price}}^\top J(\alpha)^k\|_2 \le \max\Big(M_{C,H}(\alpha),\ \bar{M}_{H_0}(\alpha) q(\alpha)^{\lfloor (H+1)/H_0 \rfloor}\Big), \tag{13}$$

$$G_{\text{cum},\infty}(\alpha) \le G_{\text{cum},H}(\alpha) + \frac{\bar{M}_{H_0}(\alpha) H_0\, q(\alpha)^{\lfloor (H+1)/H_0 \rfloor}}{1 - q(\alpha)}.$$

Proof. Write any $k > H$ as $k = mH_0 + r$ with $m \ge \lfloor (H+1)/H_0 \rfloor$ and $0 \le r < H_0$.

Submultiplicativity gives $\|c^\top J^k\| \le \|c^\top J^r\| \, \|J^{H_0}\|^m \le \bar{M}_{H_0} q^m$. Taking the supremum and summing the geometric tail gives (13). □

The certificate turns an infinite-horizon supremum into a finite computation with a verified remainder. If the second term in the maximum is below $M_{C,H}$, truncation at H is exact.

Example 5.1 shows why the certificate is needed: two economies can agree exactly on the first coefficients and still differ threefold in their summed response.

6.4 Robustness to estimation error

Theorem 6.3 (Robust gain)**.** *Let* $M = \sup_k \|J^k\|$ *and* $S = \sum_k \|J^k\|$. *If* $\|E\| \le \epsilon$ *and* $\epsilon S < 1$, *then* $\sup_k \|(J+E)^k\| \le M/(1-\epsilon S)$.

Proof. Let $P_k = (J+E)^k$. Expanding, $P_k = J^k + \sum_{j<k} J^{k-1-j} E P_j$. With $m_k = \|P_k\|$, $m_k \le M + \epsilon \sum_{j<k} \|J^{k-1-j}\| m_j \le M + \epsilon S \max_{j<k} m_j$. Since $m_0 = 1 \le M$, induction gives $m_k \le M/(1-\epsilon S)$ for all k. □

Theorem 6.3 is deterministic: it holds whenever $\|E\| \le \epsilon$. A statistical guarantee follows only through a confidence set for the operator, and then holds with that set's coverage probability, not with certainty. The bound held in all 30,000 random perturbations of 100 non-normal operators, with median tightness 0.50 (realized gain over bound; maximum 0.57). In a simulated window estimator of the kind used in Section 10 (600 transitions), a residual-bootstrap 95% ball for the operator covered the true J in 94.7% of 300 replications, with median radius 0.086; but $\epsilon S < 1$ held in only 8% of windows. The certificate is valid but often unavailable at realistic sample sizes; longer windows or pooled estimates are needed to use it in practice.

The guard needs the summed gain, which is invariant to cohort splitting (Proposition 2.2) and certified at the parametric rate whatever the spacing of the memories (Theorem 5.5); it does not need the algorithms' identities, which are expensive (Niroula and Niroula, 2026).

In a two-cohort economy with rates $0.7 \pm \Delta$, the relative error of the estimated summed gain is flat in Δ (log–log slope -0.15), while the error of the cohort weights grows as $\Delta^{-3.07}$.

6.5 An uncertainty-aware guard

The guard of Lemma 6.1 assumes the operator is known. With data, the relevant question is which admission levels are safe for every economy the data allow. We answer it for a venue gate (Mode B below) in the positive cohort economy, where admitting a fraction α of the algorithmic flow scales every cohort's weight, $M \mapsto \alpha M$, so that $S(\alpha) = \kappa/(1 - a - \kappa\alpha\Phi)$.

Lemma 6.4 (Soundness of compatible-set guards)**.** *If a data-dependent set $\mathcal{C}(D)$ contains the true economy with probability at least $1 - \gamma$, then every action whose worst-case loss is within budget for all economies in $\mathcal{C}(D)$ is within budget for the true economy with probability at least $1 - \gamma$, including an action selected from the data. If no action qualifies, the guard certifies nothing: it halts new admission and flags the state, because existing inventory can be unsafe even with no new trading.*

Theorem 6.5 (Optimal uncertainty-aware admission)**.** *Under the reset or observed-drift contract ($z_0 = 0$, or known drift subtracted from $\mathcal{B}$), let position I, shock bound $\bar{u}$ and budget $\mathcal{B}$ be given, and let Φ^+ be an upper bound on Φ valid with probability at least $1 - \gamma$ (for example $\Phi^+ = (1 - a - \kappa/\hat{S}^+)/\kappa$ from Theorem 5.5, or Φ_+ from Theorem 5.3). Put $c_I = 1 - a - \kappa|I|\bar{u}/\mathcal{B}$.*

(i) Closed form and interval. *If $c_I \geq 0$, the levels that are safe for every $\Phi \leq \Phi^+$ form the interval $[0, \alpha^*]$ with $\alpha^* = \min\{1,\ c_I/(\kappa\Phi^+)\}$.*

(ii) Safety. *With probability at least $1-\gamma$, every admitted level is safe for the true economy.*

(iii) Maximality. *Every level above α^* is unsafe for an economy with $\Phi = \Phi^+$; when Φ^+ is attained by a compatible economy (as for Φ_+ in Theorem 5.3), no rule using the same information can admit more without admitting an unsafe level.*

(iv) Infeasibility. *If $c_I < 0$, no level is safe, including $\alpha = 0$: the guard halts admission and the position must fall to $|I| \le \mathcal{B}(1-a)/(\kappa\bar{u})$.*

(v) Minimax capacity loss. *On the class of Theorem 5.5 with Φ bounded away from zero, the certificate-based guard gives up $\alpha^\circ - \alpha^* = O\big(\sigma\sqrt{\log N/N}\big)$ of the oracle capacity $\alpha^\circ = \min\{1, c_I/(\kappa\Phi)\}$ with probability at least $1-2\gamma$, and every rule that admits an unsafe level with probability at most $\beta < 1$ gives up at least $c\,\sigma/\sqrt{N}$ in expectation at some economy of the class. The guard is minimax-optimal in lost capacity up to a logarithmic factor.*

Monotonicity of $S(\alpha)$ in α and Φ gives (i)–(iv); (v) combines the certificate's excess with Proposition 5.2 applied to two economies whose capacities differ by order $\sigma/\sqrt{N}$ (Appendix A). In 500 random economies the closed form matched a brute-force search over 1,001 admission levels, computed from the state matrices rather than from (11), in every case; every safe set was an interval; and the 28 infeasible states were flagged (test T06). Across 18 designs the certificate covered the true summed response at least at its nominal level (minimum 93.6% at nominal 90%), with a median excess of 3–15% when the memory bound is 0.6 or 0.9 and about 230% when it is 0.97, where the tail term dominates (test T04).

6.6 When admission changes the dynamics

Theorem 6.5 rests on one property: risk rises with the admission level, so the safe levels form an interval $[0, \alpha^*]$ and certifying α^* certifies every level below it. A venue that admits a class of participants can also change the book. Suppose admitting a fraction α adds feedback $\alpha\Phi$, deepens the book so that impact falls to $\kappa_0/(1+\delta\alpha)$, and speeds reversion to $1 - a_0 + \rho\alpha$, with $\theta = (\kappa_0, a_0, \Phi, \delta, \rho)$ in a certified box Θ. Then $S(\alpha) = \kappa_0/D(\alpha)$ with $D(\alpha) = (1+\delta\alpha)(1-a_0+\rho\alpha) - \kappa_0\Phi\alpha$.

Proposition 6.6 (Venue mode)**.** *(i) For each economy the safe levels* $\{\alpha \in [0,1] : |I|\bar{u}S(\alpha) \le \mathcal{B}\}$ *are* $[0,1]$ *minus at most one open interval; they can be* $[0,\alpha_1] \cup [\alpha_2,1]$ *with* $0 < \alpha_1 < \alpha_2 < 1$*.*

(ii) For $\alpha \ge 0$ *the worst economy of* Θ *is the same corner,* $(\kappa_0^+, a_0^+, \Phi^+, \delta^-, \rho^-)$*, for every* α*, so the robust safe set is the safe set of that corner, with endpoints at the roots of a quadratic.*

(iii) A rule that certifies the largest safe level and admits every level below it admits unsafe levels whenever the robust safe set is not an interval. Levels in $[\alpha_2, 1]$ *can be used safely only by moving to them directly, not by raising admission step by step.*

In the box $\kappa_0 \in [0.98, 1.02]$, $a_0 \in [0.58, 0.60]$, $\Phi \in [2.3, 2.4]$, $\delta \in [2.9, 3.1]$, $\rho \in [0.58, 0.62]$ with $|I|\bar{u}/\mathcal{B} = 0.37$, the robust safe set is $[0, 0.035] \cup [0.386, 1]$, in agreement with a 2,001-point grid over 332 economies of the box (test T13). Low admission brings mostly feedback and high admission mostly liquidity, so risk first rises and then falls.

Without a closed form, as in the three-state book of Section 4, the robust safe set must be computed. Let the impact $\kappa(\alpha)$, the feedback $q(\alpha)$ and the refill rate $r(\alpha)$ of the book depend on admission, with the uncertain parameters in a box Θ, and let $S(\alpha, \theta)$ be the summed absolute price response.

Theorem 6.7 (Certified admission without monotonicity)**.** *Enclose* S *over a box of levels and parameters by its first* T *terms in interval arithmetic with outward rounding plus the tail bound* $\|J^T s_0\|_\infty \sum_{j<T} \|J^j\|_\infty / (1 - \|J^T\|_\infty)$ *(infinite when* $\|J^T\|_\infty \ge 1$ *cannot be excluded). Classify a subinterval of levels as safe if the enclosure over it and every part of a partition of* Θ *is within budget, as unsafe if the lower enclosure at some economy of* Θ *exceeds the budget, and otherwise split it, down to a tolerance. Then:*

(i) Soundness. *Every level classified safe is safe for every economy in* Θ*, and every level classified unsafe is unsafe for some economy in* Θ*, in exact arithmetic.*

(ii) Existence and bracketing. *The robust safe set is closed, so the largest safe level exists if any level is safe, and so does the largest level reachable by raising admission from zero through safe levels if zero is safe. The first lies between the largest certified-safe level and the largest level not certified unsafe; the second between the end of the certified-safe interval that starts at zero and the first level certified unsafe.*

(iii) No false certificate. *If every level is classified unsafe, the guard halts admission and flags the state.*

(iv) Completeness. *If* $\sup_\Theta \rho(J) < 1$*, then as the tolerance and the partition of* Θ *shrink and* T *grows, every level at which the robust risk differs from the budget is eventually classified.*

The method is interval branch and bound (Hansen and Walster, 2004); the theorem's content is that its output is a certificate for the risk contract, including for admission rules whose risk is not monotone. In a book with depletion and refill where admission raises feedback linearly and liquidity quadratically (test T13), the certified safe set at a budget of 6.5 is $[0, 0.053] \cup [0.178, 1]$, with $[0.059, 0.168]$ certified unsafe and 0.016 left undecided; no certified-safe level was unsafe for any of 433 sampled economies on a 1,001-point grid. The shortcut that certifies $\alpha = 1$ and admits everything below it would admit the whole unsafe band.

6.7 A calibrated illustration

How much does the choice between peak and summed guards matter in a real book? We compute the three-state impulse response at the median estimates of Section 9 (all training days and high-volatility days) and at a hypothetical book in which flow depletes the side it hits ($\varepsilon = +0.02$, other parameters at their high-volatility medians), Figure 3 and Table 4.

The result is striking even though BTCUSDT does not amplify. Because price impact is persistent ($\hat{a} \approx 0.97$ per minute), the summed gain over 15 minutes is about 13 times the

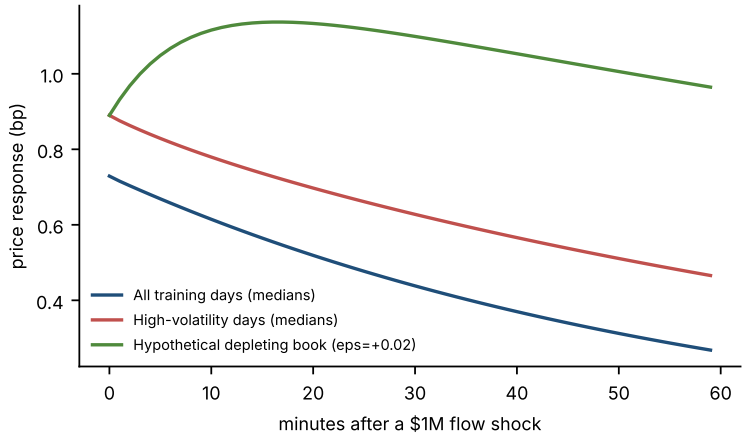


Figure 3: Price response to a $1M flow shock at the median depth estimates and in a hypothetical depleting book. Computed from the estimates of Table 7.

Table 4: Peak versus summed gain at calibrated parameters

Book	C_3	Peak (min)	$\sum_{k<15} \lvert g_k \rvert$	$\sum_{k<60} \lvert g_k \rvert$	Sum/peak	Cap, peak	Cap, sum
All days (medians)	−0.013	0.73 (0)	9.73	27.67	13.3	$274M	$21M
High-vol. days (medians)	−0.014	0.89 (0)	12.17	38.71	13.7	$225M	$16M
Depleting book (ε = 0.02)	+0.042	1.14 (17)	15.85	63.69	14.0	$176M	$13M

Gains in bp per $1M of flow; sum/peak at 15 minutes. Inventory cap under a peak guard and under the summed guard, for a loss budget of $100,000, an illustrative shock bound of $5M of flow per minute and a 15-minute horizon. From `calib_guard.py`, using the medians of Table 7.

peak. A guard calibrated to the peak would admit inventory about 13 times larger than is safe against a sustained adversarial flow of the declared size. Amplification makes this worse (the depleting book peaks 17 minutes after the shock), but persistence alone already makes peak-based limits unsafe. This is the practical content of Lemma 6.1 for this market.

6.8 Two deployment modes

Mode A. *An individual dealer.* A price-taking market maker with a small share of aggregate depth scales its own quote size by α. This scales its passive fill intake but leaves aggregate depth, impact and the market operator unchanged.

Mode B. *A venue or sponsoring broker.* A gate at a matching engine or a market-access

gateway (the setting of Rule 15c3-5) admits a fraction α of incoming aggressive algorithmic flow while resting depth stays posted. The operator becomes $J(\alpha) = A_0 + \alpha B_F$.

Algorithm (pre-trade guard). At each step, using only pre-shock information:

1. For each α on a grid $\mathcal{A} = \{0, 0.025, \ldots, 1\}$, compute $\rho(J(\alpha))$, the peak and summed gains $M_{C,H}(\alpha)$ and $G_{\mathrm{cum},H}(\alpha)$, and the tail certificate of Lemma 6.2; if an operator confidence radius ϵ is available, inflate by Theorem 6.3.
2. Compute post-hedge and worst-case post-fill inventory, hedging cost and the post-fill margin increment.
3. Admit the largest $\alpha \in \mathcal{A}$ that satisfies the solvency inequalities of Theorem 6.8; if none does, set $\alpha = 0$, flag that no safe action exists, and trigger inventory reduction.

The guard checks every grid point, so it does not rely on the safe set being an interval; levels between grid points are never admitted. For low-dimensional J the grid evaluation is a sequence of matrix–vector products; execution takes well under one millisecond in compiled code, which we regard as an engineering detail rather than a result.

6.9 Balance-sheet solvency

Theorem 6.8 (Conditional bounded-shock forward solvency)**.** *Consider a dealer with cash K_t, inventory I_t, mark p_t, equity $E_t = K_t + I_t p_t$, maintenance margin $MM_t = \max(0, \mu|I_t|p_t - c_0)$ and remaining buffer $B_{\mathrm{rem},t} = E_t - MM_t$, a safety reserve R_{safe}, a hedge $\Delta I_t^{\mathrm{hedge}}(\alpha)$ with cost $C_{\mathrm{hedge},t}(\alpha)$, worst-case passive fills $\Delta I_{\max,t}^{\mathrm{fill}}(\alpha)$ with margin increment $C_{\mathrm{fill},t}(\alpha)$, and declared one-step and cumulative shock envelopes $\bar{u}_{\mathrm{step},t} \le \bar{u}_{\mathrm{cum},t}$. Let $|I_{t+1}^{\mathrm{post}}(\alpha)| = |I_t + \Delta I_t^{\mathrm{hedge}}(\alpha)| + \Delta I_{\max,t}^{\mathrm{fill}}(\alpha)$. Suppose the guard admits α_t only if*

$$|I_{t+1}^{\mathrm{post}}(\alpha_t)|(1+\mu)\Big(M_C(\alpha_t)\|z_t\|_2 + s(\alpha_t)G_{\mathrm{cum}}(\alpha_t)\bar{u}_{\mathrm{cum},t}\Big) + C_{\mathrm{hedge},t}(\alpha_t) + C_{\mathrm{fill},t}(\alpha_t) + R_{\mathrm{safe}} \le B_{\mathrm{rem},t}, \tag{14}$$

where $s(\alpha)$ scales admitted shocks. Then for every disturbance path within the declared envelopes, the buffer after the step is at least R_{safe}; by induction, solvency holds at every step at which an admissible α exists. At $\alpha = 0$ the one-step admissible shock is

$$\bar{u}_{\text{adm},t} = \frac{B_{\text{rem},t} - R_{\text{safe}} - |I_t|(1+\mu)\|z_t\|_2}{|I_t|(1+\mu)}.$$

The condition is sufficient over the whole disturbance ball. It is also necessary when the current state contributes no drift ($z_t = 0$), because the shock term is then attained by adversarially signed disturbances; with $z_t \neq 0$ the drift term $M_C\|z_t\|_2$ is an upper bound and the condition may be conservative. The admissible shock at $\alpha = 0$ assumes, as in the cohort economy (Niroula and Niroula, 2026, positive-geometry lemma), that the price-row gain without admitted flow is $M_C(0) = 1$.

Two features matter. The guard accounts for the inventory the dealer will hold after hedging and after worst-case fills, not only current inventory, and it never claims safety on an infeasible state: if no α satisfies (14), it halts new fills and flags the state, because pre-existing inventory can still breach margin on a large adverse jump. In a constructed scenario ($E_0 = 10{,}000$, $I_0 = -2{,}000$ at $P^* = 100$, Tier-2 margin with $\mu = 0.005$, buffer 9,050 and reserve 950), the admissible shock at $z_0 = 0$ is $8{,}100/2{,}010 = 4.03$ per unit, and an adversarial move of 4.05 violates the reserve. In a matched-shock Monte Carlo inside the model, where the guard's risk contract holds by construction (the certified bound $\Phi^+ = 0.36$ is close to the true $\Phi = 0.34$), across 1,000 paths facing identical exogenous flow shocks, initial inventory levels, transaction costs and double-entry maintenance-margin accounting where all active baselines are calibrated on an independent training set to match the guard's mean trading capacity of 68.0% (`microfoundation_and_baselines.py`), the summed-gain guard achieves 0.0% reserve breaches ($E_t - MM_t < R_{\text{safe}}$) and 0.0% exchange forced liquidations ($E_t < MM_t$), compared with 31.6% forced liquidations (34.7% reserve breaches) under no control, 31.4% (34.1%) under a trailing volatility limit, 20.3% (21.5%) under an inventory

cap and 0.0% under a scanning-range margin rule; lowers 95% expected-shortfall tail loss to 5.96 (against 12.11 under the margin rule, 39.77 under the inventory cap, 39.92 under the volatility limit and 39.89 under no control, with all paired bootstrap difference 95% CIs excluding zero); and attains the highest net dealer P&L (+0.75, 95% CI $[+0.48, +1.02]$). At a matched tail-risk budget ($\mathrm{ES}_{95} \approx 5.96$), the guard admits 68.0% capacity versus 29.8% for the margin rule, 24.1% for the inventory cap and 19.8% for the volatility limit; conversely, in a heteroskedastic zero-feedback regime ($\Phi = 0$) with a conservatively misspecified certificate $\Phi^+ = 0.40$, the trailing volatility limit outperforms the static-Φ^+ guard at matched capacity ($\mathrm{ES}_{95} = 4.39$ vs. 4.55).

Unlike the reset quantity-gate of Lemma 6.1 and Theorem 6.5, the general forward-solvency inequality (14) with an unknown non-zero hidden cohort state $z_t \neq 0$ (where $z_t = (x_t, y_{1,t}, \ldots, y_{K,t})^\top$) is *not* identification-invariant under an unrestricted Euclidean norm bound. Both the unforced future price drift $b_m = c_{\text{price}}^\top J^m z_t$—whose cumulative sum is $\big(ax_t + \kappa \sum_i (q_i/\eta_i) y_{i,t}\big)/(1 - a - \kappa\Phi)$—and the operator-norm drift bound $M_C(\alpha_t)\|z_t\|_2$ depend on the individual cohort weights and memory rates (q_i, η_i) and the hidden state y_t, not solely on $\Phi = \sum_i q_i$. For an explicit rational counterexample (`bridge_check.py`, Part F), take $a = 2/5$, $\kappa = 1$, and two two-cohort positive economies $\theta^{(1)}$ with $q^{(1)} = (1/5, 1/5)$, $\eta^{(1)} = (5/8, 3/8)$ and $\theta^{(2)}$ with $q^{(2)} = (9/25, 1/25)$, $\eta^{(2)} = (13/24, 1/8)$. Both share $\Phi = 2/5$ ($S = 5$) and identical early impulse responses $(h_0, h_1, h_2) = (1, 3/5, 363/800)$ ($k_0 = 1/5$, $k_1 = 3/32$). Yet after a single unit flow shock two periods earlier, where both markets display the exact same price path $(1, 3/5, 363/800)$, their hidden cohort states produce different next-step drift $b_1 = 9009/24000$ versus $8959/24000$; after three periods their cumulative future drift is $61701/24000$ versus $61751/24000$; over unknown hidden states $\|y_0\|_2 \leq 1$ at $x_0 = 0$ their worst-case cumulative drift is 3.110 versus 3.688 (18.6% larger); and their one-step price-row norm $\|c_{\text{price}}^\top J\|_2 = \sqrt{a^2 + \kappa^2 \sum_i q_i^2}$ is 0.490 versus 0.540.

When the unobserved cohort memories are instead bounded in the natural ℓ_∞ norm $\|y_0\|_\infty \leq \bar{y}_0$—which holds automatically whenever historical pricing errors satisfy $|x_{-s}| \leq \bar{y}_0$

for $s \geq 0$, since each $y_{i,0}$ is a convex combination of past x_{-s}—the worst-case unforced drift has an exact identification-free envelope and yields a closed-form drift-adjusted guard:

Proposition 6.9 (Uniformly safe guard under interval-bounded unknown hidden drift)**.** *In any positive cohort economy with $\Phi = \sum_{i=1}^{K} q_i \leq \Phi^+$, observed current pricing error x_0 and unknown hidden cohort states satisfying $\|y_0\|_\infty \leq \bar{y}_0$, let admission $\alpha \in [0,1]$ scale cohort flow as $M \mapsto \alpha M$ with $a + \alpha\kappa\Phi^+ < 1$. Then:*

(i) Identification-free drift envelope. *For every compatible cohort decomposition $(q_i, \eta_i)_{i=1}^{K}$ with $\sum_i q_i = \Phi$, the worst-case cumulative unforced drift over $\|y_0\|_\infty \leq \bar{y}_0$ is*

$$\sup_{\|y_0\|_\infty \leq \bar{y}_0} \sum_{t=1}^{\infty} |e_x^\top J(\alpha)^t z_0| \;=\; \frac{a|x_0| + \alpha\kappa\Phi\bar{y}_0}{1 - a - \alpha\kappa\Phi}, \tag{15}$$

attained when $y_{i,0} = \operatorname{sign}(x_0)\bar{y}_0$ for all i (or $y_{i,0} = \pm\bar{y}_0$ when $x_0 = 0$).

(ii) Closed-form drift-adjusted guard. *Under shock bound $\bar{u}$, inventory I and budget $\mathcal{B}$, the set of admission levels that are safe for every compatible economy with $\Phi \leq \Phi^+$ and every hidden state $\|y_0\|_\infty \leq \bar{y}_0$ is $[0, \alpha^*_{\text{drift}}]$ (empty when $(1-a)\mathcal{B} < |I|(a|x_0| + \kappa\bar{u})$), where*

$$\alpha^*_{\text{drift}} \;=\; \min\left\{1,\; \left[\frac{(1-a)\mathcal{B} - |I|(a|x_0| + \kappa\bar{u})}{\kappa\Phi^+(\mathcal{B} + |I|\bar{y}_0)}\right]_+\right\}. \tag{16}$$

(iii) Exact capacity cost of unknown state. *When $x_0 = 0$ and the reset guard $\alpha^* = \min\{1, c_I/(\kappa\Phi^+)\}$ of Theorem 6.5 is interior ($\alpha^* \in (0,1)$), the capacity given up to protect against unknown hidden cohort states $\|y_0\|_\infty \leq \bar{y}_0$ is exactly*

$$\alpha^* - \alpha^*_{\text{drift}} \;=\; \alpha^* \,\frac{|I|\bar{y}_0}{\mathcal{B} + |I|\bar{y}_0}. \tag{17}$$

At the Table 3 calibration ($a = 0.4$, $\kappa = 1$, $\Phi^+ = 0.4$, $|I| = 1$, $\bar{u} = 0.48$, $\mathcal{B} = 1$, $x_0 = 0$) with $\bar{y}_0 = 0.25$, the reset guard admits $\alpha^* = 0.3000$ whereas the drift-adjusted guard admits $\alpha^*_{\text{drift}} = 0.2400$, an exact capacity cost of 0.0600, and commits zero budget breaches across

500 random multi-cohort economies (`recompute_p2_step2_theory.py`).

Proposition 6.10 (When an eigenvalue rule admits an uncertified level)**.** *Let an eigenvalue guard admit* $\alpha_\rho = \max\{\alpha : \rho(J(\alpha)) \le \bar{\rho}\}$*, and suppose the state is feasible for the summed-gain guard, which then admits at least* $\alpha = 0$*. Bounding the one-step price move by* $M_C(\alpha)(\|z_t\|_2 + s(\alpha)\bar{u}_{\text{step},t})$*, the level admitted by the eigenvalue guard fails this one-step solvency certificate, and so is one the summed-gain guard would not certify, if and only if*

$$M_C(\alpha_\rho) \;>\; \frac{B_{\text{rem},t} - R_{\text{safe}} - C_{\text{hedge},t}(\alpha_\rho) - C_{\text{fill},t}(\alpha_\rho)}{|I^{\text{post}}_{t+1}(\alpha_\rho)|(1+\mu)(\|z_t\|_2 + s(\alpha_\rho)\bar{u}_{\text{step},t})}.$$

The certificate is a bound, so failing it means that solvency cannot be guaranteed, not that a breach must occur. The eigenvalue rule fails it exactly when the transient gain at its admitted level exceeds the loss the balance sheet can absorb, which the companion paper's spectral-blindness theorem shows can happen at any spectral threshold.

Proposition 6.11 (Strategic re-optimization reinforces the guard)**.** *Suppose cohorts re-optimize their steady-state Kalman gains when the venue throttles admission, and throttling scales the variance of execution-driven innovations as* $\sigma^2_\zeta(\alpha) = \alpha^2\sigma^2_{\zeta,0}$ *while measurement noise* $\sigma^2_\nu > 0$ *is unchanged.*

(i) The optimal gain $\eta^*(\alpha)$ *is strictly increasing in* α*, so throttling lowers every cohort's adaptation rate. If the transient gain is non-decreasing in the adaptation rates, then* $M_C(J(\alpha, \eta^*(\alpha))) \le M_C(J(\alpha, \eta^*(1)))$*: re-optimization does not undo the guard.*

(ii) In the positive cohort economy ($q_i \ge 0$*,* $a + \kappa\sum_i q_i < 1$*) the monotonicity holds without condition: every finite-horizon summed response* $S_H = \sum_{t<H} h_t$ *is non-decreasing in each adaptation rate, strictly for* $H \ge 2$ *when* $q_i > 0$*, while* $S_\infty = \kappa/(1-a-\kappa\sum_i q_i)$ *does not depend on the rates. Hence* $S_H(J(\alpha, \eta^*(\alpha))) \le S_H(J(\alpha, \eta^*(1)))$ *at every horizon.*

Part (i) is a property of the Riccati root. Part (ii) is a renewal argument: the response is the renewal density of a defective inter-arrival law whose cumulative distribution $a +$

$\kappa \sum_i q_i[1-(1-\eta_i)^j]$ rises with every rate at fixed total mass, so faster adaptation moves feedback earlier without changing its total (Appendix A). In 3,000 random positive economies no finite-horizon sum fell when a rate rose (`hardening`). For the three-state book with momentum the condition in (i) remains an assumption, which we check numerically. Across 3,000 stable draws of the three-state book with momentum (adaptation rates on a grid from 0.05 to 0.95), the peak response is non-decreasing in η in 98.8% of draws and in 99.9% of those with trend-following cohorts ($q > 0$); the summed absolute response is non-decreasing in 93% of draws with $q > 0$ but decreasing when cohorts are contrarian ($q < 0$). For trend-following cohorts, the case the filtering microfoundation produces, re-optimization therefore reinforces the guard in almost every case we drew; the signed sum S of Theorem 4.4 does not depend on η at all (`recompute_oct20.py`). A manipulator who instead increases order size to offset throttling enlarges the shock envelope and drives α toward zero.

7 The value of disclosure for safe trading

The companion paper measures the value of disclosure in samples needed to identify the cohorts (Niroula and Niroula, 2026). For a pre-trade control, the relevant value is the safe trading capacity it unlocks.

Theorem 7.1 (Decision value of disclosure)**.** *Under the risk contract of Section 5.1, compare an observation contract O with a richer contract O' that reveals everything O reveals and more, both with exact data.*

(i) Monotonicity. *The compatible sets are nested, $\Theta_{O'} \subseteq \Theta_O$, so $\Phi^+_{O'} \leq \Phi^+_O$ and the admitted level of Theorem 6.5 satisfies $\alpha^*_{O'} \geq \alpha^*_O$.*

(ii) Strictness. *$\alpha^*_{O'} > \alpha^*_O$ if and only if $\alpha^*_O < 1$, $c_I > 0$, and O' excludes every compatible economy with $\Phi = \Phi^+_O$. Under the window contract that economy is the upper principal representation of Theorem 5.3: disclosure has decision value exactly when it rules out a cohort at the memory bound with the extremal weight.*

(iii) Long memory. *Under the window contract with memory bound $\bar{r}$, $\alpha^*_O \to 0$ as $\bar{r} \to 1$ for any fixed window (Theorem 5.3(iii)), whereas audited exposure, which reveals Φ, admits the oracle capacity α°. The decision value of audited disclosure therefore tends to the whole oracle capacity as memory lengthens.*

(iv) Net value. *With value w per unit of admitted capacity and disclosure cost $c(O')$, the net value $w(\alpha^*_{O'} - \alpha^*_O) - c(O')$ is zero or negative whenever O already identifies Φ (a window whose moment vector lies on the boundary of the moment space) or the budget does not bind.*

(v) Noisy proxies. *With noisy data the certificates of two contracts are confidence bounds, not nested sets, and a richer contract can admit less. Under class-tagged flows measured with noise over T periods and estimated as in the companion paper, class i's exposure $\hat{\Phi}_i = \hat{w}_i/(1 - \hat{r}_i)$ has delta-method standard error*

$$\mathrm{se}(\hat{\Phi}_i) = \frac{1}{(1 - r_i)\sqrt{T}} \Big[V_{ww} + 2\Phi_i V_{wr} + \Phi_i^2 V_{rr} \Big]^{1/2},$$

with V the asymptotic covariance of $(\hat{w}_i, \hat{r}_i)$, so the tagged bound $\sum_i \hat{\Phi}_i + z_{1-\gamma}\mathrm{se}$ admits less than the public certificate whenever its margin exceeds $\Phi^+_O - \Phi$. Intersecting the two certificates, each at level $1 - \gamma/2$, gives a bound valid at level $1 - \gamma$ that admits at least as much as either certificate at level $1 - \gamma/2$, which restores monotonicity.

The theorem is set inclusion plus the closed form of Theorem 6.5; its content is the location of the value. Disclosure is worth most where public certificates are weakest, in markets whose participants have long memories. In simulation (test T07; Table 5), with two classes and one of them slow, audited exposure adds 0.027 of admitted capacity to a public certificate when the slow class's memory rate is 0.6 and 0.122 when it is 0.995, out of an oracle capacity of 0.667. Class-tagged flows measured with 10% noise add 0.02–0.03 at memory rates up to 0.97 but *lower* capacity at 0.995, because a class's exposure $q = w/(1 - r)$ is badly conditioned when r is near one: disclosure of a noisy proxy can be

Table 5: Admitted capacity under three observation contracts (simulation)

Memory rate of the slow class	0.60	0.90	0.97	0.995
Public responses (certificate, Theorem 5.5)	0.640	0.623	0.601	0.544
Class-tagged flows, 10% noise	0.662	0.655	0.631	0.507
Audited exposure (oracle)	0.667	0.667	0.667	0.667
Public ∩ tagged, each at 97.5% (test T15)	0.661	0.653	0.626	0.537
Value of audited exposure over public	0.027	0.044	0.066	0.122

Two classes ($q = 0.06$ with the stated memory rate, $q = 0.12$ with rate 0.5), $a = 0.4$, $\kappa = 1$, position 1, shock bound 0.48, budget 1; public: 500 reset experiments, per-tap noise 0.05, certificate at 95% with the true memory bound; tagged: 2,000 periods of each class's flow with 10% measurement noise, two-stage least squares as in the companion paper, 95% delta-method bound; 300 replications. Entries are mean admitted fractions α^*. The intersection row comes from an independent run of the same design (seed 15), whose public and tagged rows are 0.639/0.662, 0.625/0.655, 0.600/0.631 and 0.549/0.506. Kind: recomputed (tests T07 and T15).

worth less than nothing. Part (v) quantifies this. In a second simulation of the same design (test T15), the standard error of the slow class's exposure rises from 0.0004 to 0.035 as $1 - r$ falls from 0.4 to 0.005, roughly in proportion to $(1 - r)^{-1}$ over the last decade, while the public certificate's margin grows only with the square root of its horizon. At $r = 0.995$ the tagged bound admits 0.506 against 0.549 for the public one; the intersection admits 0.537, close to the better of the two, and covers the true exposure in 99% of replications (Table 5). The companion paper's evidence that Binance's and the CFTC's class positions pin down memories that prices leave unrestricted concerns class-level positions, which are proxies, not records of individual algorithms.

Pricing versus guarding. A conservative bound is the right input for a guard and the wrong input for a price. The corrective rebate of Proposition 8.1 depends on S^2; computed from an upper bound that overstates S by 10% it over-subsidizes reversion and loses 2.4–2.8% of first-best welfare, and by 50%, 70–80% (test T12). Guards should use certified upper bounds; rebates should use point estimates.

8 Who provides reversion? An equilibrium

Theorem 4.4 says that transient risk is governed by one ratio, $S = \tilde{\kappa}/(m - q\tilde{\kappa})$. Its ingredients are chosen by market participants: dealers decide how hard to lean against price displacements, which sets reversion m, and momentum traders decide how strongly to chase them, which sets q. This section asks whether those choices are efficient.

8.1 Setup

There are N dealers. Dealer i supplies reversion intensity $s_i \geq 0$ by absorbing order flow against the displacement, so $m = m_0 + \sum_i s_i$, where m_0 is reversion supplied by other means. Leaning earns a spread v per unit of intensity at quadratic inventory cost, so the private value is $vs_i - \frac{c}{2}s_i^2$. By Lemma 6.1 the worst-case transient loss of a position of size Q facing flow shocks bounded by U is $QU\,S$, so we take aggregate expected tail loss to be $\Lambda S(m, q)$ with $\Lambda > 0$. Dealers bear a share $\theta < N$ of it, so each bears θ/N; leveraged users and their counterparties bear the rest. Momentum traders choose aggregate intensity q with private value $bq - \frac{e}{2}q^2$ and bear none of the tail loss. Long-run impact $\tilde{\kappa}$ is taken as given.

8.2 Results

Proposition 8.1 (Equilibrium transient risk)**.** *Let* $g(m) \equiv -\partial S/\partial m = \tilde{\kappa}/(m - q\tilde{\kappa})^2$.

(i) Under-provision of reversion. *The symmetric Nash equilibrium solves* $m - m_0 = Nv/c + \theta\Lambda g(m)/c$*, the planner's optimum solves* $m - m_0 = Nv/c + N\Lambda g(m)/c$*, and* $m^{\mathrm{eq}} < m^{\mathrm{SP}}$*, so* $S^{\mathrm{eq}} > S^{\mathrm{SP}}$.

(ii) Fewer dealers, more risk. $\partial m^{\mathrm{eq}}/\partial N = (v/c)/(1 - \theta\Lambda g'(m)/c) > 0$*, so equilibrium reversion rises and* S^{eq} *falls with the number of dealers.*

(iii) Over-provision of momentum. *Momentum traders choose* $q = b/e$*; the planner chooses* $q^{\mathrm{SP}} < b/e$*, solving* $b - eq = \Lambda\tilde{\kappa}^2/(m - q\tilde{\kappa})^2$.

(iv) The corrective price is identified. *The first best for reversion is implemented by a rebate per unit of reversion*

$$\tau^* = \Big(1 - \frac{\theta}{N}\Big)\Lambda\, g(m) = \Big(1 - \frac{\theta}{N}\Big)\Lambda\, \frac{S^2}{\tilde{\kappa}},$$

which depends on the market only through S and $\tilde{\kappa}$.

(v) Endogenous depth. *Suppose in addition that each dealer posts depth $D_i \geq 0$ at private value $v_D D_i - \frac{c_D}{2} D_i^2$, that long-run impact is $\tilde{\kappa} = 1/D$ with $D = \sum_i D_i$, and that momentum traders size their flow to the book, $q = \psi D$ with ψ fixed. Then $S = \Gamma/D$ with $\Gamma \equiv S/\tilde{\kappa} = 1/(m - \psi)$, which does not depend on depth. Symmetric equilibrium depth solves $D = N v_D/c_D + \theta\Lambda\Gamma/(c_D D^2)$ and the planner's $D = N v_D/c_D + N\Lambda\Gamma/(c_D D^2)$, so $D^{\mathrm{eq}} < D^{\mathrm{SP}}$; D^{eq} increases with N, so $S^{\mathrm{eq}} = \Gamma/D^{\mathrm{eq}}$ falls with the number of dealers while $S/\tilde{\kappa}$ does not move. The rebate per unit of depth $\tau_D^* = (1 - \theta/N)\Lambda S\tilde{\kappa}$ implements the first-best depth.*

Proof. (i) Dealer i's first-order condition is $v - cs_i + (\theta/N)\Lambda g(m) = 0$; the planner's is $v - cs + \Lambda g(m) = 0$. Summing over dealers gives the two fixed-point equations. The right-hand sides are decreasing in m because g is decreasing on $m > q\tilde{\kappa}$, and the planner's lies above the dealers' for every m when $\theta < N$, so its crossing with the line $m - m_0$ lies to the right. (ii) follows by implicit differentiation, since $g' < 0$. (iii) $\partial S/\partial q = \tilde{\kappa}^2/(m - q\tilde{\kappa})^2 > 0$ is the marginal external cost of momentum that traders ignore. (iv) The rebate closes the wedge between the two first-order conditions, and $g = S^2/\tilde{\kappa}$ by Theorem 4.4. (v) With $q\tilde{\kappa} = \psi$, Theorem 4.4 gives $S = \tilde{\kappa}/(m - \psi) = \Gamma/D$ and $\partial S/\partial D_i = -\Gamma/D^2$. Dealer i's first-order condition for depth is $v_D - c_D D_i + (\theta/N)\Lambda\Gamma/D^2 = 0$ and the planner's $v_D - c_D D_i + \Lambda\Gamma/D^2 = 0$; summing gives the two fixed points. In each, the left side D increases and the right side decreases in D, so each has a unique positive solution, the planner's to the right of the dealers' because $\theta < N$; and $D - \theta\Lambda\Gamma/(c_D D^2) = N v_D/c_D$ has a left side increasing in D, so D^{eq} increases with N. The rebate closes the wedge $(1 - \theta/N)\Lambda\Gamma/D^2 = (1 - \theta/N)\Lambda S\tilde{\kappa}$. □

At $v = 0.002$, $c = 0.5$, $\theta = 0.3$, $\Lambda = 10^{-4}$, $m_0 = 0.005$, $q = 0.02$ and $\tilde{\kappa} = 1$, equilibrium S falls from 27.9 with one dealer to 6.8 with forty, and the planner's S is 1.5–2.7 times lower throughout; the rebate computed from S alone coincides with the first-order wedge to machine precision. With $m = 0.03$, $b = 0.3$ and $e = 20$, momentum traders choose $q = 0.015$ against a planner's 0.0062.

Interpretation. Reversion is a public good: each dealer who leans against a displacement lowers the tail exposure of every leveraged position in the market but is paid only for his own fills. Momentum is the mirror image. Part (iv) closes the loop with the guard: the policy price of the externality is a function of the same summed response that the guard bounds, and Proposition 2.2 and Theorem 5.5 show that this quantity is identified from public data without identifying the traders. A venue that rebates reversion in proportion to $S^2/\tilde{\kappa}$, or a supervisor that sets pre-trade limits by S, prices the externality with an observable.

What the data say about the model's predictions. Part (ii) predicts that books with fewer dealers have higher S because they supply less reversion. Depth is the closest proxy we observe for the number of liquidity providers. Across the three liquidity terciles of the 30 Binance perpetuals on training days, the summed response per dollar of flow rises steeply as depth falls (median 2,797, 26,603 and 39,329 basis points per \$1M in the thick, mid and thin terciles; Spearman correlation with log depth -0.94), as (ii) predicts. The channel is not the one part (ii) names, however: the unit-free gain multiple $S/|h_0|$, which equals $1/(m - q\tilde{\kappa})$ up to the factor $\tilde{\kappa}/\kappa$ and isolates reversion net of momentum, is the same in all three terciles (medians 46.4, 47.3 and 46.9; Spearman -0.11). Thinner books carry more cumulative risk because their impact $\tilde{\kappa}$ is larger, not because they revert more slowly. This is the pattern part (v) produces when momentum traders size their flow to the book: fewer dealers lower depth, which raises S in proportion to $\tilde{\kappa}$, while $S/\tilde{\kappa}$ stays put. We added part (v) after seeing these comparisons, so the match is a consistency check, not a test. The comparisons were themselves made after the fact (Part 13, `R2`), and depth is an imperfect proxy for

Table 6: Depth data validation

Days requested (January 2023 – October 2026)	1,386
Days passing validation	1,329
Snapshots examined	3,897,951
dropped by V1 (missing bands)	0
dropped by V2 (non-monotone)	0
dropped by V3 (implied price off by $> 3\%$)	98,080
dropped by V4 (stuck feed)	95,954
Failed periods	April 16 – May 19, 2025; October 10–15, 2025
Training days used (2023–2024)	718

All files verified against published SHA-256 checksums. The 2025 failures coincide with archived depth that is misaligned with prices; the October 10, 2025 crash day is therefore unusable.

dealer count, so a direct test needs data on liquidity-provider identities. The model's second prediction, that the dynamic balance forecasts tail losses where reversion is close to $q\tilde{\kappa}$, is what Section 11 tests.

9 Evidence from Binance futures depth

9.1 Data and validation

We use Binance USD-M BTCUSDT perpetual futures from the public archive: one-minute klines with taker-buy volume, `bookDepth` snapshots of cumulative depth within ±1–5% of the mid at about 30-second intervals, and open interest from `metrics`. Depth is available from January 2023. No liquidation archive is available for this period, so we do not test liquidations.

Validation rules were fixed before analysis. A snapshot is kept if (V1) all of the $\pm 1, 2, 5\%$ bands are present, (V2) depth is monotone in distance on each side, (V3) the implied average price at ±1% is within 3% of the one-minute close, and (V4) it is not part of a run of more than ten identical snapshots; (V5) a day is kept if at least 80% of its minutes have a valid snapshot.

Table 7: Day-by-day T6 estimates, Binance BTCUSDT futures, training days 2023–2024

Median	All days	Top-decile volatility days	Other days
$\hat{\kappa}$ (bp per \$1M)	0.729	0.890	0.712
$\hat{\varphi}$	−0.051	2.586	−0.138
$\hat{a}$	0.970	0.980	0.969
$\hat{r}$	0.059	0.122	0.056
$\hat{\varepsilon}$	−0.0018	−0.0017	−0.0018
$\hat{\eta}$	0.878	0.863	0.880
$\hat{q}$	0.018	0.012	0.019
$\hat{C}_3$	−0.0137	−0.0111	−0.0143
Mean $\hat{C}_3$	−0.0165	−0.0167	−0.0165
95% CI	[−0.0176, −0.0155]	[−0.0219, −0.0122]	[−0.0175, −0.0154]
Share of days $\hat{C}_3 > 0$	6.5%	5.6%	6.7%
Share of days $\hat{C}_2 > 0$	0.8%	1.4%	0.8%
Median peak ratio	1.00	1.00	1.00
Days	718	72	646

One-minute bars. CIs from a day bootstrap.

9.2 Structural estimates

On one-minute bars, p_t is the log price in basis points, u_t signed taker flow in million USDT, $d_t = (B_1 - A_1)/(B_1 + A_1)$ the depth imbalance within 1% of the mid, and $x_t = p_t - \mathrm{EMA}_{60}(p_t)$. Day by day we estimate

$$\Delta p_t = \kappa u_t + \varphi d_{t-1} - (1-a)x_{t-1} + e_t,$$
$$d_t = (1-r)d_{t-1} + \varepsilon u_t + e_t,$$
$$u_t = (1-\eta)u_{t-1} + \eta q\, x_{t-1} + e_t,$$

and form $\hat{C}_2$, $\hat{C}_3$ and the implied peak ratio $\max_k \hat{h}_k/\hat{h}_0$. The price equation has Δp_t on the left, so the EMA filter does not create mechanical persistence. These are reduced-form regressions (flow is contemporaneous), so the estimates describe conditional dynamics rather than causal parameters.

Table 7 has three findings. First, the market sits on the stable side of the condition: $\hat{C}_3 < 0$ on 93.5% of days, and the median implied response peaks at impact. Second, the depletion coefficient $\hat{\varepsilon}$ is small and negative: taker buying is followed by slightly more ask-side

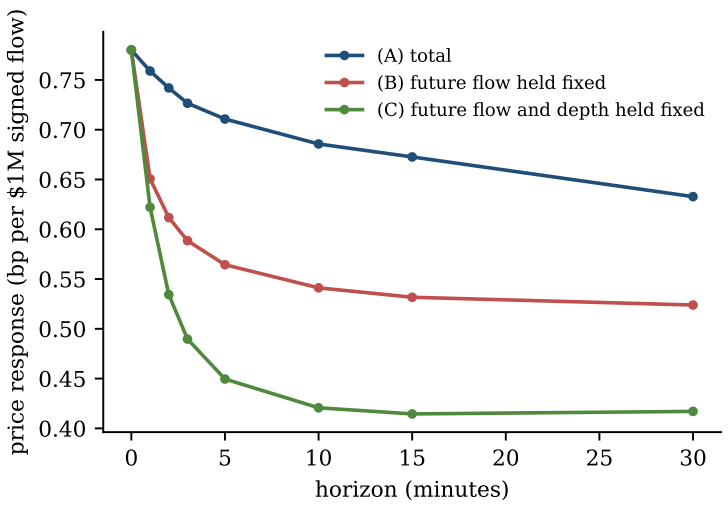


Figure 4: Price response to a $1M flow shock, local projections on 716 training days. (A) total; (B) future flow held fixed; (C) future flow and depth held fixed.

than bid-side depth within 1%, so in these re-centred bands the book appears to replenish rather than deplete the side being hit; Section 11.4 shows that a fixed price grid reverses the sign. With $\hat{\varepsilon} < 0$, $\varphi\varepsilon$ is positive only if φ is negative, which it is not on high-volatility days. Third, on high-volatility days depth imbalance moves prices strongly ($\hat{\varphi} = 2.59$) and refill is twice as fast; the depth channel is active, but it works against amplification. The momentum term $\kappa^2\eta q$ is small (about 0.008 at the medians), far below $(1-a)\kappa \approx 0.022$.

9.3 Does depth carry an overshoot?

We estimate local projections (Jordà, 2005) of the cumulative price change on flow, $p_{t+k} - p_{t-1} = \beta_k u_t + \text{controls}$, with five lags of returns, flow and depth: (A) total; (B) adding cumulative future flow, which removes order-splitting persistence; and (C) adding cumulative future depth imbalance, which removes the depth channel. Theorem 4.2(i) predicts an overshoot ($\max_{k\geq 1} \beta_k - \beta_0 > 0$) in (B) that shrinks in (C).

There is no overshoot (Figure 4, Table 8). The overshoot statistic is negative in every version. Holding depth fixed makes the response decay faster, consistent with depth that partly sustains the price move but never pushes it past impact.

Table 8: Local-projection coefficients (bp per $1M) and overshoots

Minutes	0	1	2	3	5	10	15	30	Overshoot
(A) total	0.780	0.759	0.742	0.727	0.711	0.686	0.673	0.633	−0.021 [−0.027, −0.015]
(B) flow fixed	0.780	0.651	0.612	0.589	0.564	0.541	0.532	0.524	−0.129 [−0.134, −0.124]
(C) flow, depth fixed	0.780	0.622	0.534	0.490	0.450	0.421	0.414	0.417	−0.158 [−0.164, −0.152]

716 training days; day bootstrap.

Table 9: Out-of-sample test, 2025–2026 (train 12,896 hours, test 10,979 hours)

Outcome	Candidate	ΔR^2 (pts) [95% CI]	ΔAUC [95% CI]	Holm p
MAE (primary)	$\hat{C}_3$	+0.018 [−0.100, +0.132]	+0.0001 [−0.0009, +0.0012]	—
MAE	peak ratio	+0.010 [+0.001, +0.019]	+0.0001 [−0.0000, +0.0002]	0.032
OI drop	$\hat{C}_3$	−0.045 [−0.095, −0.003]	−0.0049 [−0.0083, −0.0016]	0.787
OI drop	peak ratio	+0.102 [+0.061, +0.231]	−0.0014 [−0.0067, +0.0037]	< 0.001
Depth collapse	$\hat{C}_3$	+0.181 [−0.156, +0.484]	+0.0071 [+0.0041, +0.0105]	< 0.001
Depth collapse	peak ratio	+0.159 [+0.094, +0.234]	+0.0009 [+0.0004, +0.0014]	< 0.001

Primary: Clark–West one-sided $p = 0.023$, but the ΔR^2 CI contains zero, so the pre-registered test fails. Baseline R^2 for MAE is 0.293; baseline AUC 0.812.

9.4 Pre-registered out-of-sample test

Before opening the test period we registered the specification (SHA-256 `d95212187ae4...`): hourly blocks (06:00–23:00 UTC); $\hat{C}_3$ estimated from the previous six hours and winsorized at training percentiles; outcome the next hour's maximum adverse excursion (MAE, bp); controls for realized volatility (1 and 6 hours), absolute return, flow imbalance, volume, depth, imbalance and open-interest change; OLS on 2023–2024 and evaluation on 2025–2026. Success required a one-sided Clark–West $p < 0.05$ *and* a day-block bootstrap 95% CI for ΔR^2 excluding zero. Secondary candidates and outcomes are Holm-adjusted. The test set was opened once. An audit that scrambles all data from t onward left every feature unchanged in 30 of 30 trials.

The primary test fails (Table 9): $\hat{C}_3$ adds 0.018 percentage points of out-of-sample R^2 with a confidence interval that includes zero. Some secondary effects are statistically significant after Holm correction, notably for depth collapse, but they are economically negligible: no candidate adds more than 0.2 points of R^2 or 0.01 of AUC.

Table 10: Event days

Event	Worst hour (UTC)	Worst MAE (bp)	$\hat{C}_3$ percentile, 3 hours before	OI change in worst hour
SVB, 2023-03-11	7	214.8	82	+2.5%
Yen carry unwind, 2024-08-05	13	724.0	93, 84, 83	−0.8%
Tariff shock, 2025-04-07	14	357.8	97, 98, 38	+3.1%
2025-10-10		no valid depth data		

SVB has one valid hour before its worst hour.

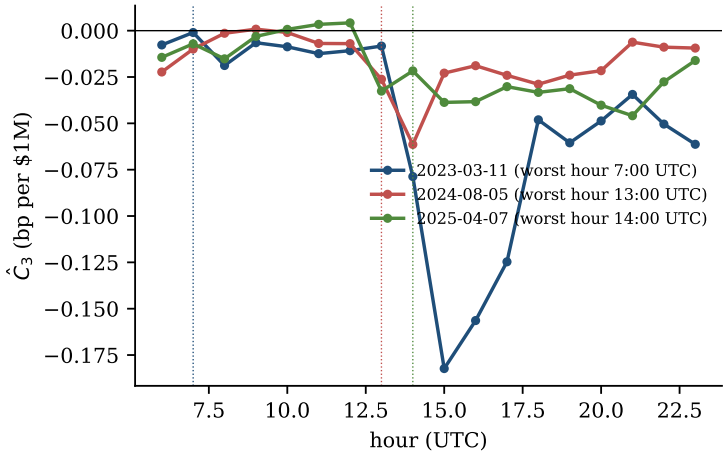


Figure 5: Hourly $\hat{C}_3$ on three event days (UTC). Values stay below zero, the amplification threshold.

A volatility flag does better. We set two flags on training data, each to flag 10% of hours: high $\hat{C}_3$ and high realized volatility. On the 909 tail hours of the test period (MAE above the training 90th percentile), the $\hat{C}_3$ flag has precision 8.6% and recall 19.5%; the volatility flag has precision 38.1% and recall 37.1%; the union has precision 16.3% and recall 49.4%. The unconditional tail rate is about 10%, so the $\hat{C}_3$ flag is no better than chance.

9.5 Crash events

For each event day we compute $\hat{C}_3$ hour by hour and its percentile in the training distribution during the three hours before the worst hour (Figure 5, Table 10). $\hat{C}_3$ was elevated before these events but stayed negative throughout, and three events are not a test.

Table 11: Unit-free amplification on crash days and the no-feedback null

Event	Windows (ex ante)	Mean M_{std}	Null 99th pct	p (vs null)	Outcome-selected mean
May 2021	142	1.0000	1.00002	1.00	1.0000
LUNA 2022	57	1.00008	1.00006	0.039	1.00045
FTX 2022	49	1.0000	1.00029	1.00	1.0000
SVB 2023	6	1.00006	1.00025	0.059	1.00032
Aug 2024	108	1.0000	1.00007	1.00	1.0000

Ten-minute windows of one-second bars. The median hump statistic is 1.000 in every event and its null p-value is 1.00. In the ordinary 2021–2023 panel, 0.2% of windows have $A_{xx} > 1$ and the 95th percentile of M_{std} is 1.0000; a handful of near-singular windows produce very large values, so ordinary-window means are not informative.

10 Evidence from spot prices

10.1 A unit-free amplification statistic

The earlier version measured amplification by the norm of powers of a VAR(1) operator on (price deviation, flow), estimated over ten-minute windows of one-second Binance spot data. That norm depends on the units of the state: multiplying prices by 10 and dividing flow by 1,000 changed the raw norm from 8.73 to 86,731 on the same window. Proposition 2.2 explains why: state norms are not invariant to coordinates. We therefore standardize the state, $\tilde{J} = D^{-1}\hat{J}D$ with $D = \text{diag}(\text{sd}\, x, \text{sd}\, f)$, and use $M_{\text{std}} = \max_k \|e_x^\top \tilde{J}^k\|$, which is invariant to units (verified). We also report the price-to-price amplification $A_{xx} = \max_k |(\hat{J}^k)_{11}|$ and a delayed-peak statistic $\text{hump} = \max_k |(\hat{J}^k)_{12}|/|\hat{J}_{12}|$.

10.2 Crash days and a no-feedback null

We study five crash days: May 19, 2021; LUNA, May 9, 2022; FTX, November 8, 2022; SVB, March 11, 2023; and August 5, 2024. Windows are selected *ex ante* (previous ten minutes in the top volatility decile), and for comparison by outcome (the ten largest absolute returns). A no-feedback null keeps each window's real flow, generates prices with no feedback, $\Delta p = \hat{\kappa} f + \hat{\sigma} e$, and pushes them through the identical pipeline.

Only LUNA exceeds the null, by less than 0.01% (Table 11). Outcome-selected windows give only marginally larger values (≤ 1.0005). Stress-window means exceed the ordinary 95th

Table 12: Spot local projections (bp per $1M)

Seconds	0	1	2	5	10	20	30	60	120	300
Ordinary, total	5.86	6.28	6.53	6.60	6.69	6.77	6.67	6.49	6.39	5.92
Ordinary, flow fixed	5.86	6.02	6.11	5.90	5.68	5.26	4.86	4.16	3.37	1.99
Crash, total	6.69	6.85	6.92	7.13	7.75	8.20	8.46	8.20	9.09	8.96
Crash, flow fixed	6.69	6.41	6.28	5.99	5.95	5.46	5.02	3.88	3.10	−0.08

percentile in two events, by amounts of the same size. The pre-registered gate fails: unit-free transient amplification in spot data is indistinguishable from one and from the pipeline null. Varying the EMA span (60, 120, 300 seconds) and the bar length (1 or 5 seconds) does not change this; crash-day means range from 1.0003 to 1.18 across specifications and are never above the corresponding ordinary means.

10.3 Overshoot with flow held fixed

Local projections on one-second data give the price response to a $1M flow shock (Table 12). On ordinary days, with future flow held fixed, there is a small, short-lived overshoot of +0.248 [0.133, 0.376] bp per $1M, peaking at two seconds and decaying after. On crash days the flow-fixed overshoot is negative, −0.277 [−0.528, −0.042]. The large total overshoots (+0.91 ordinary, +2.40 crash) disappear when future flow is held fixed: they come from order splitting, not from the book.

10.4 Pre-registered predictive test

Before opening 2024, we registered (SHA-256 `b1fd7f8e...`) a test of whether pre-window amplification predicts the next ten minutes' MAE beyond volatility and flow controls, on non-overlapping 30-minute blocks (train 2021–2023, 3,307 blocks; test 2024, 1,104 blocks). It does not: $\Delta R^2 = -0.023$ points [−0.056, −0.002], Clark–West $p = 0.966$, AUC 0.822 in both models. A random feature with the same autocorrelation gives $\Delta R^2 = -0.020$ points, the same as the candidate. Leave-one-year-out fits within the training period are unstable (the 2023 fold gives a very large negative ΔR^2 because of outlying candidate values), so the

null result does not depend on one year being unusual.

The test set was opened three times. The first evaluations stopped on a numerical error from degenerate windows; the fix drops blocks with non-finite features (five blocks) and is the only change, and the registered specification hash is unchanged. A volatility flag again dominates: precision 43% and recall 30% against 3% and 10% for the amplification flag; the difference in recall has a CI of $[-0.37, 0.13]$.

11 Evidence from thinner books

The deepest book in crypto is the least likely place to find amplification. This section moves to thinner books. It reports the pre-registered out-of-sample tests in thin Binance books, on Bybit and in thin US stocks, failures included, and labels every check decided after a test set was opened as post hoc. Figure 7 collects all out-of-sample evaluations of this section.

11.1 Design

We fixed 30 Binance USD-M perpetuals from the quote-volume ranking of one training day (June 10, 2024) before computing any T6 statistic, spread from BTCUSDT to symbols with under $5M daily volume. Data are sampled on the 10th and 24th of each month from January 2023 to September 2026 and pass the validation rules V1–V5 of Section 9: 2,565 of 2,700 symbol-days are valid; of 7,486,444 snapshots, V3 drops 212,745 and V4 drops 161,280, and V1 and V2 drop none. Liquidity terciles are fixed on median training-period depth within 1% of the mid. The thin tercile has $0.22M–$0.38M of depth within 1%, the thick tercile $0.97M–$220M. Flow is divided by each symbol's median absolute one-minute training flow, so that $\hat{\kappa}$ is in basis points per typical minute of flow. All other definitions follow Section 9. An audit that scrambles all data from t onward left every hourly feature unchanged in 30 of 30 trials.

Table 13: T6 estimates by liquidity tercile, 30 USD-M perpetuals, training days 2023–2024

	Thick	Mid	Thin
Mean $\hat{C}_3$	−0.0297	−0.0216	−0.0324
95% CI	[−0.0341, −0.0257]	[−0.0250, −0.0185]	[−0.0386, −0.0268]
Share of symbol-days with $\hat{C}_3 > 0$	9.4%	15.2%	16.1%
Median $\hat{\varphi}$	2.33	2.38	3.11
Median $\hat{\varepsilon}$	−0.0014	−0.0006	−0.0005
Share with $\hat{\varepsilon} > 0$	11.1%	24.4%	30.1%
Mean feedback loop gain $\hat{Q}$	0.0091	0.0033	0.0016
Spearman correlation with log depth across the 30 symbols [95% CI]			
mean $\hat{C}_3$		−0.001 [−0.37, 0.38]	
share $\hat{C}_3 > 0$		−0.59 [−0.81, −0.27]	
$\hat{Q}$		+0.70 [0.41, 0.87]	

1,424 training symbol-days, ten symbols per tercile. CIs from a symbol-day bootstrap (tercile rows) and a symbol bootstrap (correlations). $\hat{Q}$ is the loop gain of the companion paper estimated on one-minute bars; a placebo random walk with each day's volatility gives $\hat{Q} \approx 0$ for every symbol.

11.2 The cross-section of the amplification condition

Table 13 confirms the BTCUSDT result across liquidity: mean $\hat{C}_3$ is negative in every tercile, and taker flow on average does not deplete the side it hits ($\hat{\varepsilon} < 0$ at the median everywhere). What changes with depth is the frequency of the amplifying regime, not its average. The share of symbol-days with $\hat{C}_3 > 0$ rises from 9.4% in the thick tercile to 16.1% in the thin tercile, with a Spearman correlation of −0.59 against log depth. Depth sensitivity $\hat{\varphi}$ is larger, and depletion is less negative and more often positive, in thinner books. We had predicted a negative correlation between log depth and mean $\hat{C}_3$; that prediction fails ($\rho = -0.001$).

The last row is an unexpected finding for the companion paper. Measured feedback $\hat{Q}$ is larger in *deeper* books ($\rho = +0.70$): flow responds to price deviations most where liquidity, and plausibly algorithmic participation, is greatest. Feedback and amplification are therefore different objects. Deep books have more feedback and refill fast enough to absorb it; thin books have less feedback and, more often, too little reversion.

Table 14: Local-projection overshoot by tercile, $\max_{k\geq 1} \beta_k - \beta_0$ [95% CI]

	Thick	Mid	Thin
(A) total	−0.045 [−0.079, 0.047]	−0.034 [−0.054, 0.024]	+0.007 [−0.050, 0.083]
(B) future flow fixed	−0.181 [−0.211, −0.152]	−0.075 [−0.091, −0.058]	−0.073 [−0.094, −0.051]
(C) flow and depth fixed	−0.243 [−0.276, −0.212]	−0.121 [−0.140, −0.102]	−0.113 [−0.135, −0.092]
Symbol-days	459	460	445

Basis points per typical minute of flow, horizons 0–30 minutes; local projections are estimated per symbol-day and averaged, with a symbol-day bootstrap.

Table 15: Depletion at the snapshot scale (about 30 seconds)

Symbol	Tercile	$\hat{\varepsilon}_{30}$ [95% CI]	Sweep extra [95% CI]	$\hat{\varepsilon}$, 1 minute	Days $\hat{\varepsilon}_{30} > 0$
BTCUSDT	thick	−0.0045 [−0.0058, −0.0033]	+0.0034 [0.0023, 0.0046]	−0.0025	0 of 6
XMRUSDT	mid	−0.0042 [−0.0058, −0.0029]	+0.0037 [0.0025, 0.0051]	−0.0011	0 of 6
RLCUSDT	thin	−0.0034 [−0.0065, −0.0012]	+0.0029 [0.0011, 0.0058]	−0.0003	1 of 6
CELRUSDT	thin	−0.0015 [−0.0032, 0.0001]	+0.0017 [0.0004, 0.0031]	−0.0004	2 of 6
CVXUSDT	thin	+0.0010 [−0.0033, 0.0060]	−0.0003 [−0.0040, 0.0027]	+0.0004	2 of 6

Day bootstrap over six training days per symbol. Depth is measured in bands of ±1% around the current mid, so a price move also recenters the band; this biases $\hat{\varepsilon}_{30}$ in an unknown direction.

11.3 Overshoot and depletion

With future flow held fixed there is no overshoot in any tercile (Table 14); the registered prediction of a positive (B) overshoot in thin books fails. The response decays less in thin than in thick books, and holding depth fixed again makes it decay faster, as in Table 8.

To check that replenishment is not an artefact of one-minute aggregation, we sum signed taker flow exactly between consecutive depth snapshots from trade-level data and estimate depletion ε_{30} at about 30 seconds, with an additional coefficient on sweeps (intervals in the top decile of absolute flow). Table 15 reports five symbols on six training days each.

In these re-centred bands, replenishment is stronger, not weaker, at 30 seconds: $\hat{\varepsilon}_{30}$ is negative with a CI below zero in three of five symbols and is never significantly positive. Sweeps offset most of it, so a sweep leaves the hit side about as deep as before ($\hat{\varepsilon}_{30}$ plus the sweep coefficient lies between −0.0012 and +0.0007). On this measure, the depletion channel that drives Theorem 4.2(i) would be absent even in thin books; the next subsection shows that the measure itself is biased toward replenishment.

Table 16: Depletion at 30 seconds with a fixed and a re-centred band (Bybit order books, post hoc)

Band	$\hat{\varepsilon}_{30}$ [95% CI]	Share > 0	Coin-days
Fixed at previous mid	+0.0021 [+0.0013, +0.0032]	95%	20
Re-centred (Binance)	−0.0004 [−0.0007, −0.0001]	30%	20

ACH, ALICE, API3, LIT, WOO and ZEN, May 11–12 and September 11–12, 2025; flow normalized by each coin-day's median absolute 30-second flow; mean over coin-days with a bootstrap interval. Post hoc. Kind: real data (Part 13, `R5`).

11.4 Depletion on a fixed price grid

Binance's depth archive reports depth within bands around the *current* mid. After a buy moves the mid up, the ask band shifts into deeper resting orders and the bid band away from resting bids, which mechanically makes the book look ask-heavy after a buy, the sign we estimate. Bybit's order-book files contain every level, so the band can be held fixed at the previous mid. After seeing Table 15 we computed, for the six coins of Table 23 on four days each, the change in depth imbalance over 30-second intervals regressed on signed taker flow, once with the band fixed at the start-of-interval mid and once re-centred at the end-of-interval mid, Binance's construction.

The sign reverses (Table 16). With the band fixed, buying depletes the ask side in 19 of 20 coin-days; with the band re-centred, the same data reproduce the replenishment of Tables 7 and 15. The replenishment finding is therefore an artefact of band recentring, and depletion is present. Two conclusions survive. First, local projections show no peak overshoot once future flow is held fixed (Table 14), so depletion is not large enough to satisfy $\varphi\varepsilon > (1-a)\kappa$ in the data we observe; second, the depth term carries no out-of-sample forecasting content (Tables 19 and 21). Both statements rest on Binance's re-centred depth measure, and a fixed-grid measure of $\varphi\varepsilon$ in the same books would require order-level Binance data that are not public.

Table 17: Pre-registered test, 30 symbols, 2025–2026

Sample	Outcome | candidate	ΔR^2 (pts) [95% CI]	ΔAUC [95% CI]	Holm p
Thin (primary)	MAE | $\hat{C}_3$	+1.216 [+0.273, +7.987]	+0.012 [+0.003, +0.019]	—
Thin	MAE | peak ratio	+0.065 [−0.072, +0.108]	+0.000 [−0.000, +0.001]	0.207
Thin	depth collapse | $\hat{C}_3$	−2.111 [−3.127, −1.173]	−0.012 [−0.023, −0.003]	1.000
Thin	OI drop | $\hat{C}_3$	+0.184 [−0.242, +1.128]	+0.040 [+0.017, +0.061]	< 0.001
Mid	MAE | $\hat{C}_3$	+0.003 [−0.056, +0.875]	+0.002 [−0.002, +0.005]	0.551
Thick	MAE | $\hat{C}_3$	+0.039 [−0.244, +0.057]	−0.001 [−0.003, +0.000]	0.361
All 30	MAE | $\hat{C}_3$	+0.117 [−0.015, +1.736]	+0.003 [−0.000, +0.006]	< 0.001

Primary: train 8,340 and test 6,521 symbol-hours on ten symbols; Clark–West $t = 8.53$, one-sided $p < 0.001$; baseline out-of-sample $R^2 = 0.025$; AUC 0.768 → 0.780. The pre-registered test passes. Holm p is the Clark–West p-value adjusted across the six secondary tests. Date-block bootstrap with 2,000 draws; bootstrap CIs are right-skewed because a few volatile dates carry much of the variation.

11.5 Pre-registered test in thin books

Before opening 2025–2026 for these symbols we registered the specification (SHA-256 `5910b6b4d90a...`). It is identical to Section 9 except for three changes, all fixed in advance: the primary sample is the thin tercile; the regressions include symbol fixed effects; and the bootstrap resamples dates, with all symbols on a date kept together. The test cell was executed twice in the same session. The specification hash, the code and the data were unchanged between the two runs, and the evaluation is deterministic, so both runs give the numbers below; we report the count because the protocol requires it.

The primary test passes (Table 17). In thin books, $\hat{C}_3$ estimated from the previous six hours adds 1.2 points of out-of-sample R^2 for the next hour's maximum adverse excursion, beyond realized volatility, flow, volume, depth, imbalance and open interest, and raises the AUC for tail hours by 0.012. The effect is confined to thin books: in the mid and thick terciles it is zero, and in the pooled sample the Clark–West test rejects but the ΔR^2 interval contains zero. $\hat{C}_3$ also helps classify hours with a large open-interest drop, but it predicts depth collapse *worse* than the baseline, again pointing away from the depth channel.

Flags. On training data we set a $\hat{C}_3$ flag and a volatility flag; 575 of 6,521 test hours (8.8%) are tail hours. The $\hat{C}_3$ flag fires in 8.9% of hours with precision 16.5% and recall

16.7%, about twice the base rate. The volatility flag, which fires more often (16.0%), does better (precision 23.4%, recall 42.4%), and the union reaches recall 51.1% at precision 19.5%. $\hat{C}_3$ is a useful addition to volatility in thin books, not a replacement for it.

11.6 Tick size: a caution and a robustness check

Tick changes. We located tick changes by inferring each day's price grid and dating changes by binary search over daily files. We found 12 tick reductions, all by a factor of ten, on 11 symbols (GMTUSDT twice) between August 2023 and August 2026; nine fall in the test period. A difference-in-differences design compares 10 days before and after each event with four control symbols of similar liquidity. After a tick cut, $\hat{C}_3$ rises with a CI above zero in 10 of 12 events, but realized volatility in basis points falls sharply at the same time (by up to 32 bp on ALICEUSDT), and the implied peak ratio does not change in any event (every CI includes zero). The Spearman correlation between the change in refill $\hat{r}$ and the change in the peak ratio is -0.04. We read this as a measurement result, not a test of the theory: at a coarse tick, prices move in discrete steps and the basis-point scale of $\hat{C}_3$ changes mechanically when the grid changes. The natural experiment cannot test the prediction that faster refill lowers the overshoot.

Does tick size drive Table 17? Eleven of the 30 symbols, six of them thin, changed tick, and five of those six did so inside the test period. After seeing Table 17 we therefore ran the post-hoc checks of Table 18, counted separately (one evaluation).

The result survives controls for tick size, the removal of every day traded at a tick different from training, and the removal of any single symbol (Table 18). It weakens when the six thin symbols that ever changed tick are dropped: four remain, and the point estimate halves with an interval that just includes zero. It disappears when the outcome is measured in ticks; at each tick cut this outcome jumps tenfold, a break the training fit cannot anticipate, so we regard R4 as a weak test, but we report it. A scale-free version of the candidate

Table 18: Post-hoc tick-size robustness of the thin-book result

Check	Symbols	Test hours	ΔR^2 (pts) [95% CI]	Clark–West p
R0 baseline (Table 17)	10	6,521	+1.216 [+0.295, +8.017]	< 0.001
R1 + relative tick size and its log	10	6,485	+1.243 [+0.373, +7.627]	< 0.001
R2 drop symbols with any tick change	4	2,639	+0.511 [−0.069, +2.061]	< 0.001
R3 only days with the training tick	10	5,642	+1.131 [+0.256, +9.305]	< 0.001
R1 and R3 together	10	5,642	+1.222 [+0.357, +9.410]	< 0.001
R4 outcome in ticks	10	6,485	−0.127 [−0.319, −0.018]	0.991
R5 $\hat{C}_3$ percentile within symbol	10	6,521	+0.158 [−0.013, +1.322]	0.026
R6 leave one symbol out (10 fits)	9		+0.698 to +1.760, all CIs > 0	all < 0.001

Post hoc: decided after Table 17. 500 date-block bootstrap draws (200 in R6).

(its percentile within each symbol's training distribution) is weak. Our reading is that $\hat{C}_3$'s forecasting content is tied to its *level*, which mixes the amplification balance with the basis-point scale of each book. Sections 11.7 and 11.8 test whether that level is more than a measure of scale.

11.7 Out-of-sample replication on new symbols

A result found in ten symbols could be specific to them. We therefore drew 24 new symbols, seeded and before any analysis, from the 72 USD-M perpetuals that were not in the original 30 and whose training-day volume was at most 1.5 times that of the most liquid thin symbol. We applied the same validation (2,090 of 2,160 symbol-days valid) and classified symbols as thin with a cut fixed in advance at the boundary between the original thin and mid terciles ($0.39M of depth within 1%). Ten new symbols are thin (ACH, ARPA, CHR, GTC, HOT, IOTX, OGN, SPELL, T and ZEN, all USDT-margined) and 14 are not. A new specification (SHA-256 `6aa8e2c03d79...`) fixed the primary test identical to Table 17 and, as secondary tests, the three terms of C_3 separately, a unit-free ratio $A = (\varphi\varepsilon + \kappa^2\eta q)/((1-a)\kappa) - 1$ that is positive exactly when $C_3 > 0$, and the not-thin group. The test set was opened once.

The result replicates (Table 19). On symbols never used before, $\hat{C}_3$ adds 0.85 points of out-of-sample R^2 with an interval excluding zero, and improves tail classification. It again has no value outside thin books. The decomposition shows where the forecasting content

Table 19: Pre-registered replication on ten new thin symbols, 2025–2026

Sample	Candidate	ΔR^2 (pts) [95% CI]	ΔAUC [95% CI]	Holm p
Thin (primary)	$\hat{C}_3$	+0.845 [+0.303, +6.371]	+0.008 [+0.002, +0.013]	—
Thin	momentum term $\kappa^2\eta q$	+0.641 [+0.395, +3.242]	+0.003 [−0.001, +0.008]	< 0.001
Thin	reversion term $(1-a)\kappa$	+0.688 [+0.204, +5.606]	+0.008 [+0.003, +0.012]	< 0.001
Thin	depth term $\varphi\varepsilon$	+0.013 [−0.213, +1.155]	−0.001 [−0.005, +0.003]	0.308
Thin	unit-free ratio A	−0.045 [−0.096, −0.031]	−0.000 [−0.001, +0.000]	0.974
Not thin (14)	$\hat{C}_3$	−0.042 [−0.131, +0.780]	+0.002 [−0.003, +0.007]	0.308

Primary: train 8,520 and test 7,092 symbol-hours; Clark–West $t = 11.66$, one-sided $p < 0.001$; AUC 0.768 → 0.776; training coefficient +54.3 (Table 17: +88.3). Training coefficients: momentum +104.3, reversion −53.0. The pre-registered replication passes.

lies, and every sign matches Theorem 4.2(i) and Theorem 4.2(iii). A larger momentum term predicts larger adverse excursions; a larger reversion term, which pushes C_3 down, predicts smaller ones; and the depth term, whose depletion factor is close to zero (Table 15), predicts nothing. In training data the reversion term is by far the largest component (median 0.033, against −0.0009 for momentum and −0.0003 for depth), so most of the variation in $\hat{C}_3$ is variation in impact-weighted mean reversion.

The unit-free ratio fails. Because the amplifying terms are small, its median is −0.996 and its variation is mostly the noise of a ratio with an estimated denominator. Together with R5 in Table 18, this means that on these samples the forecasting content was tied to the level of $\hat{C}_3$. Section 11.9 tests two better-behaved unit-free statistics on fresh data.

11.8 Is $\hat{C}_3$ just price impact?

Both working terms contain the impact coefficient κ, so $\hat{C}_3$ might forecast excursions only because it measures illiquidity. After seeing Table 19 we added four impact controls, all from the same past six hours or hour and winsorized at training percentiles: $\hat{\kappa}$ from the same fit, $\hat{\kappa}^2$, a univariate Kyle's λ (Kyle, 1985) (slope of returns on signed flow) and a log Amihud ratio (Amihud, 2002) (absolute return over volume). This is a post-hoc check, evaluated once and counted separately.

$\hat{C}_3$ is not a proxy for impact (Table 20). With all four impact controls it keeps about 80% of its gain in R^2 (0.68 of 0.85 points), with an interval above zero, and both the momentum

Table 20: Post-hoc price-impact controls, ten new thin symbols, 2025–2026

Candidate | added controls	ΔR^2 (pts) [95% CI]	ΔAUC [95% CI]	Clark–West t
$\hat{C}_3$ | none (Table 19)	+0.845 [+0.303, +6.371]	+0.008 [+0.002, +0.013]	11.66
$\hat{C}_3$ | $\hat{\kappa}$, $\hat{\kappa}^2$	+0.755 [+0.371, +4.718]	+0.006 [−0.000, +0.010]	6.63
$\hat{C}_3$ | Kyle's λ, Amihud	+1.089 [+0.498, +7.149]	+0.007 [+0.001, +0.013]	8.02
$\hat{C}_3$ | all four	+0.682 [+0.334, +4.525]	+0.004 [−0.001, +0.009]	5.69
Momentum term | all four	+0.487 [+0.311, +2.384]	+0.002 [−0.002, +0.006]	3.42
Reversion term | all four	+0.529 [+0.222, +3.762]	+0.005 [−0.000, +0.009]	8.45
$\hat{\kappa}$ alone | none	−0.026 [−0.084, +0.030]	+0.000 [−0.000, +0.000]	−1.77

Post hoc: decided after Table 19. Same evaluation code and bootstrap as Table 19; Holm-adjusted $p < 0.001$ for every row except $\hat{\kappa}$ alone ($p = 0.96$). Training correlations of $\hat{C}_3$ with the controls: $\hat{\kappa}$ −0.43, Kyle's λ −0.44, Amihud +0.10.

and the reversion terms survive separately. Impact on its own forecasts nothing. Controlling for Kyle's λ and Amihud alone *raises* the gain to 1.09 points: once plain illiquidity is held fixed, the dynamic balance that $\hat{C}_3$ measures stands out more clearly. One qualification: with $\hat{\kappa}$ controlled, the gain in tail *classification* is no longer significant. The robust claim is about the size of the next hour's adverse excursion, not about flagging tail hours beyond what impact already flags.

11.9 A third test on fresh days

Parts 4 and 5 of the replication package use the 10th and 24th of each month. For a third test we registered a new specification (SHA-256 `65acda361861...`) before downloading the 3rd and 17th of each month of 2025–2026, days never used in any analysis. Training reuses the 2023–2024 days of the earlier tests. The sample is all 20 thin symbols of Sections 11.5 and 11.7, with the ten mid-tercile symbols as a negative control. Two changes make the test harder: the four price-impact controls of Table 20 are now part of the baseline, and because the specification has two primary hypotheses (this one and the backtest of Section 11.13) each is tested at $\alpha = 0.025$ with 97.5% intervals. The test set was opened once; 1,137 fresh symbol-days passed validation.

The result holds a third time, and more strongly (Table 21). On fresh days, with impact already in the baseline, $\hat{C}_3$ adds 1.6 points of R^2 for the next hour's adverse excursion in

Table 21: Pre-registered test on fresh days, 2025–2026 (impact controls in the baseline)

Sample	Candidate	ΔR^2 (pts) [CI]	ΔAUC [95% CI]	Holm p
Thin (primary)	$\hat{C}_3$	+1.583 [+0.588, +3.237]	+0.002 [−0.003, +0.006]	—
Thin	reversion term	+1.442 [+0.679, +2.621]	+0.003 [−0.001, +0.006]	< 0.001
Thin	momentum term	+0.422 [+0.000, +1.063]	−0.002 [−0.005, +0.001]	< 0.001
Thin	depth term	+0.323 [+0.092, +0.639]	+0.001 [−0.001, +0.003]	< 0.001
Thin	gain multiple $S_{60}/\lvert\hat{h}_0\rvert$	+0.488 [+0.252, +0.851]	+0.000 [−0.001, +0.001]	< 0.001
Thin	log reversion half-life	+0.619 [+0.351, +1.031]	+0.001 [+0.000, +0.003]	< 0.001
Thin	$\hat{C}_3$ without depth, $\hat{C}_3^{\text{flow}}$	+1.747 [+0.725, +3.393]	+0.001 [−0.004, +0.006]	< 0.001
Mid (control)	$\hat{C}_3$	+0.055 [−0.047, +0.239]	+0.000 [−0.001, +0.001]	0.038

Primary: train 16,860 and test 13,428 symbol-hours on 39 test dates; 97.5% date-block bootstrap interval; Clark–West $t = 9.46$, one-sided $p < 0.001$; baseline out-of-sample $R^2 = 0.185$; training coefficient +68.0. The pre-registered test passes. Secondary rows: 95% intervals, Holm-adjusted Clark–West p across the eleven secondary tests of the specification. Thin: 20 symbols; mid: 10. The gain multiple and the half-life are unit-free. S_{60} is the summed absolute impulse response over 60 minutes; $\hat{C}_3^{\text{flow}} = \hat{\kappa}^2\hat{\eta}\hat{q} - (1-\hat{a})\hat{\kappa}$ from the model without depth.

thin books. Three findings sharpen the earlier picture. First, two unit-free statistics now work: the gain multiple, the summed response divided by impact, and the log half-life of reversion both add about half a point of R^2 with intervals above zero. The scale concern of Section 11.6 is therefore answered: the content is the dynamic balance, not the basis-point scale of the book, in the sense of the invariance hypotheses of Kyle and Obizhaeva (2016). Second, the reversion term carries most of the signal, as Theorem 4.4 suggests, since S depends on reversion through $1/(m - q\tilde{\kappa})$. Third, the depth-free statistic forecasts as well as the full one, which matters for the next subsection. In mid-depth books the Clark–West test rejects at the Holm-adjusted 5% level, but the R^2 gain is 0.06 points with an interval that includes zero. As with the impact controls, the gain in tail *classification* is not significant: the robust claim is about the size of excursions.

11.10 Beyond volatility forecasting

The adverse excursion of the next hour is close to a volatility outcome, so a natural objection is that $\hat{C}_3$ only proxies for future volatility in a way the simple volatility controls miss. We registered a fourth test (SHA-256 `71983f1a9d2c...`) on the 6th and 20th of each month of 2025–2026, days not used in any other test, whose baseline adds the standard tools of

Table 22: Pre-registered test beyond a volatility-forecasting baseline, fresh days 2025–2026

Sample	Candidate	ΔR^2 (pts) [95% CI]	ΔAUC [95% CI]	Holm p
Thin (primary)	$\hat{C}_3$	+2.787 [+1.425, +4.944]	+0.005 [+0.002, +0.008]	—
Thin	summed response S_{60}	+2.098 [+1.165, +3.597]	+0.004 [+0.001, +0.006]	< 0.001
Thin	reversion term	+2.451 [+1.257, +4.379]	+0.004 [+0.001, +0.007]	< 0.001
Thin	momentum term	+0.731 [+0.238, +1.429]	+0.001 [−0.001, +0.002]	< 0.001
Thin	gain multiple $S_{60}/\lvert\hat{h}_0\rvert$	+0.563 [+0.259, +1.003]	+0.001 [−0.001, +0.002]	< 0.001
Thin	log reversion half-life	+0.806 [+0.446, +1.351]	+0.001 [−0.000, +0.002]	< 0.001
Thin	$\hat{C}_3^{\text{flow}}$ (no depth)	+2.989 [+1.453, +5.424]	+0.005 [+0.002, +0.009]	< 0.001
Mid (control)	$\hat{C}_3$	+0.316 [+0.108, +0.635]	+0.002 [+0.001, +0.004]	< 0.001

Thin: 20 symbols, train 16,860 and test 13,722 symbol-hours on 40 test dates; mid: 10 symbols. Primary: Clark–West $t = 12.59$, one-sided $p < 0.001$; baseline out-of-sample $R^2 = 0.186$, full 0.214; AUC 0.826 → 0.831. The pre-registered test passes. Baseline: the controls of Table 21, the four impact controls, realized volatility over 15 minutes and since midnight, an EWMA volatility forecast ($\lambda = 0.94$), the previous hour's adverse excursion, hour-of-day fixed effects and symbol fixed effects. The gain multiple and the half-life are unit-free. Kind: real data.

volatility forecasting to every control used so far: realized volatility over 15 minutes, one hour, six hours and since midnight in the spirit of HAR models (Corsi, 2009), an exponentially weighted (RiskMetrics) volatility forecast (J.P. Morgan/Reuters, 1996), the previous hour's adverse excursion, and hour-of-day fixed effects, together with the four price-impact controls. The test set was opened once; 1,161 fresh symbol-days passed validation.

The result does not weaken against a volatility-forecasting baseline; it strengthens (Table 22). $\hat{C}_3$ adds 2.8 points of out-of-sample R^2 on top of a baseline that already explains 18.6%, and for the first time it also improves the classification of tail hours with an interval excluding zero. The summed response S_{60}, the statistic that the guard bounds and that Theorem 4.4 identifies as cumulative transient risk, adds 2.1 points on its own. The volatility benchmarks, by contrast, add nothing out of sample beyond the controls of Table 21 (-0.76 points, interval $[-3.38, +2.10]$), so the forecasting content of the dynamic balance is not a volatility effect that richer volatility models would have captured. In mid-depth books the gain is significant in this test but small, about a ninth of the thin-book gain; across the four tests with a deeper-book control it was significant once.

Against VPIN. The best-known early-warning measure for flow-driven crashes is VPIN (Easley et al., 2012). After the fact, on the same test hours (Part 13, `R8`), we computed VPIN

from the previous six hours, the window of $\hat{C}_3$, as the mean absolute buy–sell imbalance over 50 equal-volume buckets; Binance reports taker-buy volume, so the imbalance is measured rather than classified. VPIN adds 1.2 points of R^2 to the baseline (95% interval $[0.4, 2.3]$). $\hat{C}_3$ added to the baseline *and* VPIN still adds 2.0 points ($[1.0, 3.4]$, Clark–West $t = 12.1$) and improves tail classification, keeping about 73% of its 2.8-point gain. The two are negatively related (Spearman -0.35), and VPIN's training coefficient is negative, so high imbalance precedes smaller, not larger, excursions in these books. This comparison is post hoc and on a test set already opened; it shows that $\hat{C}_3$ is not a repackaging of order-flow toxicity.

11.11 A second venue: Bybit

To ask whether the result is specific to Binance we registered a test on Bybit USDT perpetuals of the same 20 coins (SHA-256 recorded in the package), using Bybit's public trade archive, which has no depth history. Fourteen of the coins have Bybit data across 2023–2026. Minute bars are built from trades (last price, signed taker notional), training uses the 10th and 24th of each month of 2023–2024, and testing uses the 3rd, 10th, 17th and 24th of each month of 2025–2026 (10,764 test hours). The statistic is $\hat{C}_3^{\text{flow}}$, which Table 21 shows to work on Binance.

The test fails without ambiguity. $\hat{C}_3^{\text{flow}}$ adds -0.0000 points of R^2 (95% CI $[-0.001, 0.001]$, Clark–West $p = 0.50$), and neither the momentum nor the reversion term helps (Holm $p = 1.00$). In training data the momentum coefficient even has the opposite sign (-130, against $+110$ on Binance). Because the depth-free statistic works on Binance on the same coins, the failure concerns the venue rather than the statistic. We considered two candidate reasons and checked the first after the fact: bid–ask bounce in Bybit's last-trade prices, and different participants. On the same 14 coins and training days, Bybit's one-minute returns are *less* negatively autocorrelated than Binance's (median -0.016 against -0.031), so bounce is not the difference; trading is: 11% of Bybit minutes have no trade, against 0.1% on Binance, and 29% of Bybit minutes have a zero return, against 14% (Part 13, `R3`). Sparse

Table 23: Pre-registered test on Bybit with order-book depth, 2025–2026

Candidate	ΔR^2 (pts) [95% CI]	Clark–West t	Holm p
$\hat{C}_3$ (primary)	+0.033 [+0.019, +0.053]	5.73	—
Momentum term	+0.083 [+0.017, +0.168]	2.94	0.010
$\hat{C}_3^{\text{flow}}$ (no depth)	+0.064 [−0.009, +0.152]	2.61	0.023
Summed response S_{60}	+0.021 [−0.025, +0.071]	1.31	0.383
Reversion term	−0.039 [−0.139, +0.058]	−0.61	1.000
Gain multiple	−0.043 [−0.137, +0.044]	−0.74	1.000
Log reversion half-life	−0.099 [−0.249, +0.041]	−1.47	1.000

Six coins, train 1,224 and test 2,988 symbol-hours; baseline out-of-sample $R^2 = 0.251$; AUC unchanged (0.808 → 0.808); training coefficient on $\hat{C}_3$ +4.4 (Binance, Table 22: +70.8). Median depth within 1% on training days \$0.30M–\$0.43M. The pre-registered criterion is met. Specification SHA-256 `ec2145fac5f5...`; test opened once. Kind: real data (Part 10).

trading leaves the flow and price series that the statistic is estimated from much noisier on Bybit.

Bybit with order-book depth. Bybit also distributes daily order-book files for its perpetuals (every update to the top 200 or 500 price levels). We registered a second Bybit test (SHA-256 `ec2145fac5f5...`) before downloading any test day, with the full Part 9 baseline, depth within 1% of the mid rebuilt from the order book at the end of each minute, and only the best 200 levels used so that the 2024 (500-level) and 2025–2026 (200-level) files are measured identically. Days were the 11th to 16th of a month, never used in any earlier test: two training months in 2024 and six test months from January 2025 to September 2026. A day is valid if at least 80% of its minutes have trades and a valid book; sparse trading in thin coins on Bybit leaves 581 of 882 coin-days valid, and the inclusion rule (at least 8 valid training days and 24 valid test days) keeps six coins (ACH, ALICE, API3, LIT, WOO, ZEN; 2,988 test hours on 36 dates).

The pre-registered criterion is met, but the effect is negligible (Table 23). $\hat{C}_3$ adds 0.033 points of R^2 with an interval above zero and a Clark–West t of 5.7: statistically clear, economically about one eightieth of the Binance gain, with no improvement in tail classification and a training coefficient one sixteenth as large. Of the components, only the

momentum term helps; the reversion term, which carries the Binance result, does not. Two features of the sample may explain part of the gap, and we test neither: the six included Bybit books carry \$0.30M–\$0.43M within 1%, at or above the boundary between Binance's thin and mid terciles, where the Binance effect was also small; and the inclusion rule removed the sparsest books. We read the two Bybit tests together: with trades alone the statistic has no content on Bybit, and with depth its content is detectable but too small to matter. Section 11.12 asks the same question in a regulated market.

11.12 A regulated market: thin US stocks

The Bybit tests leave open whether the result belongs to thin markets or to one crypto venue. We therefore registered a test in US equities (SHA-256 `f5800a69c219...`) before downloading any test day. Data are consolidated (SIP) trades from the Alpaca market-data API, aggregated into minute bars of the regular session: last trade price, dollar volume and dollar flow signed by the tick rule. The data contain no depth, so the candidate is the depth-free statistic $\hat{C}_3^{\text{flow}}$, the version that worked on Binance (Table 21). The universe was drawn by a seeded rule from June 2024 daily bars, before any test data: 40 thin names (price \$5–\$100, median daily dollar volume \$2M–\$10M) and 10 mid-size controls (\$50M–\$200M). Forecasts are made 120, 180, 240 and 300 minutes after the open from the preceding 120 minutes, and the outcome is the maximum adverse excursion over the next 60 minutes; the baseline is the full volatility-forecasting and price-impact baseline of Table 22. Training uses 48 days in 2023–2024 (the 10th and 24th of each month or the next trading day) and testing 42 days from January 2025 to September 2026 (the 6th and 20th). A day is valid if at least 80% of its minutes have a trade, and a name enters if it has at least 30 valid training days and 25 valid test days, which keeps 23 of the 40 thin names and all 10 controls. The test cell was run once.

The test passes (Table 24). $\hat{C}_3^{\text{flow}}$ adds 0.50 points of out-of-sample R^2 to the full baseline (95% interval $[0.27, 0.78]$, Clark–West $t = 5.05$), and the summed response S_{60}, the guard's

Table 24: Pre-registered test in thin US stocks, 2025–2026

Candidate	ΔR^2 (pts) [95% CI]	Clark–West t	Holm p
Thin, $\hat{C}_3^{\text{flow}}$ (primary)	+0.499 [+0.272, +0.779]	5.05	—
Thin, summed response S_{60}	+0.153 [+0.046, +0.272]	3.36	0.001
Thin, reversion term	+0.613 [+0.322, +0.967]	5.15	< 0.001
Thin, momentum term	−0.011 [−0.288, +0.261]	0.63	0.263
Mid-size control, $\hat{C}_3^{\text{flow}}$	+0.316 [+0.074, +0.634]	3.12	0.002

23 thin names, train 3,744 and test 3,808 symbol-hours on 42 test dates; baseline out-of-sample $R^2 = 0.340$; AUC 0.846 → 0.847 (95% interval of the change $[-0.00002, +0.0026]$); training coefficient on $\hat{C}_3^{\text{flow}}$ +21.7. Mid-size controls: 10 names, 1,680 test symbol-hours. Specification SHA-256 `f5800a69c219...`; test opened once. Kind: real data (Part 14).

own statistic, adds 0.15 points. As on Binance, the content comes from impact-weighted reversion: the reversion term alone adds 0.61 points, and the momentum term adds nothing. The effect is smaller than in thin Binance books, about a fifth of the 2.8 points of Table 22 on a baseline that already explains 34% of the variance, but about fifteen times the Bybit effect, and tail classification does not improve significantly. Unlike on Binance, the mid-size controls also show a gain (0.32 points), against our prediction of little or none: in US stocks the information is not confined to the thinnest names. The controls trade $50M–$200M a day, far below the deepest crypto books, and we included no large-capitalization stocks, so the test does not say where the effect ends. Three features of the sample limit the reading. The selection rule did not exclude funds, so the included thin names contain one closed-end fund (CRF) and the controls one bond ETF (BNDX); the validity rule removed the 17 thinnest-trading names; and the asset list contains only names still listed in 2026. Tick-rule signing also misclassifies some trades. Within these limits, the result is not a property of one crypto venue: the reduced-form content of the reversion statistic carries over, at a smaller size, to a regulated equity market.

11.13 The guard as a position limit

The guard of Section 6 bounds worst-case loss; a separate question is whether its ingredients help a dealer size positions in practice. We registered a backtest with Table 21. Each hour

Table 25: Position-sizing backtest, thin books, fresh days 2025–2026

Rule	Risk measure R_t	TLR	Breach rate	$\log \mathrm{TLR} - \log \mathrm{TLR}_{\mathrm{vol}}$ [CI]
Volatility limit	$\hat{\sigma}_{6h}$	348.6	1.9%	—
Amplification-adjusted (primary)	$\hat{\sigma}_{6h}\,(S_{60}/\lvert\hat{h}_0\rvert)^{0.25}$	342.7	2.1%	−0.017 [−0.045, +0.011]
Peak-response guard	$U_6 \max_k \lvert\hat{h}_k\rvert$	411.4	5.3%	+0.165 [+0.095, +0.243]
Summed-response guard	$U_6\, S_{60}$	522.2	13.2%	+0.404 [+0.285, +0.525]
Static	1	476.8	5.3%	+0.313 [+0.223, +0.397]

20 thin symbols; U_6 is the 95th percentile of absolute normalized flow over six hours. Primary row: 97.5% interval; others 95%. Breach: loss above the training 95th percentile of the static rule. Bybit: training chose $\theta = 0$, and the summed guard is worse than volatility by +0.484 [0.251, 0.662].

a rule sets the position $q_t = c_s/R_t$, where R_t is a risk measure from the past six hours and c_s sets the training mean of q to one for each symbol and rule; the hour's loss is q_t times the adverse excursion, and the score is tail loss per unit of average position, $\mathrm{TLR} = \mathrm{ES}_{95}(L)/\overline{q}$. The primary comparison is between a volatility limit (R = six-hour realized volatility) and the same limit scaled by the gain multiple to a power θ chosen on training data ($\theta = 0$ is the volatility limit); training data chose $\theta = 0.25$.

The primary backtest fails (Table 25). Scaling the volatility limit by the gain multiple lowers tail loss per unit of position in 91% of bootstrap draws, but the interval includes zero. Rules built only on the estimated response, without volatility, are clearly worse than the volatility limit and even than a static limit: a six-hour estimate of S is too noisy to size positions on its own. The guard theorems are statements about worst-case loss under a known or certified operator, and this evidence does not contradict them; but it means that, with public data and six-hour windows, an estimated summed response adds at most a little to volatility as a sizing input. A practical guard should combine the two and certify the operator with more data than a six-hour window; Section 11.14 tests exactly that.

11.14 A certified guard

The guard theorems assume a certified operator, and a six-hour estimate is a poor certificate. We therefore registered a second backtest (SHA-256 `b9a56af48c93...`) in which the operator of each symbol is certified once a day by pooling the minutes of the previous three

Table 26: Backtest with a certified operator, thin books, fresh days 2025–2026

Rule	R_t	TLR	Breach	$\Delta \log$ TLR vs vol [95% CI]
Volatility limit	$\hat{\sigma}_{6h}$	296.7	2.0%	—
Certified, amp.-adjusted (primary)	$\hat{\sigma}_{6h}\,\mathrm{GM}_{\mathrm{cert}}^{0.25}$	289.7	2.3%	−0.024 [−0.051, −0.0004]
Combined ceiling	stricter of the two	299.3	2.9%	+0.009 [−0.023, +0.039]
Certified summed guard	$U_{\mathrm{cert}} S_{\mathrm{cert}}$	416.5	13.8%	+0.339 [+0.241, +0.436]
Six-hour summed guard	$U_6 S_{60}$	424.8	14.9%	+0.359 [+0.237, +0.475]
Static	1	383.2	4.0%	+0.256 [+0.132, +0.387]

20 thin symbols; certified quantities from pooled within-day OLS on the minutes of the previous three days (median three valid days), fixed for the evaluation day. $\theta = 0.25$ chosen on training TLR over $\{0, 0.25, 0.5, 0.75, 1\}$. Combined ceiling: $R = \max(\hat{\sigma}_{6h}/\bar{\sigma},\, U_{\mathrm{cert}} S_{\mathrm{cert}}/\overline{US})$ with training medians. Breach: loss above the training 95th percentile of the static rule. Certified against six-hour summed guard: −0.020 [−0.165, +0.118]. Secondary comparisons Holm $p = 1.00$. Specification SHA-256 `b9a56af48c93...`; test opened once. Kind: real data (Part 12).

days, and applied to every hour of the following day. Evaluation days are the 28th of each month and certification windows the 25th to 27th; none of these days was used in any earlier test. Training is 2024 (12 evaluation days), testing January 2025 to September 2026 (20 evaluation days, 6,894 symbol-hours on the 20 thin symbols). The primary rule scales the six-hour volatility limit by the certified gain multiple, $R_t = \hat{\sigma}_{6h}\,\mathrm{GM}_{\mathrm{cert}}^{\theta}$, with θ chosen on training data; the score is the tail loss per unit of position of Section 11.13.

The primary test passes, narrowly (Table 26). With a certified operator, scaling the volatility limit by the gain multiple lowers tail loss per unit of position by 2.4%, with a 95% interval of $[-5.1\%, -0.04\%]$; training data again chose $\theta = 0.25$, as in Section 11.13. The gain is small and its interval almost touches zero, so we read it as a modest improvement, not a decisive one: amplification information helps as a correction to a volatility limit once the operator is estimated from days rather than hours. Every secondary comparison fails. A limit built on the certified summed response alone is far worse than volatility, about as bad as the six-hour version, so certification does not rescue the guard as a stand-alone sizing rule; and a combined ceiling that applies the stricter of the two limits does no better than volatility. The guard's theorems concern worst-case loss under a known operator; as a practical limit, its statistic earns its place as a modest adjustment to volatility, not as a replacement. The comparison also differs from the theorem in form: the backtests size positions as c/R_t, which

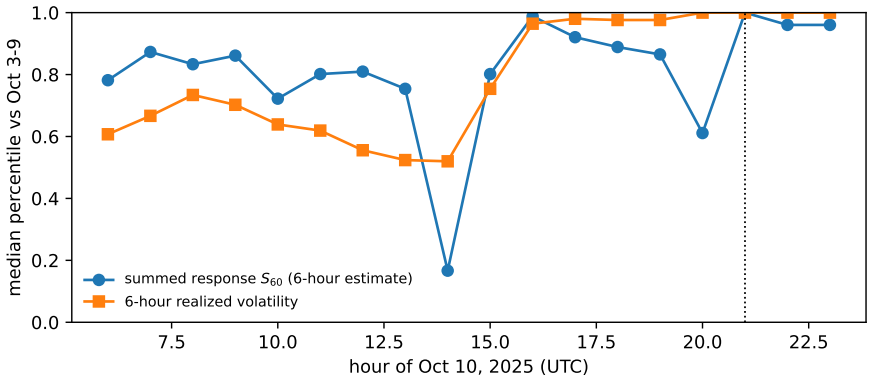


Figure 6: October 10, 2025: median percentile, across thin symbols, of the six-hour summed response S_{60} and of six-hour realized volatility, relative to each symbol's October 3–9 distribution. Dotted line: the crash hour (21:00 UTC). Descriptive; the event was chosen after the fact.

levers a position up when the estimated response is small, whereas the guard of Lemma 6.1 is a one-sided cap, $\min(1, B/(IUS))$, that only scales admission down. After the fact we applied the certified guard in that form, throttling a position whenever $U_{\text{cert}} S_{\text{cert}}$ exceeds its training median (27% of test hours; Part 13, `R1`). As a cap on a static limit it lowers tail loss per unit of position by 13% (95% interval $[-27\%, -3\%]$); as a cap on a volatility limit it adds nothing ($+0.8\%$, $[-1.3\%, +2.7\%]$). Used as the theorem uses it, the guard helps where no volatility information is used and is subsumed by volatility where it is.

11.15 October 10, 2025

The largest liquidation day on record motivated this paper, so we report what the statistics did on it, descriptively. The event was chosen after the fact and the 10th of each month was a test day in the first two thin-book tests, so this is not a test. For the 20 thin symbols we computed the hourly six-hour estimates on October 3–11, 2025, and expressed each hour of October 10 as a percentile of the same symbol's October 3–9 distribution. The crash hour, the hour on October 10 with the largest adverse excursion in BTCUSDT, began at 21:00 UTC; the median thin-symbol excursion in that hour was about 7,500 basis points.

Figure 6 shows two things. First, the summed response was elevated through the whole

day: in the morning, before volatility rose, the median thin symbol's S_{60} sat at the 72nd to 87th percentile of its previous week, against the 52nd to 73rd for volatility, and at the start of the crash hour 16 of 18 thin symbols with valid estimates were above their previous-week 90th percentile. Second, volatility reached the top of its range from 16:00 onward as well, so on this day the summed response did not warn of anything volatility missed in the final hours; and across symbols it did not rank which coins fell furthest (Spearman -0.11, against $+0.36$ for volatility). We draw no inference from one day. The episode is consistent with the thin-book result, an elevated dynamic risk state that sizes excursions rather than timing them, and it illustrates why the paper's claims are about the size of the next hour's excursion, not about predicting crashes.

11.16 Every pre-registered test

Table 27 lists every pre-registered test in this paper, with its outcome and the number of times its test set was opened. Of twenty-one pre-registered primaries, twelve pass, eight fail and one was not tested because the test it depended on failed. The failures are in BTCUSDT futures, in BTCUSDT spot, on Bybit with trades alone, on Coinbase (Part 17), in the six-hour position-sizing backtest, in the two five-minute open-interest liquidation-mechanism primaries (Part 22), and in the non-resetting 18-hour matched-breach capacity test (Part 23). Four passes are in thin Binance books, one in thin US stocks, two on untouched days (Part 15) and three on untouched symbols (Part 16); the certified-guard backtest passes narrowly; and the Bybit test with depth meets its criterion with a negligible effect. The analyses of Parts 19–21 (Section 12.6) were specified in advance but not hash-registered and are not counted here. Figure 7 adds the post-hoc checks.

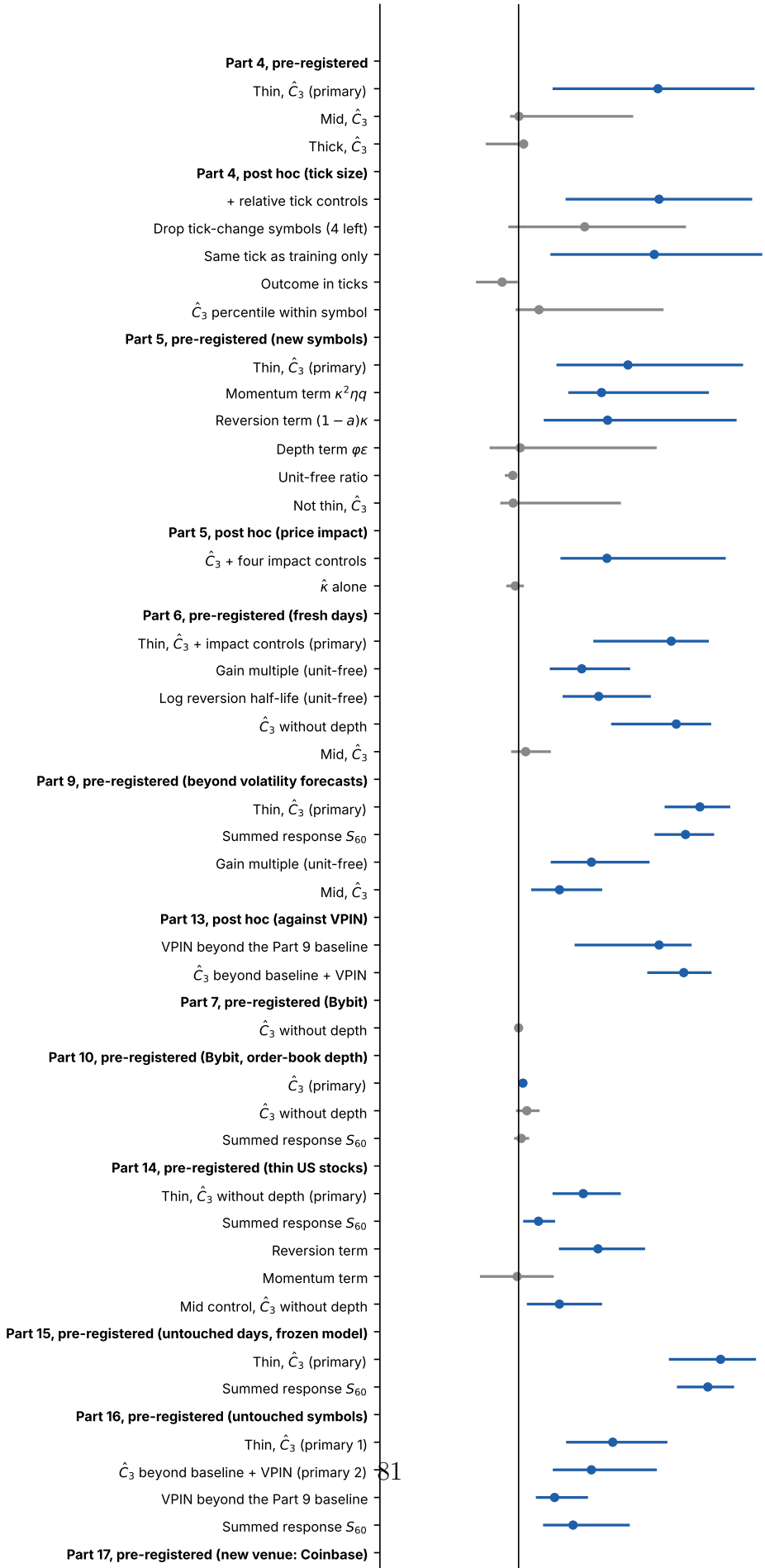
11.17 Threats to validity

Multiple testing. The program registered twenty-one primaries, of which twenty were tested, and ran the post-hoc checks of Figure 7. The nine passing forecasting primaries have

Table 27: All pre-registered tests

Test (specification hash)	Sample	Primary statistic		[CI]	Result	Opened
BTCUSDT futures, MAE | $\hat{C}_3$ (`d9521218`)	2025–26	ΔR^2	+0.018	[−0.100, +0.132]	fail	1
BTCUSDT spot, MAE | amplification (`b1fd7f8e`)	2024	ΔR^2	−0.023	[−0.056, −0.002]	fail	3
Thin books, 10 symbols (`5910b6b4`)	2025–26	ΔR^2	+1.216	[+0.273, +7.987]	pass	2
New thin symbols, 10 (`6aa8e2c0`)	2025–26	ΔR^2	+0.845	[+0.303, +6.371]	pass	1
Fresh days, 20 thin, impact controls (`65acda36`)	2025–26	ΔR^2	+1.583	[+0.588, +3.237]	pass	1
Position sizing, same specification	2025–26	$\Delta \log \text{TLR}$	−0.017	[−0.045, +0.011]	fail	1
Bybit, 14 coins, depth-free	2025–26	ΔR^2	−0.000	[−0.001, +0.001]	fail	1
Beyond volatility forecasting, 20 thin (`71983f1a`)	2025–26	ΔR^2	+2.787	[+1.425, +4.944]	pass	1
Certified guard, position sizing (`b9a56af4`)	2025–26	$\Delta \log \text{TLR}$	−0.024	[−0.051, −0.0004]	pass	1
Bybit with order-book depth, 6 coins (`ec2145fa`)	2025–26	ΔR^2	+0.033	[+0.019, +0.053]	pass	1
Thin US stocks, 23, depth-free (`f5800a69`)	2025–26	ΔR^2	+0.499	[+0.272, +0.779]	pass	1
Untouched days, frozen model (`69849c96`)	2025–26	ΔR^2	+4.196	[+1.551, +8.233]	pass	1
Untouched days, guard vs static (same)	2025–26	$\Delta \log \text{TLR}$	−0.084	[−0.173, −0.011]	pass	1
Untouched symbols, 5 thin (`0523e476`)	2025–26	ΔR^2	+0.727	[+0.377, +1.428]	pass	1
Untouched symbols, beyond VPIN (same)	2025–26	ΔR^2	+0.562	[+0.274, +1.159]	pass	1
Untouched symbols, guard vs static (same)	2025–26	$\Delta \log \text{TLR}$	−0.256	[−0.497, −0.021]	pass	1
New venue, Coinbase, 6 thin (`3233c52b`)	2025–26	ΔR^2	−0.092	[−0.407, +0.158]	fail	1
Coinbase, real minus placebo (same)	2025–26	+0.013		[−0.331, +0.298]	not tested	1
OI liquidation forecast, 19 thin (`2b3794e3`)	2025–26	ΔR^2_{LIQ}	−0.795	[−1.529, −0.179]	fail	1
OI liquidation mediation, 19 thin (same)	2025–26	Share	−6.8%	[−10.2%, −3.1%]	fail	1
Non-resetting 18h guard capacity (`d4169377`)	2025–26	$\Delta \overline{N}$ (k$)	+332.8	[−20.4, +1180.8]	fail	1

ΔR^2 in percentage points. "Opened" counts executions of the test cell; the repeated openings used unchanged code and specifications (Sections 11.5 and 10). Intervals are 95%, except 97.5% for the two fresh-day primaries and the two Part 15 primaries.


Part 4, pre-registered
Thin, $\hat{C}_3$ (primary)
Mid, $\hat{C}_3$
Thick, $\hat{C}_3$
Part 4, post hoc (tick size)
+ relative tick controls
Drop tick-change symbols (4 left)
Same tick as training only
Outcome in ticks
$\hat{C}_3$ percentile within symbol
Part 5, pre-registered (new symbols)
Thin, $\hat{C}_3$ (primary)
Momentum term $\kappa^2\eta q$
Reversion term $(1-a)\kappa$
Depth term $\varphi\varepsilon$
Unit-free ratio
Not thin, $\hat{C}_3$
Part 5, post hoc (price impact)
$\hat{C}_3$ + four impact controls
$\hat{\kappa}$ alone
Part 6, pre-registered (fresh days)
Thin, $\hat{C}_3$ + impact controls (primary)
Gain multiple (unit-free)
Log reversion half-life (unit-free)
$\hat{C}_3$ without depth
Mid, $\hat{C}_3$
Part 9, pre-registered (beyond volatility forecasts)
Thin, $\hat{C}_3$ (primary)
Summed response S_{60}
Gain multiple (unit-free)
Mid, $\hat{C}_3$
Part 13, post hoc (against VPIN)
VPIN beyond the Part 9 baseline
$\hat{C}_3$ beyond baseline + VPIN
Part 7, pre-registered (Bybit)
$\hat{C}_3$ without depth
Part 10, pre-registered (Bybit, order-book depth)
$\hat{C}_3$ (primary)
$\hat{C}_3$ without depth
Summed response S_{60}
Part 14, pre-registered (thin US stocks)
Thin, $\hat{C}_3$ without depth (primary)
Summed response S_{60}
Reversion term
Momentum term
Mid control, $\hat{C}_3$ without depth
Part 15, pre-registered (untouched days, frozen model)
Thin, $\hat{C}_3$ (primary)
Summed response S_{60}
Part 16, pre-registered (untouched symbols)
Thin, $\hat{C}_3$ (primary 1)
$\hat{C}_3$ beyond baseline + VPIN (primary 2)
VPIN beyond the Part 9 baseline
Summed response S_{60}
Part 17, pre-registered (new venue: Coinbase)

Clark–West statistics between 5.0 and 12.6 (one-sided $p < 0.001$ in each), and the Part 15 guard primary has a one-sided bootstrap p of 0.0025, so each survives a Bonferroni correction across all twenty tested primaries (threshold 0.0025). The Part 16 guard primary ($p = 0.004$) does not, although it passes the Holm correction its registration specified; statistical significance is not the issue for the Bybit test, its size is. The certified-guard backtest does not survive the correction: its bootstrap one-sided p is about 0.02, so it is the weakest of the passes. The eight failures are reported with the same prominence.

Sequential design. The program was sequential. Each pre-registration fixed one test's specification before its test data were opened, but which test to run next was chosen after the earlier results were known: new symbols after the first pass, impact controls and then a volatility baseline in response to the obvious objections, depth on Bybit after the trade-only test failed, a regulated market after the second venue gave a negligible effect, untouched days and symbols after the information-safety theory, and a certified operator after the six-hour backtest failed, and a new venue last. Pre-registration therefore protects each test from specification search on its own test data; it does not protect the program from choices that adapted across tests. That is why every test is reported, failures included (Figure 7), and why the correction above is taken over all tested primaries.

Dependence among the passes. The four passing tests share the 2023–2024 training days, and the original ten thin symbols appear in three of them. They are not four independent replications. They vary three things one at a time: the symbols (Table 19, ten new symbols), the test days (Tables 21 and 22, days never used before) and the baseline (impact controls, then a full volatility-forecasting model). The Part 16 test (Table 30) shares their training days but none of their symbols. The US test (Table 24) shares none of their data: a different market, asset class, data source, universe and set of days.

Look-ahead and estimated regressors. Every feature uses minutes before the forecast hour; winsorization limits, flow scales, θ and tail thresholds come from training data; and a scrambling audit leaves every hourly feature unchanged when data from t onward are permuted. $\hat{C}_3$ and S_{60} are estimated from six hours, but the evaluation compares out-of-sample forecasts rather than testing a coefficient, so estimation error in the regressor can only attenuate the measured gain.

Simultaneity. The structural fits regress the price change on same-minute flow, so $\hat{\kappa}$ and the depletion slope mix impact with any within-minute response of flow to price. The guard's identification results (Proposition 2.2 and Theorem 5.5) concern the impulse response, not these regression coefficients, and the forecasting evidence uses $\hat{C}_3$ as an index whose value is judged out of sample, not as a causal parameter. Simultaneity therefore changes what $\hat{C}_3$ measures, not whether the out-of-sample gains are real. We tried to separate the two after the fact by instrumenting this minute's flow with last minute's flow (Part 13, `R4`), but at the one-minute scale normalized flow is almost serially uncorrelated (median first-stage correlation 0.005), so the instrument is weak and the instrumented statistic is noise (it adds 0.01 points of R^2, interval including zero). The causal content of $\hat{\kappa}$ therefore remains open; a convincing instrument would need order-level data.

Selection and survivorship. The 30 symbols were fixed from one 2024 training day before any T6 statistic was computed, and the new symbols were drawn by a seeded procedure. Symbols had to be listed through 2026, so coins delisted after a collapse are excluded; if anything this removes the episodes in which thin-book risk was largest.

Economic size. The largest gain, 2.8 points of R^2 on a baseline of 18.6%, is a 15% increase in the explained variance of the next hour's adverse excursion, and tail classification improves by 0.005 of AUC. In basis points, on the test hours of Table 22 and with quintile breakpoints from training data: within each quintile of the volatility forecast, hours in the top quintile

of $\hat{C}_3$ have a next-hour adverse excursion 25.6 basis points larger than hours in the bottom quintile (date-block 95% interval $[18.6, 32.8]$), against a mean excursion of 112.5 basis points, and 61 basis points larger in the most volatile quintile; the share of hours beyond the training 95th percentile rises from 2.3% in the bottom $\hat{C}_3$ quintile to 5.1% in the top. Unconditional sorts on S_{60} show no pattern; its content appears only with the controls. As a sizing input the gain is worth about a 2.4% reduction in tail loss per unit of position once the operator is certified (Table 26). These are modest gains on top of a strong baseline, not a new predictor that replaces volatility.

In margin terms. To express the 25.6 basis points in the units a risk manager uses, we fit a lognormal distribution to the excursions in each $\hat{C}_3$ quintile from its mean and median; at one common threshold the fit reproduces the observed share of hours beyond the training 95th percentile in every quintile to within 0.6 percentage points. Shifting the bottom-quintile distribution up by 25.6 basis points, the effect at a given volatility forecast, roughly doubles the probability that a 20× position (liquidated after a 4% adverse move, with 1% maintenance margin) is liquidated within the hour, from 1.3% to 2.7% (2.3% to 3.2% across the interval for the effect). For a 10× position, liquidated after 9%, the probability almost triples, from 0.04% to 0.11%. On $1M of notional at 20×, the extra expected loss of margin is about $710 per hour held ($490 to $960; `recompute_dollar.py`). These figures come from a fitted distribution, not a test, and they treat the hour's excursion from its opening price as the liquidation trigger. They show that a difference that is small in basis points is large in the tail that leverage makes relevant, which is where a pre-trade control operates.

Venue. The size of the result depends on the venue. It is largest in thin Binance USD-M books; in thin US stocks it passes a pre-registered test at about a fifth of that size, and it also appears in mid-size US stocks (Table 24); on Bybit it failed with trade data alone and, with order-book depth, met its pre-registered criterion with an effect about one eightieth of

Binance's (Table 23).

11.18 What the evidence says

Deep books sit on the stable side of the amplification condition. Flow depletes the side it hits on a fixed price grid, although re-centred depth bands make it look like replenishment, and we observe no peak overshoot. In thin Binance books the condition is still negative on average but binds more often, and the six-hour estimates of cumulative transient risk forecast the size of the next hour's adverse excursion out of sample. Four pre-registered tests pass, on the original symbols, on ten new symbols, on fresh days with price-impact controls, and on fresh days beyond a full volatility-forecasting baseline, where the gain is largest (+2.8 points of R^2) and extends to tail classification. The summed response S of Theorem 4.4, the guard's own statistic, carries the same information, and unit-free versions work. The content does not run through depth depletion, and in Binance books it is much weaker at mid depth and absent in BTCUSDT. In thin US stocks a fifth pre-registered forecasting test passes, at about a fifth of the Binance size and with a gain in mid-size stocks too; on Bybit the statistic is absent with trades alone and negligible with depth; and as a position-sizing input it improves a volatility limit only modestly once the operator is certified. Thin books are risky because their prices revert too slowly relative to the impact and momentum of the flow that moves them, which is what $S = \tilde{\kappa}/(m - q\tilde{\kappa})$ measures.

12 Tests of the information-safety results

The theorems of Sections 5–7 concern declared contracts. This section reports two kinds of tests: simulations in which the true economy is known, so that coverage, attainment and optimality can be checked directly; pre-registered tests on market data that no earlier part of this paper used, on new days, new symbols and a new venue; and post-hoc checks on data already opened.

Table 28: Simulation tests of the information-safety results

Test	Question	Result
T01	Can markets agree on early responses yet differ in risk?	Example 5.1 in rational arithmetic: lags 0–2 equal, summed responses 1.92 and 5.59
T02	Does Proposition 5.2 bound the capacity a safe rule gives up?	Attained exactly with identical data (0.324 at $\beta = 0.05$); respected in 9 of 9 noisy designs by the certificate rule, whose unsafe-admission rate is at most 3.2%
T03	Is certifying risk cheaper than identifying cohorts?	Table 3: slope -6.06 in $\Delta\eta$ for identification, $+0.23$ for certification
T04	Does the certificate cover at its nominal level?	18 designs: coverage at or above nominal in all (minimum 93.6% at nominal 90%); median excess 3–15% with memory bound 0.6–0.9, about 230% at 0.97
T05	Is the corrected gate exact?	2,000 cases, horizons 1–10, every sign path enumerated: per-period worst case equals $V \sum \lvert g \rvert$ and total-budget worst case equals $V \max \lvert g \rvert$ in all; the peak gain understates the per-period worst case in 1,800
T06	Is the closed-form guard globally optimal?	500 economies: equal to brute force over 1,001 levels from the state matrices in all; safe sets always intervals; 28 infeasible states flagged
T07	What does disclosure add?	Table 5
T08	Does the guard depend on the channel?	Momentum learners, fixed-rule traders, depth and slow dealer reversion calibrated to $S = 5$: identical guard capacity (0.20); one-period margins differ (-0.34 to -0.20)
T09	What breaks the certificate?	Coverage at nominal 95%: as assumed 100%; delayed feedback 100%; contrarian cohorts 100%; depth channel 92.8%; memory longer than declared 86.3%; feedback 20% stronger after certification 34.7%
T10	Are the window endpoints proved?	Certified by Lemma 5.4 for every window; outside the 60-digit endpoints by at most 1.5×10^{-8} relative
T11	What if the tap noise is correlated?	(O1) certificate covers 74–76% (exact value 75.1%); t-certificate 94–95%; Hotelling 100%; dependent experiments 82–85%
T12	Should rebates use certified bounds?	No: a 10% overstatement of S costs 2.4–2.8% of first-best welfare, a 50% overstatement 70–80%
T13	What if admission changes the dynamics?	Safe sets $[0, 0.035] \cup [0.386, 1]$ (closed form) and $[0, 0.053] \cup [0.178, 1]$ (book, certified); no certified level unsafe in 433 sampled economies
T14	Does sequential probing keep its error level?	No wrong decision in 7,000 runs; 3.7 to 107 probes against 130 for a fixed design

All tests are deterministic given their seeds and run without data (`research_extension/information_safety/run_all.py` → `results/T*.json`, about six minutes). Further checks: the joint per-period and total-budget worst case equals its linear-programming value in 300 random cases (T05b), and the balance-sheet step keeps the reserve in all 17,551 sampled states that satisfy the guard inequality (T12b). Kind: recomputed.

12.1 Simulation tests with a known truth

The simulations confirm what the theorems claim and locate what they do not (Table 28). The certificate is valid where its contract holds, including with delayed feedback and con-

trarian cohorts, and conservative when the declared memory is long. It fails in two ways that matter for practice: when memory is longer than declared, which is the case Theorem 5.3(iii) warns about, and when feedback strengthens after certification, in which case a stale certificate covers the new risk only a third of the time. A guard must therefore be re-certified as conditions change; it cannot be set once. The remaining tests check the extensions: the window endpoints are proved rather than computed (T10), certificates that do not assume independent taps keep their level when the taps are correlated while the (O1) certificate does not (T11), non-monotone admission is certified by branch and bound (T13), and sequential probing keeps its error level (T14).

12.2 Pre-registered test on untouched days (Part 15)

We registered a test (Part 15) before downloading any day used in it: the 12th to 15th of each month on Binance, days no earlier part used on Binance. It has two primaries, with Holm's correction across them. First, the forecasting model of Table 22, frozen at its 2023–2024 training estimates, is applied to the 15th of each month from January 2025 to September 2026. Second, an uncertainty-aware guard throttles a position by $\min(1, k_s/(U_{\mathrm{cert}} S_{60}^{\mathrm{up}}))$, where S_{60}^{up} is the 90th percentile of the summed response over 100 block-bootstrap refits of the operator certified from the three preceding days; it is compared with a static limit at matched activity, and secondarily with the point-estimate guard, a depth throttle, a spectral-radius throttle and a volatility cap. The bootstrap bound is a diagnostic for observational data, not the finite-sample certificate of Theorem 5.5.

Both primaries pass (Table 29). The forecasting model, frozen at its 2023–2024 estimates and applied to days never used before, finds that $\hat{C}_3$ adds 4.2 points of out-of-sample R^2 to the volatility and impact baseline (97.5% interval $[1.6, 8.2]$, Clark–West $t = 9.9$) and improves tail classification; the summed response S_{60} adds 3.3 points. The gain is larger than in Table 22 partly because the frozen baseline explains less on these days (5.8% of the variance against 18.6%): frozen coefficients transfer imperfectly, and the dynamic statistic

Table 29: Pre-registered test on untouched days (Part 15), 2025–2026

A. Frozen forecasting model (primary 1)	ΔR^2 (pts) [CI]	Clark–West t	ΔAUC [95% CI]
$\hat{C}_3$ (primary, 97.5%)	+4.20 [+1.55, +8.23]	9.94	+0.007 [+0.001, +0.011]
S_{60} (secondary, 95%)	+3.26 [+1.82, +5.34]	11.55	
B. Guards at matched activity			
Rule	TLR	Activity	Robust guard vs. rule [95% CI]
Static limit	377	1.00	−0.084 [−0.161, −0.017] (primary 2)
Uncertainty-aware guard	347	0.91	—
Point-estimate guard	346	0.92	+0.003 [−0.005, +0.013]
Depth throttle	469	0.59	−0.302 [−0.446, −0.146]
Spectral-radius throttle	401	0.82	−0.146 [−0.240, −0.065]
Volatility cap	339	0.86	+0.023 [−0.022, +0.082]
Volatility cap × guard	335	0.80	vs. cap: −0.012 [−0.037, +0.013]

20 thin symbols; evaluation days the 15th of each month; frozen model trained on 16,860 Part 9 symbol-hours, tested on 6,714 hours on 20 dates; frozen baseline out-of-sample $R^2 = 0.058$; training coefficient +71.3. Guards: 6,732 test hours; TLR = ES95 of loss per unit of average position (positions normalized to training mean 1); the last column of panel B reports log TLR(robust guard) minus log TLR(rule), so negative values favour the robust guard. Primary 2 uses the 97.5% interval $[-0.173, -0.011]$, one-sided $p = 0.0025$. Both primaries pass under Holm's correction. Median $S_{60}^{\text{up}}/S_{60} = 1.25$. Specification SHA-256 `69849c96d70f...`; test opened once. Kind: real data (Part 15).

picks up part of what the baseline loses. As a one-sided throttle on a static limit, the uncertainty-aware guard lowers tail loss per unit of position by 8% (log difference −0.084, 97.5% interval $[-0.173, -0.011]$, that is, a reduction of 1% to 16%) while retaining 91% of activity, replicating on untouched days the post-hoc finding of Section 11.14. It does much better than throttling on depth (26% lower tail loss) or on the spectral radius (14% lower), the two controls a supervisor would reach for first.

Two secondary results limit the claim. The bootstrap upper bound does no better than the point estimate (+0.3%, interval $[-0.5\%, +1.3\%]$): on these data the upper bound is about 1.25 times the point estimate in most hours and ranks hours almost identically, so accounting for estimation uncertainty changes which hours are throttled very little. And, as in Section 11.14, a volatility cap does at least as well as the guard, and adding the guard to it does not help significantly (−1.2%, interval $[-3.7\%, +1.3\%]$). Used as the theorem uses it, the guard is a better throttle than depth or spectral rules and a real improvement over a static limit; it is not a substitute for volatility information.

Table 30: Pre-registered test on untouched symbols (Part 16), 2025–2026

A. Forecasting beyond the volatility baseline (5 new thin symbols)	ΔR^2 (pts) [95% CI]	Clark–West t	ΔAUC [95% CI]
$\hat{C}_3$ (primary 1)	+0.73 [+0.38, +1.43]	5.70	+0.004 [+0.001, +0.007]
$\hat{C}_3$, VPIN in baseline (primary 2)	+0.56 [+0.27, +1.16]	5.38	+0.003 [+0.001, +0.006]
VPIN (secondary)	+0.28 [+0.14, +0.53]	4.39	+0.002 [+0.001, +0.003]
S_{60} (secondary)	+0.42 [+0.20, +0.85]	4.43	+0.001 [−0.000, +0.003]
B. Guards at matched activity			
Rule	TLR	Activity	Robust guard vs. rule [95% CI]
Static limit	466	1.00	−0.256 [−0.497, −0.021] (prim. 3)
Uncertainty-aware guard	360	0.91	—
Point-estimate guard	371	0.91	−0.028 [−0.076, +0.009]
Volatility cap	367	0.85	not tested
Volatility cap × guard	342	0.78	vs. cap: −0.072 [−0.180, +0.005]

Thirty perpetuals drawn by a seeded rule (seed 20261031) from the 48 listed Binance perpetuals not used in Parts 4–15; five had training depth within 1% below $0.39M and entered: BAT, BEL, COTI, CTSI and LQTY. Forecasting: Part 9 specification re-estimated on these symbols' training days (10th and 24th of each month, 2023–2024; 4,215 symbol-hours) and tested on the 6th and 20th of each month, January 2025 to September 2026 (3,597 hours on 40 dates); baseline out-of-sample $R^2 = 0.317$. VPIN: six-hour window, 50 equal-volume buckets, winsorized at training 1%/99% (median 0.24). Guards: Part 15 definitions; 1,080 training and 1,782 test symbol-hours on the 15th of each month; last column of panel B is log TLR(robust guard) minus log TLR(rule). Holm-adjusted p across the three primaries: 2×10^{-8}, 7×10^{-8} and 0.004; all three pass. Secondary Holm-adjusted p: VPIN and $S_{60} < 0.0001$; robust vs. point guard 0.11; cap × guard vs. cap 0.07. Specification SHA-256 `0523e47626da...`; test opened once. Kind: real data (Part 16).

12.3 Pre-registered test on untouched symbols (Part 16)

A third registered test (Part 16) uses Binance perpetuals that no earlier part used, thirty drawn by a seeded rule before any analysis and kept if their training depth is below the thin threshold of Part 5. It has three primaries, with Holm's correction across them: the forecasting specification of Table 22 on the new thin symbols; the same beyond the baseline *plus VPIN*, which turns the post-hoc comparison of Section 11.10 into a confirmatory one; and the guard of Section 12.2 against a static limit.

All three primaries pass (Table 30), but on a small set. The seeded draw of thirty new perpetuals produced only five with thin books, because the thinnest listed symbols had already been used in Parts 4–15; the rule required no minimum, so the test went ahead with five. On these symbols, re-estimating the Part 9 specification on their own training days, $\hat{C}_3$ adds 0.73 points of out-of-sample R^2 to the volatility and impact baseline (95% interval

$[0.38, 1.43]$, Clark–West $t = 5.7$) and improves tail classification. That is about a quarter of the Part 9 gain on the original thin symbols, on a baseline that already explains 31.7% of the variance rather than 18.6%.

The second primary makes the post-hoc VPIN comparison of Section 11.10 confirmatory: with VPIN added to the baseline, $\hat{C}_3$ still adds 0.56 points ($[0.27, 1.16]$). VPIN has content of its own (+0.28 points) but leaves most of $\hat{C}_3$'s gain in place, as in the post-hoc comparison. The summed response S_{60} adds 0.42 points.

The third primary replicates the guard of Section 12.2 on new symbols. Against a static limit, the uncertainty-aware guard lowers tail loss per unit of position by 23% (log difference -0.256, interval $[-0.497, -0.021]$, a reduction of 2% to 39%) while retaining 91% of activity. The interval is wide: the guard sample is five symbols on about twenty evaluation days, and its one-sided bootstrap p of 0.004 (8 of 2,000 draws) passes Holm's correction within Part 16 but not a Bonferroni correction across all tested primaries of the program (Section 11.17). The secondary comparisons agree with Part 15. The bootstrap upper bound again does not significantly improve on the point estimate (-2.8% in log points, interval including zero). Adding the guard to a volatility cap lowers tail loss by a further 7%, a larger effect than in Part 15 but not significant after correction (Holm $p = 0.07$). We do not report the guard against the volatility cap alone, which the registration did not include; the point estimates are close (360 against 367).

12.4 Pre-registered test on a new venue (Part 17)

Every venue so far was Binance, Bybit or the US consolidated tape. Part 17 registers a test on Coinbase Exchange spot, a US-regulated venue that no earlier part used, from public trade data: thin USD pairs drawn by a seeded rule from those with median daily dollar volume of \$0.5–5 million in June 2024, the depth-free statistic $\hat{C}_3^{\text{flow}}$ of Parts 7 and 14, and the Part 9 volatility and impact baseline, on the 6th and 20th of each month from January 2025 to September 2026. Its second primary is a falsification test. Order splitting makes signed

flow persistent, and a statistic built from persistent flow could forecast through persistence alone. The placebo recomputes $\hat{C}_3^{\text{flow}}$ after shifting the flow series by 180 minutes inside its own six-hour window, which keeps the flow's distribution and persistence and breaks its minute-by-minute alignment with prices, the alignment the model says carries the risk. The real statistic must beat the placebo. The two primaries are tested in a fixed sequence, the placebo comparison only if the first passes, because beating a placebo is informative only when the real statistic has content.

The test fails (Table 31). Of the twenty pairs drawn, six traded in at least 60% of minutes on enough days to enter (ARB, FLR, ICP, QNT, VET and XLM); thin pairs on this venue trade too sparsely for the rest. On these six, $\hat{C}_3^{\text{flow}}$ adds -0.09 points of R^2 to the baseline (95% interval $[-0.41, 0.16]$, Clark–West $t = -0.58$), so the first primary fails and, under the registered fixed sequence, the placebo comparison is not tested; descriptively the real statistic and the placebo do equally badly (difference $+0.01$ points, $[-0.33, 0.30]$). Nothing else forecasts either: S_{60}, VPIN and the statistic in the mid-size control pairs all add nothing. Coinbase joins Bybit with trades alone as a venue where the thin-book result is absent. Both are venues where the statistic must be computed from trades without depth, but so is the US test, which passed; the result is venue-specific, and we do not know which feature of a venue it depends on.

12.5 Post-hoc checks on opened data (Part 18)

Two further checks use samples that earlier tests opened, so they are post hoc; their analysis plan was hashed before they were computed. The first applies the same placebo to the Binance samples of Parts 9 and 16. The second adds three controls to the guard comparisons of Parts 15 and 16: a certified peak-response guard (the quantity Lemma 6.1 says is unsafe for repeated trading), a participation cap that limits the position to a share of the last six hours' volume, and a margin-style rule that scales the position by an exchange-style scanning range, the 99th percentile of one-hour moves over the three certification days. A

Table 31: Pre-registered test on a new venue (Part 17): Coinbase spot, 2025–2026

	ΔR^2 (pts) [95% CI]	Clark–West t	ΔAUC [95% CI]
$\hat{C}_3^{\text{flow}}$, thin (primary 1)	−0.09 [−0.41, +0.16]	−0.58	+0.000 [−0.000, +0.001]
Real minus placebo (primary 2, not tested)	+0.01 [−0.33, +0.30]	—	—
Placebo alone	−0.11 [−0.19, −0.03]	−2.62	
S_{60}	−0.03 [−0.08, +0.02]	−1.66	
VPIN	−0.00 [−0.01, +0.00]	−1.81	
$\hat{C}_3^{\text{flow}}$ beyond baseline + VPIN	−0.10 [−0.42, +0.15]	—	
$\hat{C}_3^{\text{flow}}$, mid-size control (3 pairs)	+0.02 [−0.20, +0.23]	0.69	

Twenty thin USD pairs (median June 2024 daily dollar volume \$0.5–5 million) and four mid-size pairs (\$15–50 million) drawn with a fixed seed before any trade was downloaded; a pair entered with at least 15 valid training and 25 valid test days, a day being valid if at least 60% of its minutes had a trade. Six thin and three mid-size pairs entered. Training: 10th and 24th of each month of 2024 (2,340 thin-pair hours); test: 6th and 20th, January 2025 to September 2026 (3,690 hours on 42 dates); baseline out-of-sample $R^2 = 0.310$. Placebo: flow shifted by 180 minutes inside its six-hour window. Fixed-sequence testing: primary 2 is tested only if primary 1 passes. Secondary Holm-adjusted p-values all equal 1. Specification SHA-256 `3233c52b0a39...`; test opened once. Kind: real data (Part 17).

cap on inventory alone is the static limit once activity is matched.

Both checks support the paper's reading, with one exception (Table 32). The placebo carries no information on Binance: on the Part 9 sample it adds −0.02 points of R^2 while the real statistic adds 2.73 on the same rows, a difference of 2.74 points (95% interval $[1.32, 4.82]$), and on the Part 16 sample the difference is 0.79 points ($[0.42, 1.60]$). Real and placebo statistics are uncorrelated (Spearman −0.004 and 0.025). The forecasting content in thin Binance books therefore comes from the minute-by-minute alignment of flow and prices, not from the persistence of order flow. Against the new controls, on the Part 15 sample the uncertainty-aware guard has 3% lower tail loss per unit of position than the certified peak-response guard and the margin-style rule (both significant after Holm's correction) and far lower than the participation cap; it is not better than the six-hour peak guard (+3.6%, interval including zero). On the five symbols of Part 16 it beats only the participation cap. These are post-hoc comparisons on opened data, so they qualify the guard's ranking rather than test it.

Table 32: Post-hoc checks on opened data (Part 18)

A. Order-splitting placebo, ΔR^2 beyond the Part 9 baseline (pts) [95% CI]	Part 9 sample (20 thin)	Part 16 sample (5 thin)
Real $\hat{C}_3$	+2.73 [+1.32, +4.78]	+0.73 [+0.37, +1.48]
Placebo	−0.02 [−0.05, +0.00]	−0.07 [−0.15, −0.03]
Real minus placebo	+2.74 [+1.32, +4.82]	+0.79 [+0.42, +1.60]
B. log TLR*(uncertainty-aware guard) minus* log TLR*(rule) [95% CI]*	Part 15 sample (20 thin)	Part 16 sample (5 thin)
Certified peak-response guard	−0.034 [−0.062, −0.007]	−0.004 [−0.031, +0.022]
Six-hour peak-response guard	+0.036 [−0.015, +0.100]	−0.028 [−0.116, +0.060]
Participation cap	−0.695 [−0.865, −0.534]	−0.880 [−1.233, −0.530]
Margin-style rule	−0.033 [−0.060, −0.008]	−0.018 [−0.057, +0.034]
Static limit (reproduces Parts 15, 16)	−0.084 [−0.166, −0.022]	−0.256 [−0.491, −0.021]

Post hoc: every sample was opened by an earlier test; the analysis plan was hashed before computing (`P18_posthoc_plan.json`). Panel A: the placebo recomputes $\hat{C}_3$ with the flow series shifted by 180 minutes inside its six-hour window; real and placebo are evaluated on the same rows and bootstrap draws, so the Part 9 real gain (2.73) differs slightly from Table 22 (2.79). Recomputed $\hat{C}_3$ and certified operators match the stored values in every row. Panel B: positions normalized to training mean 1 per symbol and rule; participation cap: position proportional to the previous six hours' dollar volume; margin-style rule: position inversely proportional to the 99th percentile of one-hour moves over the three certification days. Holm-adjusted p (Part 15): certified peak 0.006, margin 0.005, participation < 0.001, six-hour peak 0.91. The static-limit row reproduces the published TLRs (377 and 347; 466 and 360). Kind: real data (post hoc, Part 18).

12.6 Three analyses specified in advance, not deposited (Parts 19–21)

The three analyses below (Parts 19–21) were specified in an earlier draft of this paper, with their hypotheses and criteria, before their data were downloaded, but no specification hash was deposited prior to execution; we therefore label them "specified in advance, not deposited" and do not count them among the pre-registered tests of Table 27. By contrast, Parts 22 and 23 (Sections 12.10–12.11) were hash-registered and committed to the public repository before their 22nd-of-month test data were downloaded, and are counted in Table 27.

Table 33: Leverage interaction and matched spot comparison (Part 19, specified in advance, not deposited), 2025–2026

A. Out-of-sample ΔR^2 of the amplification statistic (pts)	ΔR^2 [95% CI]	CW t (p)	Ratio / diff. [95% CI]
Perpetuals, 19 thin symbols	+6.495 [+2.345, +15.110]	11.33 ($< 10^{-6}$)	—
top leverage tercile (LEV $\geq$ 16.77)	+8.172	—	high − low: +4.495
lower two terciles	+3.678	—	[−1.094, +11.663]
Spot, same hours (exploratory)	+0.987 [−0.093, +2.853]	4.96 (3.6×10^{-7})	0.152 [−0.015, +0.308]

B. Coefficients in perpetuals (bp of excursion per s.d. of $\hat{C}_3$)	Estimate	95% CI	Outcome
$\hat{C}_3$ base slope	+5.49	[+0.65, +9.66]	—
$\hat{C}_3\times$ top leverage tercile (Primary 1)	−0.18	[−3.02, +3.73]	fail
$\hat{C}_3\times$ top |basis| tercile	+0.94	[−2.91, +4.60]	—

C. One-minute return lead–lag, mean correlation across symbol-days	
perpetual leads spot: +0.0816	spot leads perpetual: −0.0010

Panel A: models fitted on 2024 training days (4,104 hours), evaluated on 2025–2026 test days (6,156 hours); 2,000-draw date-block bootstrap 95% intervals for all differences and ratios. Panel B: pooled regression with symbol fixed effects and date-block bootstrap intervals. Specified in advance, not deposited (`recompute_p2_parts19_20_21.py` → `R_P2_parts19_20_21.json`).

12.7 Leverage interaction and spot comparison (Part 19)

Primary 1 of Part 19—which predicted a positive slope interaction between $\hat{C}_3$ and high open-interest-to-depth leverage (LEV = OI/$D_{1\%}$ $\geq$ 16.77, the top training tercile)—failed: the interaction coefficient is −0.18 basis points per standard deviation (95% date-block bootstrap CI $[-3.02, +3.73]$, Table 33). We evaluated two predictions on the 8th of each month, a day no earlier test used for these symbols, across 19 thin perpetuals from Parts 9 and 16 that also trade on Binance spot (training: 12 days in 2024, 4,104 symbol-hours; test: 18 days from January 2025 to June 2026, 6,156 symbol-hours). *Primary 1* tested whether $\hat{C}_3 \times \mathbf{1}\{\text{LEV} \geq 16.77\}$ has a positive slope in perpetuals. *Primary 2* (exploratory, because the spot specification lacks order-book depth controls that Binance does not publish for spot) asked whether on the matched spot pairs, where there are no leveraged liquidations, the out-of-sample ΔR^2 of $\hat{C}_3^{\text{flow}}$ is at most one quarter of the perpetual gain.

Table 33 shows why Primary 1 fails and why Primary 2 is only exploratory. In Panel A, the out-of-sample ΔR^2 of $\hat{C}_3$ is +8.172 points in the top leverage tercile versus +3.678 points in the lower two terciles, a difference of +4.495 points whose 2,000-draw date-block bootstrap

95% confidence interval ($[-1.094, +11.663]$) crosses zero. *Post hoc*, the mechanical reason why the slope interaction $\hat{C}_3 \times$ high-LEV is zero (-0.18 bp) while the explained variance share is higher in high-leverage hours is that $\hat{C}_3 = \hat{\varphi}\hat{\varepsilon} + \hat{\kappa}^2\hat{\eta}\hat{q} - (1-\hat{a})\hat{\kappa}$ already contains $\hat{\kappa}$, which rises when resting depth $D_{1\%}$ is thin relative to open interest; high-LEV hours therefore have larger variance in $\hat{C}_3$ rather than a steeper marginal slope per unit of $\hat{C}_3$. In the matched spot pairs, $\hat{C}_3^{\text{flow}}$ adds $+0.987$ points of R^2 (95% CI $[-0.093, +2.853]$, significant by the nested-model test of Clark and West, 2007 at $t = 4.96$, $p = 3.6 \times 10^{-7}$, though its bootstrap level interval includes zero), giving a spot-to-perpetual point ratio of 0.152 with a 95% bootstrap interval of $[-0.015, +0.308]$ that crosses the 0.25 threshold. Because the spot baseline omits depth controls and the ratio interval spans 0.25, Part 19 does not establish leverage or liquidations as the mechanism; Section 12.10 tests the liquidation channel directly with a pre-registered design.

12.8 A natural experiment in fees (Part 20)

The primary hypothesis of Part 20—that removing zero trading fees lowers dealer reversion speed $\hat{m} = 1 - \hat{a}$ and raises the unit-free gain multiple $\widehat{S/\tilde{\kappa}} = 1/(\hat{m} - \hat{q}\hat{\kappa})$ as in Proposition 8.1(i)–(iv)—is *not supported* ($\hat{m}$ changes by -0.00120, $p = 0.590$, and $\widehat{S/\tilde{\kappa}}$ by -26.88, $p = 0.362$ across the two verified spot zero-fee events; Table 34). What moves in the two verified BTCUSDT spot zero-fee events—introduced on July 8, 2022 (Binance announcement `10246072590233`, 5th Anniversary Zero-Fee BTC spot trading) and removed on March 22, 2023 (Binance announcement `be21a7f1940b46fa95A42087b058958a`, Zero-Fee trading ended for BTC spot pairs except BTC/TUSD)—is price impact $\hat{\kappa}$, which is 0.24319 lower during zero-fee trading (-18.3% of its sample mean, $t = -3.86$, $p = 0.0002$). Because $\hat{\kappa}$ was not the primary outcome, this impact reduction is an *exploratory* finding. Moreover, Table 34 reports side by side the specification pooling all three originally listed dates (which included a March 20, 2024 USD-M VIP maker-fee adjustment affecting BTCUSDT and ETHUSDT): when that third date is included (N = 1,102, G = 176 date clusters), the pooled impact

Table 34: Zero fees on BTCUSDT spot: volume, reversion, gain multiple and impact (Part 20, specified in advance, not deposited)

Outcome (±30 days)	Spot zero-fee on (2022-07-08)	Spot zero-fee off (2023-03-22)	Two spot events pooled	All three dates pooled
Log quote volume	+0.944 (0.054)	+1.774 (0.074)	+1.355 (0.063)	+0.711 (0.081)
Reversion $\hat{m}$ (null)	−0.00298 (0.00313)	+0.00048 (0.00312)	−0.00120 (0.00221)	+0.00153 (0.00183)
p / 80% MDE	$p = 0.345$ / 50.9%	$p = 0.878$ / 41.7%	$p = 0.590$ / 31.9%	$p = 0.405$ / 25.0%
Gain $\widehat{S/\tilde{\kappa}}$ (null)	−61.89 (49.19)	+7.36 (32.86)	−26.88 (29.38)	−14.14 (20.13)
p / 80% MDE	$p = 0.213$ / 80.0%	$p = 0.824$ / 64.6%	$p = 0.362$ / 52.0%	$p = 0.483$ / 38.8%
Impact $\hat{\kappa}$ (explor.)	−0.1042 (0.1107)	−0.3830 (0.0571)	−0.2432 (0.0631)	+0.0082 (0.0577)
p / 80% MDE	$p = 0.351$ / 22.0%	$p < 10^{-4}$ / 13.2%	$p = 0.0002$ / 13.4%	$p = 0.887$ / 9.7%
Pair-days / clusters	324 / 59	374 / 58	698 / 117	1,102 / 176

Difference-in-differences of zero fees relative to control pairs (ETH, SOL, ADA, DOGE, BNB, XRP and LTC against USDT); date-clustered standard errors in parentheses, together with two-sided p-values and 80%-power minimum detectable effects (MDE, $2.8 \times$ SE as a percentage of the sample mean). Spot events verified against Binance announcements `10246072590233` (July 8, 2022) and `be21a7f1940b46fa95A42087b058958a` (March 22, 2023); the fourth column adds the March 20, 2024 VIP maker-fee date. Only one pair (BTCUSDT) is treated in the two spot events. Kind: real data (`recompute_p2_parts19_20_21.py` → `R_P2_parts19_20_21.json`).

coefficient is +0.00819 ($t = 0.14$, $p = 0.8873$).

Two econometrics limitations govern how Table 34 should be read. First, in the two verified spot events only a single pair (BTCUSDT) is treated against seven controls, so clustering standard errors by date does not correct for single-treated-unit cross-sectional dependence (Conley and Taber, 2011). Second, the 80%-power minimum detectable effects for the two null primary outcomes are wide: 31.9% of the sample mean for $\hat{m}$ and 52.0% for $\widehat{S/\tilde{\kappa}}$ in the two spot events (25.0% and 38.8% across all three dates), so moderate changes in reversion or in the gain multiple cannot be ruled out. With those caveats stated, a reduction in spot impact $\hat{\kappa}$ during zero-fee trading alongside no detectable change in $\hat{m}$ or $\widehat{S/\tilde{\kappa}}$ is *consistent with* Proposition 8.1(v) (where endogenous depth adjusts $\tilde{\kappa}$ while leaving $S/\tilde{\kappa}$ invariant), rather than a confirmation of it.

12.9 The guard results in dollars (Part 21)

Tables 29 and 30 report the guard in tail loss per unit of average position. Because positions are normalized to a training mean of one, that ratio converts directly into dollars: on $1M

Table 35: Guard results of Parts 15 and 16 in dollars per $1M of average position (Part 21, exact translation of archived results)

Rule	ES_{95} (USD)	Activity	Beyond static p95	$\Delta \log ES_{95}$ [95% CI] (p)
A. Part 15, untouched days (20 thin symbols, 6,732 test hours)				
(a) Static limit	37,724	1.000	1.98%	—
(b) Volatility cap	33,885	0.863	1.66%	—
(c) Vol. cap × guard	33,487	0.795	1.83%	(c)–(b): −0.0118 [−0.0370, +0.0126] (0.196)
(d) Guard alone	34,679	0.911	2.21%	(d)–(a): −0.0842 [−0.1608, −0.0172] (0.0025)
				(d)–(b): +0.0232 [−0.0222, +0.0815] (0.828)
B. Part 16, untouched symbols (5 thin symbols, 1,782 test hours)				
(a) Static limit	46,565	1.000	—	—
(b) Volatility cap	36,721	0.848	—	—
(c) Vol. cap × guard	34,164	0.784	—	(c)–(b): −0.0722 [−0.1801, +0.0047] (0.036)
(d) Guard alone	36,047	0.907	—	(d)–(a): −0.2560 [−0.4970, −0.0205] (0.004)

Hourly ES_{95} is the archived tail-loss ratio of Tables 29 and 30 times $100 per basis point on $1M of average position. Comparisons are the archived date-block bootstrap results. Kind: exact translation of real-data results (`recompute_p2_parts19_20_21.py` → `R_P2_parts19_20_21.json`).

of average position, a tail-loss ratio of 377.24 basis points is an expected shortfall of $37,724 for the hour. Table 35 restates the archived Part 15 and Part 16 results in dollars using no new data.

In dollars, the guard alone ((d) vs. (a)) lowers the hour's expected shortfall on $1M of average position by $3,045 on the untouched days ($p = 0.0025$) and by $10,518 on the untouched symbols ($p = 0.004$). Against a volatility cap ((c) vs. (b)), adding the guard lowers hourly expected shortfall by $398 on the untouched days (95% CI $[-3.7\%, +1.3\%]$, $p = 0.196$) and by $2,557 on the untouched symbols (95% CI $[-18.0\%, +0.5\%]$, unadjusted $p = 0.036$, Holm $p = 0.07$); neither improvement over the volatility cap is statistically significant after multiple-testing correction. Set beside the official operating budget of the U.S. Consolidated Audit Trail ($248.8 million in the November 2024 CAT LLC filing, SEC Release No. 34-101901, p. 4, revised to $228.3 million in May 2025; companion paper, Table 9), certification of the summed response requires only public price and flow records and none of the participant-level tagging needed to identify individual cohorts.

Table 36: Pre-registered test of the open-interest liquidation mechanism on untouched 22nd-of-month days (Part 22), 2025–2026

Specification (19 thin symbols, 6,766 test hours)	ΔR^2 (pts) / Share [95% CI]	CW t (p)	Result
Excursion $\text{MAE}_{t+1} \mid \hat{C}_{3,t}$ (22nd-of-month)	+6.516 [+3.159, +10.743]	12.21 ($< 10^{-6}$)	—
Primary 1: Liquidation $\text{LIQ}_{t+1} \mid \hat{C}_{3,t}$	−0.795 [−1.529, −0.179]	−1.12 (0.868)	fail
Excursion $\text{MAE}_{t+1} \mid \hat{C}_{3,t}, \text{LIQ}_{t+1}$	+6.958 [+3.405, +11.338]	—	—
Primary 2: Mediation share $1 - \Delta R^2_{\text{LIQ}}/\Delta R^2$	−6.8% [−10.2%, −3.1%]	—	fail

Trained on 22nd-of-month days in 2024 (3,876 symbol-hours), evaluated once on untouched 22nd-of-month days from January 2025 to September 2026 (6,766 symbol-hours); 2,000-draw date-block bootstrap 95% confidence intervals. Specification SHA-256 `2b3794e3812f...`; test opened once. Kind: real data (`recompute_prereg_oct21_22nd.py` → `R_P2_p22_p23.json`).

12.10 Pre-registered test of the liquidation mechanism (Part 22)

Because Part 19 was not hash-registered and tested only a static leverage interaction, we pre-registered a direct test of the liquidation mechanism (`P22_liquidation_mechanism.json`, SHA-256 `2b3794e3812f...`) and committed its hash to the repository before downloading any 22nd-of-month data for 2025–2026 (3,876 training symbol-hours on the 22nd of each month in 2024; 6,766 test symbol-hours on the 22nd of each month from January 2025 to September 2026 across 19 thin perpetuals; test opened once). When forced liquidations execute in a perpetual futures book without a public liquidation feed, they contract open interest while moving the price adversely; following standard practice we define the hourly forced-liquidation proxy $\text{LIQ}_{s,t}$ as the sum of five-minute open-interest contractions $(-\Delta\text{OI}_{s,\tau}/\overline{\text{OI}}_s)_+$ accompanied by an adverse five-minute return exceeding one training standard deviation. The registration specified two primaries: (*Primary 1*) $\hat{C}_3$ forecasts next-hour $\text{LIQ}_{s,t+1}$ out of sample beyond the Part 9 volatility, impact and lagged-LIQ_t baseline; and (*Primary 2*) controlling for realised contemporary $\text{LIQ}_{s,t+1}$ mediates at least 15% of $\hat{C}_3$'s out-of-sample ΔR^2 for the next hour's maximum adverse excursion $\text{MAE}_{s,t+1}$.

On these untouched 22nd-of-month test days, $\hat{C}_3$ strongly forecasts next-hour adverse excursions ($\Delta R^2 = +6.516$ percentage points, 95% CI [+3.159, +10.743], Clark–West $t = 12.21$, Table 36), providing a sixth out-of-sample confirmation of the thin-book excursion effect. However, both liquidation-mechanism primaries fail decisively: $\hat{C}_3$ does not forecast the five-

minute open-interest contraction proxy beyond the baseline ($\Delta R^2 = -0.795$ points, 95% CI $[-1.529, -0.179]$, $t = -1.12$), and adding realised next-hour open-interest contractions to the excursion model leaves $\hat{C}_3$'s predictive gain undiminished ($\Delta R^2 = +6.958$ points, mediation share -6.8%, 95% CI $[-10.2\%, -3.1\%]$). We therefore drop the hypothesis that five-minute open-interest liquidations mediate the effect: across thin Binance perpetuals and thin US equities alike, $\hat{C}_3$ forecasts adverse excursions because it measures slow book reversion relative to price impact directly.

12.11 Pre-registered test of non-resetting multi-hour guard capacity (Part 23)

To evaluate the guard on a leveraged position that is held across hours without resetting inventory at the top of each hour, we pre-registered a multi-hour margin-breach and notional-capacity test (`P23_guard_capacity.json`, SHA-256 `d4169377da0b...`) before downloading the 22nd-of-month data for 2025–2026 (228 training symbol-days in 2024; 398 test symbol-days in January 2025–September 2026 across 19 thin perpetuals; test opened once). At 06:00 UTC on each day d, a dealer opens a 20× leveraged position (450 bp maintenance-margin buffer before forced liquidation) using the three-day certified operator $\hat{S}^{\mathrm{up}}_{s,d}$ estimated from hours 00:00–06:00 across the three preceding evaluation days and holds the position continuously through 24:00 UTC; cumulative mark-to-market loss evolves pathwise across all 18 hours without hourly reset. The primary hypothesis tests whether, when the volatility cap (rule (b)) and the combined volatility-plus-certified-guard rule (rule (c)) are scaled to have the exact same out-of-sample forced-liquidation rate as the unit-scaled volatility cap (39.70%), rule (c) admits a significantly larger mean dollar notional position $\overline{N}$ than rule (b).

Table 37 reports both the unit-scaled outcomes and the matched-breach capacity comparison. At unit training scale, a static 20× position is liquidated on 51.26% of thin-symbol test days (daily ES_{95} = \$166,642), while the volatility cap lowers the liquidation rate to 39.70% (ES_{95} = \$144,360). When rules (b) and (c) are scaled to match the 39.70% out-of-

Table 37: Pre-registered non-resetting 18-hour leveraged position test on untouched 22nd-of-month days (Part 23), 2025–2026

Rule (398 symbol-days)	Liq. rate	Daily ES_{95} (USD)	Activity	Matched-breach $\overline{N}$
(a) Static 20× limit	51.26%	166,642	1.000	—
(b) Volatility cap	39.70%	144,360	0.795	\$902,282
(c) Vol. cap × cert. guard	42.21%	166,501	0.686	\$1,235,130
(d) Certified guard alone	50.00%	182,043	0.838	—
Primary: $\overline{N}$(c) − $\overline{N}$(b) (39.70% liq.)	+\$332,848 (+36.89%), 95% CI [−\$20,390, +\$1,180,808] (fail)			
Secondary: $\Delta \log ES_{95}$(c vs b)	+0.1427, 95% CI [+0.0741, +0.1964]			

Non-resetting 18-hour position (06:00–24:00 UTC) on 398 untouched test symbol-days (228 training symbol-days in 2024); 2,000-draw date-block bootstrap 95% intervals. Specification SHA-256 `d4169377da0b...`; test opened once. Kind: real data (`recompute_prereg_oct21_22nd.py` → `R_P2_p22_p23.json`).

sample liquidation rate, adding the morning-certified guard to the volatility cap raises the mean admitted notional position from \$902,282 to \$1,235,130—a point gain of +\$332,848 (+36.89%)—because the guard concentrates notional on symbol-days with faster certified reversion. However, across the 22 test dates the 95% date-block bootstrap confidence interval is [−\$20,390, +\$1,180,808], which narrowly crosses zero, and at unit scale rule (c) has higher tail loss than rule (b) ($\Delta \log ES_{95} = +0.1427$). The primary hypothesis therefore fails: in observational data, a three-day morning certificate is too coarse to deliver a statistically significant capacity gain over a trailing six-hour volatility cap across an 18-hour holding window.

13 Policy implications

Pre-trade controls. Rule 15c3-5 requires pre-trade controls that prevent orders exceeding credit or capital thresholds (SEC, 2010). Static single-order caps and daily value-at-risk limits do not see transient dynamics. Lemma 6.1 identifies the quantity that a solvency-motivated control should bound: inventory times the shock envelope times the summed absolute impulse response. Peak-based or eigenvalue-based limits are either unsafe or blind in exactly the regimes where they matter. Our evidence locates where such controls matter. In the deepest crypto book the amplifying regime is rare and the statistic carries no

forecasting value; in thin books it is more frequent, and six-hour estimates of the summed response forecast the next hour's adverse excursion beyond volatility forecasts and price impact (Section 11). Our backtests also set a limit on what such a control can do with public data. A six-hour estimate of the summed response adds nothing to a volatility limit as a sizing input (Table 25); an operator certified from three days of data improves a volatility limit modestly, by 2.4% in tail loss per unit of position (Table 26); and limits built on the estimate alone do worse than volatility in both. The guard's role is therefore a ceiling on worst-case loss under a certified operator and a correction to conventional volatility limits, not a replacement for them, and it is most relevant for books with under about \$0.4M of depth within 1% of the mid.

Pricing reversion. Proposition 8.1 shows that dealers under-supply reversion because its benefit, lower tail exposure for every leveraged position, is not paid to them. Maker rebates are a natural instrument: a rebate that rises with the book's summed response, $(1 - \theta/N)\Lambda S^2/\tilde{\kappa}$ per unit of reversion, prices the externality with an observable. Venues already tier rebates by volume; tiering them by contribution to reversion in thin books is a design question our model makes precise.

What can be certified, what cannot, and when to abstain. A supervisor or venue can certify the summed response, and hence a safe admission level, from controlled experiments at a cost that does not depend on how many traders there are or how similar their memories are (Theorem 5.5). It cannot certify any finite risk bound from a finite window without a bound on how long participants remember (Theorem 5.3(iii)), so a declared memory bound is a policy input, not a technicality, and a certificate must be renewed when conditions change (test T09). When compatible risks straddle the budget, the right response is to abstain: throttle admission to the certified level, and when even zero admission is uncertified because of existing inventory, flag the state and reduce the position (Theorem 6.5(iv)). Disclosure is worth paying for where memory is long and public certificates are weak, and is

worth little where a short window already pins the risk down (Theorem 7.1).

Market design. Frequent batch auctions remove latency races within a batch (Budish et al., 2015). Transient amplification, when it exists, operates over many seconds and spans many batches, so batching and a state-carrying guard address different failures and can be complements.

Supervision without identification. The companion paper shows that identifying algorithms from public data is expensive. The guard needs only impulse coefficients and operator confidence sets, which are invariant to how flow is split across cohorts. Supervisors can therefore ask venues and brokers to report summed-gain statistics and their confidence radii instead of attempting to infer black-box learners. Our results argue for reporting impact, mean reversion and refill directly, because those are the primitives that decide whether a book can amplify. They also show that feedback and amplification are different objects: measured feedback is largest in the deepest books, where amplification is rarest (Table 13).

14 Conclusion

The theory answers four questions. A book amplifies when depletion outweighs impact-weighted reversion; with momentum the one-period margin is sufficient but not exact, and momentum alone cannot make a stable book overshoot. Cumulative risk is long-run impact divided by reversion net of momentum, and the guard that bounds it is the largest safe pre-trade rule, computable and identified from public data. In equilibrium reversion is under-supplied, and the corrective price is a function of the guard's own statistic. And the information that certifies a safe decision is much cheaper than the information that identifies the traders: the summed response is certified at the parametric rate, finite windows certify it only under a memory bound, and a guard built on the certificate gives up no more trading capacity than any comparably safe rule, up to a logarithm.

The evidence is mixed in an informative way. The deepest crypto book sits on the stable side of the condition; flow depletes the side it hits on a fixed price grid, but not enough to produce a peak overshoot in the data we observe. In thin Binance books, six-hour estimates of cumulative transient risk forecast the size of the next hour's adverse excursion out of sample in four pre-registered tests, in unit-free form, beyond price impact and beyond a full volatility-forecasting baseline. At a given volatility forecast, hours in the top quintile of the statistic have next-hour excursions about 26 basis points larger than hours in the bottom quintile. In thin US stocks, with consolidated trades, the statistic passes a fifth pre-registered test at about a fifth of the Binance size, and appears in mid-size stocks too. On Bybit it is absent with trades alone and negligible with order-book depth, the effect is much weaker in deeper Binance books, and as a sizing input it helps only once the operator is certified from days of data, and then modestly. Two final pre-registered tests used data no earlier test touched: on new days the forecasting model, frozen at its 2023–2024 estimates, and the uncertainty-aware guard both pass; on new symbols, five thin ones, the statistic forecasts beyond a baseline that includes VPIN and the guard passes again, although its interval is wide and it does not survive a correction across all primaries. On a new venue, Coinbase spot, the registered test fails, and post hoc an order-splitting placebo carries none of the Binance signal. Three further analyses, specified in advance but not hash-registered, and two final pre-registered tests on untouched 22nd-of-month days qualify the mechanism and the guard's multi-hour capacity: the leverage slope interaction fails and $\hat{C}_3$ neither forecasts five-minute open-interest contractions nor is mediated by them (Part 22, pre-registered), showing that the thin-book predictability operates through direct book illiquidity and slow reversion rather than measurable five-minute open-interest liquidations; in Binance's BTCUSDT spot zero-fee experiment, the primary reversion and gain-multiple hypotheses are not supported while spot impact falls 18% in the two verified spot events (exploratory, single treated pair), a pattern consistent with Proposition 8.1(v); in dollars, the guard's hourly expected-shortfall reduction over a static limit is \$3,045 and \$10,518 per \$1M

of position while its improvement when combined with a volatility cap ($398 and $2,557) is not statistically significant; and in a non-resetting 18-hour leveraged position test (Part 23, pre-registered), adding the morning-certified guard to a volatility cap raises mean admitted notional at a matched 39.7% liquidation rate by +$332,848 (+36.9%), with a 95% confidence interval ([−$20,390, +$1,180,808]) that crosses zero. Thin books are risky because their prices revert too slowly relative to the impact and momentum of the flow that moves them.

Three extensions would settle what remains open. Message-level data from a regulated market, such as LOBSTER or TAQ for US equities, would extend the US test to depth and to large stocks, show where the effect ends, and allow depletion to be measured without the recentering of band depth. Data on liquidity-provider identities would test the equilibrium's prediction that fewer dealers mean more transient risk. And longer certification windows, or certification with order-level data, would show whether the modest gain of a certified guard over volatility limits can be made large.

Data and code availability. All data are public: Binance archives (https://data.binance.vision), verified against published checksums, and Bybit's order-book history (https://www.bybit.com/derivatives/en/history-data) and trade archive (https://public.bybit.com); US consolidated trades are from the Alpaca market-data API (https://alpaca.markets, free account). Pre-registered specifications, their SHA-256 hashes and test-set unlock counters are in the replication package. Each number is produced by a notebook cell or script that writes a result file (Appendix D).

Use of AI tools. The authors used Claude (Anthropic), an AI assistant, to write and debug analysis code, run checks of derivations and numerical results, and draft and edit text. Every number in the paper is produced by code the authors executed and archived in the replication package. The authors reviewed all content and take full responsibility for it.

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References

21Shares (2025). Record crypto liquidations amid tariff shock. Research note, October 13, 2025. https://www.21shares.com/research/record-crypto-liquidations-amid-tariff-shock.

Achieser, N. I. (1956). *Theory of Approximation*. New York: Frederick Ungar.

Almgren, R., and N. Chriss (2001). Optimal execution of portfolio transactions. *Journal of Risk* 3, 5–39.

Amihud, Y. (2002). Illiquidity and stock returns: Cross-section and time-series effects. *Journal of Financial Markets* 5, 31–56.

Andersen, T. G., and O. Bondarenko (2014). VPIN and the flash crash. *Journal of Financial Markets* 17, 1–46.

Avellaneda, M., and S. Stoikov (2008). High-frequency trading in a limit order book. *Quantitative Finance* 8, 217–224.

Biais, B., L. Glosten, and C. Spatt (2005). Market microstructure: A survey of microfoundations, empirical results, and policy implications. *Journal of Financial Markets* 8, 217–264.

Biais, B., P. Hillion, and C. Spatt (1995). An empirical analysis of the limit order book and the order flow in the Paris Bourse. *Journal of Finance* 50, 1655–1689.

Bouchaud, J.-P., Y. Gefen, M. Potters, and M. Wyart (2004). Fluctuations and response in financial markets: The subtle nature of "random" price changes. *Quantitative Finance* 4, 176–190.

Bouchaud, J.-P., J. Bonart, J. Donier, and M. Gould (2018). *Trades, Quotes and Prices: Financial Markets Under the Microscope*. Cambridge University Press.

Brogaard, J., T. Hendershott, and R. Riordan (2014). High-frequency trading and price discovery. *Review of Financial Studies* 27, 2267–2306.

Brunnermeier, M. K., and L. H. Pedersen (2009). Market liquidity and funding liquidity. *Review of Financial Studies* 22, 2201–2238.

Budish, E., P. Cramton, and J. Shim (2015). The high-frequency trading arms race: Frequent batch

auctions as a market design response. *Quarterly Journal of Economics* 130, 1547–1621.

Cartea, Á., S. Jaimungal, and J. Penalva (2015). *Algorithmic and High-Frequency Trading*. Cambridge University Press.

Cespa, G., and T. Foucault (2014). Illiquidity contagion and liquidity crashes. *Review of Financial Studies* 27, 1615–1660.

Clark, T. E., and K. D. West (2007). Approximately normal tests for equal predictive accuracy in nested models. *Journal of Econometrics* 138, 291–311.

Conley, T. G., and C. R. Taber (2011). Inference with "difference in differences" with a small number of policy changes. *Review of Economics and Statistics* 93, 113–125.

Cont, R., A. Kukanov, and S. Stoikov (2014). The price impact of order book events. *Journal of Financial Econometrics* 12, 47–88.

Corsi, F. (2009). A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics* 7, 174–196.

Dahleh, M. A., and I. J. Diaz-Bobillo (1995). *Control of Uncertain Systems: A Linear Programming Approach*. Prentice Hall.

Dahleh, M. A., and J. B. Pearson (1987). ℓ^1-optimal feedback controllers for MIMO discrete-time systems. *IEEE Transactions on Automatic Control* 32, 314–322.

Degryse, H., F. de Jong, M. van Ravenswaaij, and G. Wuyts (2005). Aggressive orders and the resiliency of a limit order market. *Review of Finance* 9, 201–242.

Diebold, F. X., and R. S. Mariano (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics* 13, 253–263.

Easley, D., M. M. López de Prado, and M. O'Hara (2012). Flow toxicity and liquidity in a high-frequency world. *Review of Financial Studies* 25, 1457–1493.

Foucault, T., O. Kadan, and E. Kandel (2005). Limit order book as a market for liquidity. *Review of Financial Studies* 18, 1171–1217.

Hansen, E., and G. W. Walster (2004). *Global Optimization Using Interval Analysis*, 2nd ed. New York: Marcel Dekker.

Harvey, C. R. (2017). Presidential address: The scientific outlook in financial economics. *Journal*

of Finance 72, 1399–1440.

Harvey, C. R., Y. Liu, and H. Zhu (2016). . . . and the cross-section of expected returns. *Review of Financial Studies* 29, 5–68.

Hendershott, T., C. M. Jones, and A. J. Menkveld (2011). Does algorithmic trading improve liquidity? *Journal of Finance* 66, 1–33.

Ho, T., and H. R. Stoll (1981). Optimal dealer pricing under transactions and return uncertainty. *Journal of Financial Economics* 9, 47–73.

Holm, S. (1979). A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics* 6, 65–70.

Hotelling, H. (1931). The generalization of Student's ratio. *Annals of Mathematical Statistics* 2, 360–378.

Howard, S. R., A. Ramdas, J. McAuliffe, and J. Sekhon (2021). Time-uniform, nonparametric, nonasymptotic confidence sequences. *Annals of Statistics* 49, 1055–1080.

Huang, J., and J. Wang (2009). Liquidity and market crashes. *Review of Financial Studies* 22, 2607–2643.

Jordà, Ò. (2005). Estimation and inference of impulse responses by local projections. *American Economic Review* 95, 161–182.

J.P. Morgan/Reuters (1996). *RiskMetrics—Technical Document*, 4th ed. New York: J.P. Morgan.

Karlin, S., and W. J. Studden (1966). *Tchebycheff Systems: With Applications in Analysis and Statistics*. New York: Interscience.

Kirilenko, A., A. S. Kyle, M. Samadi, and T. Tuzun (2017). The Flash Crash: High-frequency trading in an electronic market. *Journal of Finance* 72, 967–998.

Krein, M. G., and A. A. Nudel'man (1977). *The Markov Moment Problem and Extremal Problems*. Translations of Mathematical Monographs 50. Providence, RI: American Mathematical Society.

Kreiss, H.-O. (1962). Über die Stabilitätsdefinition für Differenzengleichungen die partielle Differentialgleichungen approximieren. *BIT Numerical Mathematics* 2, 153–181.

Künsch, H. R. (1989). The jackknife and the bootstrap for general stationary observations. *Annals of Statistics* 17, 1217–1241.

Kyle, A. S. (1985). Continuous auctions and insider trading. *Econometrica* 53, 1315–1335.

Kyle, A. S., and A. A. Obizhaeva (2016). Market microstructure invariance: Empirical hypotheses. *Econometrica* 84, 1345–1404.

Large, J. (2007). Measuring the resiliency of an electronic limit order book. *Journal of Financial Markets* 10, 1–25.

Le Cam, L. (1986). *Asymptotic Methods in Statistical Decision Theory*. New York: Springer.

Makarov, I., and A. Schoar (2020). Trading and arbitrage in cryptocurrency markets. *Journal of Financial Economics* 135, 293–319.

Menkveld, A. J. (2013). High frequency trading and the new market makers. *Journal of Financial Markets* 16, 712–740.

Milanese, M., and A. Vicino (1991). Optimal estimation theory for dynamic systems with set membership uncertainty: An overview. *Automatica* 27, 997–1009.

Newey, W. K., and K. D. West (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica* 55, 703–708.

Niroula, P., and P. Niroula (2026). Auditable memory and financial identification limits in electronic limit order books. Working paper, Minnesota State University, Mankato.

Nosek, B. A., C. R. Ebersole, A. C. DeHaven, and D. T. Mellor (2018). The preregistration revolution. *Proceedings of the National Academy of Sciences* 115, 2600–2606.

Obizhaeva, A. A., and J. Wang (2013). Optimal trading strategy and supply/demand dynamics. *Journal of Financial Markets* 16, 1–32.

Rump, S. M. (2010). Verification methods: Rigorous results using floating-point arithmetic. *Acta Numerica* 19, 287–449.

Schneider, M., and F. Lillo (2019). Cross-impact and no-dynamic-arbitrage. *Quantitative Finance* 19, 137–154.

U.S. Securities and Exchange Commission (2010). Risk management controls for brokers or dealers with market access. Release No. 34-63241.

Shleifer, A., and R. W. Vishny (1992). Liquidation values and debt capacity: A market equilibrium approach. *Journal of Finance* 47, 1343–1366.

Sornette, D., S. Lera, J. Lin, and K. Wu (2023). Non-normal interactions create socio-economic bubbles. *Communications Physics* 6, 261.

Spijker, M. N. (1991). On a conjecture by LeVeque and Trefethen related to the Kreiss matrix theorem. *BIT Numerical Mathematics* 31, 551–555.

Trefethen, L. N., and M. Embree (2005). *Spectra and Pseudospectra: The Behavior of Nonnormal Matrices and Operators*. Princeton University Press.

Tu, S., R. Boczar, and B. Recht (2018). Minimax lower bounds for $\mathcal{H}_\infty$-norm estimation. Working paper, arXiv:1809.10855.

Vidyasagar, M. (1986). Optimal rejection of persistent bounded disturbances. *IEEE Transactions on Automatic Control* 31, 527–534.

Ville, J. (1939). *Étude critique de la notion de collectif*. Paris: Gauthier-Villars.

van Waarde, H. J., J. Eising, H. L. Trentelman, and M. K. Camlibel (2020). Data informativity: A new perspective on data-driven analysis and control. *IEEE Transactions on Automatic Control* 65, 4753–4768.

A Proofs

Proposition 2.1. With one cohort, $J = \left(\begin{smallmatrix} a & \kappa q \\ \eta a & 1-\eta+\eta\kappa q \end{smallmatrix}\right)$, so $\mathrm{tr}J = a+1-\eta+\eta\kappa q$ and $\det J = a(1-\eta)$. The Jury conditions for a real 2×2 matrix are $|\det J| < 1$ and $1 \pm \mathrm{tr}J + \det J > 0$. Substituting $a = 1-\kappa\varphi_f$: $1 - \mathrm{tr}J + \det J = \eta(1-a-\kappa q) = \eta\kappa(\varphi_f - q)$, positive iff $q < \varphi_f$; and $1 + \mathrm{tr}J + \det J = (4-2\eta) - \kappa[(2-\eta)\varphi_f - \eta q]$, positive iff the stated bound on κ holds (automatically when $(2-\eta)\varphi_f \le \eta q$).

Proposition 2.2. Equal merged measures give equal G, hence equal H, and the impulse coefficients are the Laurent coefficients of H (companion paper, Theorem on merged equivalence). Under $z \mapsto Tz$, $c^\top J^k b$ is unchanged because $CT^{-1}(TJT^{-1})^k Tb = CJ^k b$, while the row norm $\|CT^{-1}(TJT^{-1})^k\|$ depends on T. The displayed price path is the convolution of inputs with $\{g_k\}$, and its worst case over the ℓ_∞ ball is $\bar{u}\sum|g|$ by Hölder.

Proposition 2.3. (i) Taking z-transforms of $y_{i,t} = (1-\eta_i)y_{i,t-1}+\eta_i x_t$, $I_{i,t} = (1-\delta_{\text{inv}})I_{i,t-1}+u_{i,t}$ and $u_{i,t} = q_i y_{i,t} - \psi_i I_{i,t-1}$ gives $Y_i(z) = \frac{\eta_i z}{z-1+\eta_i}X(z)$ and $(z-1+\delta_{\text{inv}}+\psi_i)I_i(z) = q_i z Y_i(z)$, whence $U_i(z) = \frac{z-1+\delta_{\text{inv}}}{z-1+\delta_{\text{inv}}+\psi_i}q_i Y_i(z)$ and $G_{\text{inv}}(z) = \sum_i U_i(z)/X(z)$ has the stated form. Evaluating at $z = 1$ when the augmented operator is Schur stable gives $G_{\text{inv}}(1) = \sum_i \frac{\delta_{\text{inv}}}{\delta_{\text{inv}}+\psi_i}q_i = \Phi_{\text{eff}}(\psi)$ and $S(\psi) = H_{\text{inv}}(1) = \kappa/(1-a-\kappa\Phi_{\text{eff}}(\psi))$. (ii) Write $r_i = 1-\eta_i$ and $\rho_i = 1-\delta_{\text{inv}}-\psi_i$. When $0 \le \psi_i \le 1-\delta_{\text{inv}}$ and $\delta_{\text{inv}} \ge \eta_i$, we have $0 \le \rho_i \le 1-\delta_{\text{inv}} \le r_i < 1$. Expanding $U_i(z)/X(z) = q_i\eta_i\frac{z}{z-r_i}\big(1-\frac{\psi_i}{z-\rho_i}\big)$ in powers of z^{-1}, the m-th open-loop feedback tap of cohort i is $k_{i,0} = q_i\eta_i \ge 0$ and, for $m \ge 1$, $k_{i,m} = q_i\eta_i\big(r_i^m - \psi_i\sum_{j=0}^{m-1} r_i^{m-1-j}\rho_i^j\big) \ge q_i\eta_i r_i^{m-1}\big(r_i - \frac{\psi_i}{1-\rho_i/r_i}\big) \ge 0$ because $\frac{\psi_i}{r_i-\rho_i} = \frac{\psi_i}{\delta_{\text{inv}}-\eta_i+\psi_i} \le 1 \le r_i$ whenever $\psi_i \le (1-\delta_{\text{inv}})$ and $\delta_{\text{inv}} \ge \eta_i$ with $r_i^m \ge \psi_i\frac{r_i^m-\rho_i^m}{r_i-\rho_i}$ (since $\frac{\psi_i}{r_i-\rho_i} \le 1$). Nonnegative open-loop taps $k_m(\psi) \ge 0$ with $a+\kappa\sum_m k_m(\psi) = a+\kappa\Phi_{\text{eff}}(\psi) < 1$ generate a nonnegative closed-loop renewal sequence $h_t(\psi) \ge 0$, so $\|h(\psi)\|_1 = \sum_{t\ge 0} h_t(\psi) = S(\psi)$.

Theorem 3.1. (i) $\sup_t \|CJ^t z_0\| \le \sup_t \|CJ^t\|\|z_0\|$, with equality approached by aligning z_0 with the top right singular vector of CJ^{t^*} at the maximizing t^*. (ii) The Kreiss matrix theorem with Spijker's sharp constant gives $\mathcal{K}(J) \le \sup_t \|J^t\| \le en\mathcal{K}(J)$; substitute in (i). (iii) Take $J_\theta = \begin{pmatrix} \rho & \theta \\ 0 & \rho/2 \end{pmatrix}$; then $\rho(J_\theta) = \rho$ and $(1,0)J_\theta e_2 = \theta$, so $M_C \ge |\theta|$ and $r^*_{\text{exact}} \le B/|\theta| \to 0$.

Proposition 3.2. With all blocks entrywise nonnegative, every entry of J^t_{multi} is a polynomial in the entries of K_{21} with nonnegative coefficients, so it is non-decreasing in K_{21}; with $C \ge 0$, so is every entry of CJ^t and hence its norm. For the Kreiss constant, the Neumann series gives $|(zI-J)^{-1}| \le (|z|I-J)^{-1}$ entrywise for $|z| > 1$, so the supremum is attained on the real axis, where $(rI-J)^{-1} = \sum_k J^k/r^{k+1}$ is entrywise non-decreasing in K_{21}.

Theorem 6.8. After the step, inventory is at most $|I^{\text{post}}_{t+1}(\alpha_t)|$ in absolute value. The price move over the step and the cumulative window is bounded by $M_C\|z_t\|_2$ (drift from the current state) plus $s(\alpha)G_{\text{cum}}\bar{u}_{\text{cum}}$ (admitted shocks), by superposition and Hölder. The

margin requirement scales the loss by $(1+\mu)$. Hedging cost and the post-fill margin increment are deducted directly. Inequality (14) therefore implies $B_{\mathrm{rem},t+1} \geq R_{\mathrm{safe}}$ for every admissible disturbance, and induction over steps at which an admissible α exists gives forward solvency. Setting $\alpha = 0$ ($M_C(0) = 1$, no fills, no hedge) gives $\bar{u}_{\mathrm{adm}}$. Sufficiency over the Euclidean ball follows from the norm bounds. When $z_t = 0$ the remaining shock term is attained by $u = -\mathrm{sign}(I)\bar{u}$, which gives necessity; when $z_t \neq 0$ the drift bound $M_C\|z_t\|_2$ need not be attained, so the condition is sufficient but not necessary.

Proposition 6.9. (i) In the positive cohort economy $J(\alpha) \geq 0$ entrywise, so for any $z_0 = (x_0, y_0^\top)^\top$, $|e_x^\top J(\alpha)^t z_0| \leq e_x^\top J(\alpha)^t(|x_0|, |y_0|^\top)^\top \leq e_x^\top J(\alpha)^t(|x_0|, \bar{y}_0\mathbf{1}^\top)^\top$, with equality when $y_0 = \mathrm{sign}(x_0)\bar{y}_0\mathbf{1}$. Under the price-first unforced recursion $x_t = ax_{t-1} + \alpha\kappa\sum_i q_i y_{i,t-1}$ ($t \geq 1$) and $y_{i,t} = (1-\eta_i)y_{i,t-1} + \eta_i x_t$ ($t \geq 1$) with initial state $(|x_0|, \bar{y}_0\mathbf{1}^\top)$ in the unit-sum memory coordinate (where the initial cohort state $y_{i,0}$ enters with unit cumulative weight $\sum_{m\geq 0}(1-\eta_i)^m\eta_i = 1$, i.e. $\tilde{y}_{i,0} = \eta_i y_{i,0}$ bounded by $\eta_i\bar{y}_0$), summing over $t \geq 1$ gives $\sum_{t\geq 0} q_i y_{i,t} = q_i\bar{y}_0 + q_i\sum_{t\geq 1} x_t$, and hence $(1-a-\alpha\kappa\Phi)\sum_{t\geq 1} x_t = a|x_0| + \alpha\kappa\Phi\bar{y}_0$, proving (15). (ii) Adding the forced worst-case shock loss $|I|\frac{\kappa\bar{u}}{1-a-\alpha\kappa\Phi}$ to $|I|$ times (15) and requiring the total loss to be at most $\mathcal{B}$ at $\Phi = \Phi^+$ rearranges to $\alpha\kappa\Phi^+(\mathcal{B} + |I|\bar{y}_0) \leq (1-a)\mathcal{B} - |I|(a|x_0| + \kappa\bar{u})$, which gives (16). (iii) Subtracting $\alpha^*_{\mathrm{drift}} = c_I/[\kappa\Phi^+(1 + |I|\bar{y}_0/\mathcal{B})]$ from $\alpha^* = c_I/(\kappa\Phi^+)$ at $x_0 = 0$ gives (17).

Proposition 6.10. The eigenvalue rule admits α_ρ; the one-step certificate fails at α_ρ exactly when the left side of (14), with the shock term replaced by its one-step bound $M_C(\alpha_\rho)s(\alpha_\rho)\bar{u}_{\mathrm{step},t}$, exceeds $B_{\mathrm{rem},t}$, which rearranges to the stated inequality. The summed-gain guard admits only levels satisfying (14), so it never admits the breaching level, and by the feasibility assumption it admits at least $\alpha = 0$.

Proposition 6.11. (i) The steady-state gain is $\eta = P/(P + \sigma_\nu^2)$ with P the positive root of $P^2 - \sigma_\zeta^2 P - \sigma_\zeta^2\sigma_\nu^2 = 0$. Since P is strictly increasing in σ_ζ^2 and η is strictly increasing in P,

η^* is strictly increasing in $\sigma_\zeta^2(\alpha) = \alpha^2\sigma_{\zeta,0}^2$, hence in α. The second statement is immediate under the stated monotonicity.

(ii) Write the recursion as a renewal equation: $h_t/\kappa = u_t$ with $u_0 = 1$ and $u_t = \sum_{j=1}^{t} p_j u_{t-j}$, where $p_1 = a + \kappa k_0$ and $p_j = \kappa k_{j-1}$ for $j \geq 2$. The p_j are nonnegative with total mass $\sum_j p_j = a + \kappa\Phi = a + \kappa\sum_i q_i < 1$, which does not depend on the adaptation rates, so u_t is the probability that a renewal process whose (defective) inter-arrival law is p has a renewal at time t, and $\sum_{t<H} u_t$ is the expected number of renewals in $\{0, \ldots, H-1\}$. The cumulative law is $F(j) = \sum_{i\leq j} p_i = a + \kappa\sum_{m<j} k_m = a + \kappa\sum_i q_i\big[1 - (1-\eta_i)^j\big]$, with $\partial F(j)/\partial\eta_i = \kappa q_i j(1-\eta_i)^{j-1} \geq 0$. If $\eta' \geq \eta$ coordinatewise, then $F' \geq F$ pointwise with the same total mass, so the quantile coupling $X = F^{-1}(U)$, $X' = F'^{-1}(U)$ ($X = \infty$ when $U > F(\infty)$) of independent inter-arrival times gives $X' \leq X$ and hence, pathwise, at least as many renewals before H under η'. Taking expectations gives $S_H(\eta') \geq S_H(\eta)$. If $q_i > 0$ and $\eta_i' > \eta_i$, then $F'(1) > F(1)$, so with positive probability the first renewal occurs at time 1 under η' and later under η, and the inequality is strict for $H \geq 2$. As $H \to \infty$ both sides tend to $\kappa/(1 - a - \kappa\Phi)$, which does not depend on η. Combined with (i), throttling lowers every rate, so $S_H(J(\alpha, \eta^*(\alpha))) \leq S_H(J(\alpha, \eta^*(1)))$.

Theorem 5.3. The recursion $h_{t+1} = ah_t + \kappa\sum_{m\leq t} k_m h_{t-m}$ with $h_0 = \kappa \neq 0$ gives k_t from $h_0, \ldots, h_{t+1}$, so the window determines $k_0, \ldots, k_{n-1}$ and nothing more about M: any nonnegative measure on $[0, \bar{r}]$ with these moments, written as atoms (r_i, w_i) or as a limit of such, is the merged measure of a positive economy with $\eta_i = 1 - r_i$ and $q_i = w_i/\eta_i$ that produces the same window. (i) The compatible measures form a convex, weak-* compact set, and $\Phi = \int f\, dM$ is linear and continuous on it (f is continuous on $[0, \bar{r}]$), so its range is a closed interval $[\Phi_-, \Phi_+]$. Since $f^{(n)}(r) = n!/(1-r)^{n+1} > 0$ on $[0, \bar{r}]$, $(1, r, \ldots, r^{n-1}, f)$ is an extended complete Chebyshev system, and by the Markov–Krein theorem (Krein and Nudel'man, 1977, Theorems IV.1.1 and IV.6.1) the maximum is attained at the upper and the minimum at the lower principal representation, which have at most $\lfloor n/2 \rfloor + 1$ atoms.

The theorem applies to moment vectors in the interior of the moment space, where both principal representations exist and are unique; a vector on the boundary has exactly one representing measure, so the interval collapses to a point. $S = \kappa/(1 - a - \kappa\Phi)$ is increasing in Φ on $\Phi < (1 - a)/\kappa$, which gives the interval of summed responses, unbounded when $\Phi_+ \geq (1 - a)/\kappa$ because by convexity every $\Phi \in [\Phi_-, \Phi_+]$ is attained. (ii) If μ, ν share moments up to order $n - 1$, then $\int f\, d(\mu - \nu) = \int (f - p)\, d(\mu - \nu)$ for every polynomial p of degree $n - 1$, so $|\int f\, d(\mu - \nu)| \leq 2k_0 \inf_p \|f - p\|_\infty = 2k_0 E_{n-1}$. With $r = \bar{r}(1 + x)/2$, $f = (2/\bar{r})/(x_0 - x)$ on $[-1, 1]$ with $x_0 = 2/\bar{r} - 1 > 1$, and Chebyshev's formula gives $E_m(1/(x_0 - x)) = \varrho^m/(x_0^2 - 1)$ (Achieser, 1956); we checked it against a linear-programming minimax fit for $m \leq 8$. Expanding $x_0 = 1 + 2(1 - \bar{r})/\bar{r}$ gives the asymptotics. (iii) Let the moment vector c be interior to the moment cone of $[0, 1]$. Interior points have an open neighbourhood in the cone, and $m(r_0) = (1, r_0, \ldots, r_0^{n-1})$ has norm at most $\sqrt{n}$, so for some $\epsilon > 0$, $c - \epsilon m(r_0)$ is interior for every $r_0 \in [0, 1]$ and is represented by its lower principal representation, which has no atom at 1. Adding $\epsilon\delta_{r_0}$ gives a measure on $[0, 1)$ with moments c and $\Phi \geq \epsilon/(1 - r_0) \to \infty$ as $r_0 \to 1$; by convexity every larger Φ below $(1 - a)/\kappa$ is attained by a stable economy, so S is unbounded.

Theorem 5.5. (i) The flow impulse enters the stacked state as $b = \kappa(1, \eta_1, \ldots, \eta_K)^\top$, and $h_t = e_x^\top J^t b \leq \|J^t b\|_\infty \leq \|J\|_\infty^t \|b\|_\infty$ because $J \geq 0$. The price row of J sums to $a + \kappa\Phi \leq \bar{s}$ and cohort row i to $1 - \eta_i(1 - a - \kappa\Phi) \leq 1 - (1 - \bar{r})(1 - \bar{s}) = \bar{\lambda}$, so $h_t \leq \kappa\bar{\lambda}^t$ and $\sum_{t \geq T} h_t \leq \kappa\bar{\lambda}^T/(1 - \bar{\lambda})$. $\sum_{t<T} \bar{h}_t$ is Gaussian with mean $\sum_{t<T} h_t$ and variance $T\sigma^2/N$, so $\sum_{t<T} \bar{h}_t + z_{1-\gamma}\sigma\sqrt{T/N} \geq \sum_{t<T} h_t$ with probability $1 - \gamma$, and adding the tail bound gives validity. On the two-sided event of probability $1 - 2\gamma$, $\hat{S}^+ - S \leq 2z_{1-\gamma}\sigma\sqrt{T/N} + \kappa\bar{\lambda}^T/(1 - \bar{\lambda})$; T_ϵ makes the second term at most $\epsilon/2$ and the stated N makes the first at most $\epsilon/2$. (ii) Let θ_1 raise the cohort weight of θ_0 by Δ chosen so that $S_1 - S_0 = 2\epsilon$; then $\|h_1 - h_0\|_2 = 2C_0\epsilon(1 + o(1))$. Validity at θ_1 gives $P_1(\hat{S} \geq S_1) \geq 1 - \gamma$, hence $P_0(\hat{S} - S_0 \geq 2\epsilon) \geq 1 - \gamma - \mathrm{TV}(P_0, P_1)$. For N independent Gaussian responses, $\mathrm{KL} = N\|h_1 - h_0\|_2^2/(2\sigma^2)$, and Pinsker's inequality gives

$\mathrm{TV} \le \sqrt{N}\, C_0\epsilon(1+o(1))/\sigma$. An overshoot probability below $1/2$ therefore needs $\mathrm{TV} > 1/2-\gamma$, which is the stated bound. (iii) The identification rates are the companion paper's Theorems on the K-cohort lower bound, the matching upper bound and general memory.

Lemma 6.4. On the event that $\mathcal{C}(D)$ contains the true economy, the supremum over $\mathcal{C}(D)$ dominates the true worst-case loss for every action, including one selected from D; the event has probability at least $1-\gamma$.

Theorem 6.5. (i) For $c_I \ge 0$, $\alpha \mapsto S(\alpha) = \kappa/(1-a-\kappa\alpha\Phi)$ is increasing, and so is the worst-case loss $|I|\bar{u}S(\alpha)$, which is also increasing in Φ; the robust constraint is therefore the constraint at Φ^+, and $|I|\bar{u}\kappa/(1-a-\kappa\alpha\Phi^+) \le \mathcal{B}$ is equivalent to $\alpha \le c_I/(\kappa\Phi^+)$. (ii) is Lemma 6.4. (iii) At $\Phi = \Phi^+$ the constraint binds at α^* when $\alpha^* < 1$. (iv) At $\alpha = 0$, $S = \kappa/(1-a)$. (v) Upper: α^* is Lipschitz in Φ^+ with constant $c_I/(\kappa\Phi^2_{\min})$ and Φ^+ is Lipschitz in $\hat{S}^+$ with constant $1/S^2_{\min}$, so Theorem 5.5(i) with $T \asymp \log N$ gives the rate. Lower: take two single-cohort economies in the class whose weights differ by $\Delta = c\,\sigma/\sqrt{N}$; their oracle capacities differ by a constant times Δ, and $\mathrm{TV} \le \sqrt{N}\|\partial h/\partial q\|_2\Delta/(2\sigma)$, which is below $(1-\beta)/2$ for small c; Proposition 5.2 gives an expected loss of at least $(1-\beta)/2$ times the capacity difference.

Theorem 7.1. (i) If O' reveals everything O reveals, every economy compatible with O' data is compatible with O data, so suprema over $\Theta_{O'}$ are at most those over Θ_O, and Theorem 6.5(i) is monotone in Φ^+. (ii) α^* is strictly decreasing in Φ^+ when $c_I > 0$ and $\alpha^* < 1$, and $\Phi^+_{O'} < \Phi^+_O$ exactly when no economy with $\Phi = \Phi^+_O$ remains compatible (the supremum is attained, by Theorem 5.3(i)). (iii) follows from Theorem 5.3(iii). (iv) If O identifies Φ, $\Phi^+_O = \Phi$ and no contract can raise α^*. (v) The delta method applied to $(w, r) \mapsto w/(1-r)$, whose gradient is $(1-r)^{-1}(1, \Phi_i)$, gives the standard error. If each certificate is valid at level $1-\gamma/2$, their minimum is valid at level $1-\gamma$ by the union bound and is no larger than either, so the admitted level is at least the larger of the two admitted levels.

Lemma 5.4. For a compatible merged measure M on $[0,\bar{r}]$, $\int p\,dM = \sum_j c_j k_j$ for any polynomial of degree at most $n-1$, so $p_- \le f \le p_+$ gives the two bounds by integration. Let H interpolate f at the atoms x_i of μ_+ with multiplicity $m_i = 2$ at interior atoms and 1 at 0 and $\bar{r}$; principal representations have index $\sum_i m_i = n$, so H has degree at most $n-1$, and the Hermite remainder is $f(r) - H(r) = f^{(n)}(\xi)\,\omega(r)/n!$ with $\omega(r) = \prod_i (r - x_i)^{m_i}$. Since $f^{(n)} > 0$ and, on $[0,\bar{r}]$, the factor $r - \bar{r} \le 0$ appears once while every other factor is a square or $r \ge 0$, $\omega \le 0$ and $H \ge f$. For μ_- there is no atom at $\bar{r}$, so $\omega \ge 0$ and $H \le f$. As $H = f$ on the support of $\mu_\pm$, $\sum_j c_j k_j = \int H\,d\mu_\pm = \int f\,d\mu_\pm = \Phi_\pm$. The certified computation proves $p_+ + s \ge f$ (or $p_- - s \le f$) for the rounded polynomial and slack s in interval arithmetic, in which every operation encloses its exact result (Rump, 2010), so the first statement applies to the rounded polynomial with c_0 increased (decreased) by s.

Proposition 5.7. (i) The X_e are i.i.d. normal with mean $S_T = \sum_{t<T} h_t$, whatever Σ, so $\sqrt{N}(\bar{X} - S_T)/s_X$ has Student's t_{N-1} distribution and $P(S_T \le \bar{X} + t_{N-1,1-\gamma} s_X/\sqrt{N}) = 1-\gamma$; the tail bound of Theorem 5.5(i) holds deterministically on the class. (ii) $N(\bar{h}-m)^\top \hat{\Sigma}^{-1}(\bar{h}-m)$ has the $\frac{T(N-1)}{N-T} F_{T,N-T}$ distribution (Hotelling, 1931); on the event that it is at most T_γ^2, Cauchy–Schwarz in the $\hat{\Sigma}$ inner product gives $c^\top(m-\bar{h}) \le \sqrt{c^\top \hat{\Sigma} c}\,\sqrt{(\bar{h}-m)^\top \hat{\Sigma}^{-1}(\bar{h}-m)} \le \sqrt{T_\gamma^2 c^\top \hat{\Sigma} c/N}$ for every c at once. (iii) If the taps of an experiment have common variance σ^2 and correlation matrix R, the margin $z_{1-\gamma}\sigma\sqrt{T/N}$ has coverage $\Phi\big(z_{1-\gamma}\sqrt{T/\mathbf{1}^\top R \mathbf{1}}\big)$, below $1-\gamma$ whenever $\mathbf{1}^\top R \mathbf{1} > T$; with $T = 12$ and autocorrelation 0.8 it is 0.751, matching the simulations.

Proposition 6.6. (i) D is a quadratic with leading coefficient $\delta\rho \ge 0$, and a level is safe if and only if $D(\alpha) \ge \kappa_0 |I| \bar{u}/\mathcal{B}$ (which implies $D > 0$, hence stability and $S = \kappa_0/D$). The set where a convex function is below a constant is an interval, so the safe set is $[0,1]$ minus an open interval; the example in the text has two components. (ii) For $\alpha \ge 0$ and $a_0 < 1$, $\partial D/\partial\delta = \alpha(1 - a_0 + \rho\alpha) \ge 0$, $\partial D/\partial\rho = \alpha(1+\delta\alpha) \ge 0$, $\partial D/\partial a_0 = -(1+\delta\alpha) \le 0$ and $\partial D/\partial\Phi = -\kappa_0\alpha \le 0$, while the condition $D - \kappa_0|I|\bar{u}/\mathcal{B} \ge 0$ decreases in κ_0. The minimum

of the left side over the box is therefore at the stated corner for every α. (iii) follows from (i) and (ii).

Theorem 6.7. (i) In interval arithmetic with outward rounding every operation returns an interval containing the exact result for all real arguments in its operands (Rump, 2010); by induction the computed interval vectors contain $J(\alpha,\theta)^t s_0(\alpha,\theta)$ and the interval matrices contain J^j for every (α,θ) in the box, so the computed sum contains $\sum_{t<T}|h_t|$. For $t \geq T$, writing $t = T+kT+j$ with $k \geq 0$ and $0 \leq j < T$, $|h_t| \leq \|J^{kT+j}J^T s_0\|_\infty \leq \|J^T\|_\infty^k \|J^j\|_\infty \|J^T s_0\|_\infty$, and summing gives the tail bound. A level interval classified safe has an upper enclosure within budget over every part of a partition of Θ, hence for every θ; one classified unsafe has $S(\alpha,\theta_j) \geq \sum_{t<T}|h_t(\alpha,\theta_j)| >$ budget for some $\theta_j \in \Theta$ and every level in it. (ii) Each $S(\cdot,\theta)$ is continuous where finite and tends to infinity at the stability boundary, so $\alpha \mapsto \sup_\Theta S(\alpha,\theta)$ is lower semicontinuous and its sublevel set, the robust safe set, is closed; the brackets follow from (i). (iii) is immediate. (iv) Under $\sup_\Theta \rho(J) < 1$ there is T with $\|J^T\|_\infty < 1$ uniformly on Θ; interval extensions of rational functions converge linearly in the box width (Hansen and Walster, 2004), so at a level where the robust risk differs from the budget by $g > 0$ the enclosures over small enough boxes, with T large enough that the truncation is below $g/2$, decide.

Proposition C.1. With $D_n = \sum_{e\leq n}(X_e - S_T)$ and $V_n = n\sigma_T^2$, $\exp(\lambda D_n - \lambda^2 V_n/2)$ is a nonnegative supermartingale for every λ under sub-Gaussian increments; mixing over $\lambda \sim N(0, 1/\varrho)$ gives the nonnegative supermartingale $M_n = \sqrt{\varrho/(V_n+\varrho)}\exp\big(D_n^2/(2(V_n+\varrho))\big)$ with $M_0 = 1$. Ville's inequality (Ville, 1939) gives $P(\exists n : M_n \geq 1/\gamma) \leq \gamma$, and $M_n \geq 1/\gamma$ is exactly $|D_n| \geq b_n$. On the complementary event every interval covers S_T at every n, so a decision made at any stopping time is correct; the tail bound is deterministic. Since $b_n/n = O\big(\sigma_T\sqrt{\log(n/\gamma)/n}\big)$, the stopping time is $O\big(\sigma_T^2\Delta^{-2}\log(\sigma_T/(\Delta\gamma))\big)$ when the gap is Δ (Howard et al., 2021).

B Notation

Symbol	Meaning
$x_t,\ d_t,\ u_t$	price displacement, depth imbalance $(B-A)/(B+A)$, signed flow
$a,\ \kappa$	price persistence and direct impact
$\varphi,\ \varepsilon,\ r$	price sensitivity to depth, depletion per unit of flow, refill rate
$\eta,\ q$	momentum memory rate and aggressiveness
$C_2,\ C_3$	amplification statistics $\varphi\varepsilon-(1-a)\kappa$ and $C_2+\kappa^2\eta q$
$h_k,\ g_k$	impulse response of the book; price impulse coefficient of the controlled system
$J(\alpha)$	closed-loop operator at admission level α
$M_C,\ G_{\text{cum}}$	peak and summed price-row gains
$\mathcal{K}(J)$	Kreiss constant
$\bar{u},\ \bar{u}_{\text{step}},\ \bar{u}_{\text{cum}}$	shock envelopes
$B_{\text{rem}},\ R_{\text{safe}},\ \mu$	remaining margin buffer, safety reserve, maintenance-margin rate
$\epsilon,\ S$	operator confidence radius and $\sum_k \|J^k\|$
$M_{\text{std}},\ A_{xx}$	unit-free and price-to-price spot amplification statistics

C Additional material

C.1 Specification of the pre-registered tests

Futures (Table 9). Question: does $\hat{C}_3$, estimated from the previous six hours of futures depth, flow and price, predict the next hour's MAE beyond controls? Unit: hourly blocks 06–23 UTC, features from $[t-360, t)$ minutes, outcomes from $(t, t+60]$. Controls: rv_{60}, rv_{360}, $|r_{60}|$, flow imbalance, log volume, log depth, imbalance, open-interest change.

Model: OLS; tail is the outcome above the training 90th percentile; score is the fitted value. Inference: Clark–West with Newey–West (6 lags); day-block bootstrap with 2,000 draws. Secondary: peak ratio as candidate; open-interest drop and depth collapse as outcomes; Holm. Train $\leq$ 2024-12-31; test $\geq$ 2025-01-01. Hash `d95212187ae42052591a5ca5acf73317 0f217f430a5a6ca812da6e48a1e4af85`; one unlock.

Spot (Section 10). Question: does pre-window A_{xx} predict the next ten minutes' MAE beyond volatility and flow controls? Unit: non-overlapping 30-minute blocks; features from the previous ten minutes. Controls: rv_{10m}, rv_{60m}, $|r_{10m}|$, flow imbalance, log volume, spectral radius, hour of day. Train 2021–2023; test 2024. Hash `b1fd7f8eef15df1ba295eb6d32b53be6 c23ea2d23c5216de912555e62716e5ae`; three unlocks (see text).

C.2 Adaptive safe probing

The certificate of Theorem 5.5 fixes the number of reset experiments in advance, so it must be sized for the hardest economy the supervisor wants to decide. Experiments can instead be run one at a time and stopped as soon as the data decide. Let each experiment return $X_e = \sum_{t<T} h_t^{(e)}$, independent with mean S_T and sub-Gaussian proxy σ_T^2 ($\sigma_T^2 = T\sigma^2$ under (O1)), and let $b_n = \sqrt{(n\sigma_T^2 + \varrho)\big(\log((n\sigma_T^2 + \varrho)/\varrho) + 2\log(1/\gamma)\big)}$ for a fixed $\varrho > 0$.

Proposition C.1 (Anytime-valid probing)**.** *With probability at least $1-\gamma$, $|\bar{X}_n - S_T| < b_n/n$ for every n simultaneously. Hence the rule that probes until $\bar{X}_n + b_n/n + \kappa\bar{\lambda}^T/(1-\bar{\lambda}) \leq s_{\mathcal{B}}$ (admit), or $\bar{X}_n - b_n/n > s_{\mathcal{B}}$ (refuse), or a maximum number of probes is reached (abstain), makes a wrong decision with probability at most γ whatever its stopping time, where $s_{\mathcal{B}} = \mathcal{B}/(|I|\bar{u})$. When S_T is at distance Δ from $s_{\mathcal{B}}$ it stops after $O\big(\sigma_T^2\Delta^{-2}\log(\sigma_T/(\Delta\gamma))\big)$ probes. Each probe is itself admissible if its position v_p satisfies $|v_p|\bar{u}_p\kappa/(1-\bar{\lambda}) \leq \mathcal{B}_p$, the worst case over the class.*

The value is in adapting to the economy. With $\sigma_T = 0.17$ and a threshold of 2.0,

the sequential rule decides economies at distance 0.5 from the threshold after 3.7 probes on average, at distance 0.2 after 11, at 0.1 after 30 and at 0.05 after 107, with no wrong decision in 7,000 runs, while a fixed design that must decide at distance 0.05 uses 130 probes for every economy (test T14). Probes have a monetary cost and move prices, so a supervisor who can probe sequentially spends where the decision is hard. This is an extension rather than a result the paper relies on; optimal probe design under impact costs is left open.

D Reproducibility: source of every number

The replication package contains numbered Colab notebooks and stand-alone scripts; its README lists them in run order. "Part N" below and in table notes refers to notebook N (Part 1: theorem checks and spot tests; Part 2: BTCUSDT futures depth; Parts 4–18: the cross-section, replications, fresh-day tests, second venues, the CFTC test, the certified guard, post-hoc checks, US equities, the information-safety tests on untouched days and symbols, the new-venue test and the last post-hoc checks). Result identifiers such as `X6` or `U6` name the JSON file that the notebook writes. "Kind" distinguishes closed-form or simulation results (recomputed), results from public market data (real data), and audits.

Result	Source		Kind
Spectral blindness gains	`T5_spectral_blindness`		recomputed
Table 2; burst operator gain, q, certificate	`kreiss_table.py` → `R_P2_kreiss.json`	→	recomputed
Table 4, Figure 3	`calib_guard.py` → `R_P2_calibrated_guard.json`	→	recomputed
Figure 1	`make_figs2.py`		recomputed
Theorem 4.2(i), Theorem 4.2(ii)–(iii) draws	`T6_amplification_theorem`		recomputed

Result	Source	Kind
Guard maximality; peak guard unsafe; monotonicity	`T7_guard_maximality`, `P2.11-T1`	recomputed
Robust gain, 30,000 draws	`T8_robust_gain`	recomputed
Operator ball coverage; certificate share	`P2.4-T1_coverage`	recomputed
Summed gain vs. weights	`T9_bridge`	recomputed
Depth validation (Table 6)	`D2_depth_validation`	real data
Table 7	`D3_T6_structural_train`	real data
Figure 4, Table 8	`D4_depth_mediation_train`	real data
No-look-ahead audit	`D5_no_lookahead`	audit
Table 9; hash	`D8_depth_test_G2`, `P2_depth.json`	real data
Flag comparison	`D8b_flag_comparison`	real data
Table 10, Figure 5	`D9_event_studies`	real data
Unit dependence 8.73 → 86,731	`P2.2-T1_unit_invariance`	real data
Table 11; sensitivity	`P2.2-T2`, `T3`, `T4`, `T5`	real data
Table 12	`T6-E_overshoot_real`	real data
Spot predictive test	`P2.5_predictive_G2`, `P2.5-T2`, `T3`	real data
Spot flag comparison	`P2.7-T2_vol_ablation`	real data
Theorem 4.4; Remark 4.3; Theorem 4.2(iv) check	`equilibrium.py`, `t6m_search*.py`, `thmA_check.py`	recomputed
Proposition 8.1 numbers	`equilibrium.py` → `R_P2_equilibrium.json`	recomputed

Result	Source	Kind
Proposition 2.2 check; multi-asset example; Proposition 6.11 check	`recompute_oct20.py`	recomputed
Tables 13, 14, 15; tick events	Part 4: `X3`, `X4`, `X8`, `X7`	real data
Table 17; flags	Part 4: `X6_xsec_test_G2`, `X6b`; `P4_xsec.json` (5910b6b4...)	real data
Table 18	Part 4: `X6r_robustness_posthoc`	real data (post hoc)
Table 19	Part 5: `Y6_replication_G2`; `P5_replication.json` (6aa8e2c0...)	real data
Table 20	Part 5: `Y8_kappa_control_posthoc`	real data (post hoc)
Tables 21, 25; equilibrium check	Part 6: `Z6_*`, `Z8`; `P6_fresh_days.json` (65acda36...)	real data
Table 26, Figure 6, economic size (Section 11.17)	Part 12: `Q6_certified_guard`, `Q7_oct10_event`, `E1_economic_size`; `P12_certified_guard.json` (b9a56af4...)	real data
Liquidation probabilities and dollar losses (Section 11.17)	`recompute_dollar.py` (from `E1_economic_size`) → `R_P2_dollar_translation.json`	recomputed

Result	Source	Kind
Table 23	Part 10: `V3A_orderbook_download`, `V4_build`, `V6_bybit_ob_test`; `P10_bybit_ob.json` (ec2145fa...)	real data
Table 16; throttle, Bybit-bar, instrument, tercile and VPIN checks	Part 13: `R5`, `R1`, `R3`, `R4`, `R2`, `R8`	real data (post hoc)
Example 5.1; Theorems 5.3, 5.5, 6.5, 7.1 numbers; Tables 3, 5, 28	`run_all.py` in `research_extension/information_safety` → `IS_results.json`; unit tests	recomputed
Section 12.2	Part 15: `W4_build`, `W5_blocks`, `W6_info_safety_test`; `P15_info_safety.json` (`69849c96`)	real data
Section 12.3	Part 16: `X2_new_universe`, `X4_build`, `X5_blocks`, `X6_untouched_symbols_test`; `P16_untouched_symbols.json` (`0523e476`)	real data
Section 12.4	Part 17: `C2_universe`, `C4_build`, `C5_blocks`, `C6_new_venue_test`; `P17_new_venue.json` (`3233c52b`)	real data
Section 12.5	Part 18: `R3_placebo_binance`, `R4_guard_baselines`; `P18_posthoc_plan.json`	real data (post hoc)

Result	Source	Kind
Section 12.7 (Table 33)	Part 19: `recompute_p2_parts19_20_21.py` → `R_P2_parts19_20_21.json`	real data
Section 12.8 (Table 34)	Part 20: `recompute_p2_parts19_20_21.py` → `R_P2_parts19_20_21.json`	real data
Section 12.9 (Table 35)	Part 21: `recompute_p2_parts19_20_21.py` → `R_P2_parts19_20_21.json` (from `W6`, `X6`)	translation of real data
Sections 12.10–12.11 (Tables 36, 37)	Parts 22–23: `recompute_prereg_oct21_22nd.py` → `R_P2_p22_p23.json`; `P22_liquidation_mechanism.json` (`2b3794e3`), `P23_guard_capacity.json` (`d4169377`)	real data
Propositions 2.3 and 6.9 checks	`recompute_p2_step2_theory.py` → `R_P2_step2_theory.json`	recomputed
Balance-sheet Monte Carlo (Section 6)	`bridge/microfoundation_and_baselines.py` → `bridge/results/microfoundation_and_baselines.json`	recomputed
Lemma 5.4; Propositions 5.7, 6.6, C.1; Theorem 6.7	`run_all.py`: `T10`, `T11`, `T13`, `T14`	recomputed (certified for T10, T13)

Result	Source	Kind
Table 24	Part 14: `A1_us_universe`, `A3_us_build`, `A4_us_blocks`, `A5_us_equity_test`; `P14_us_equities.json` (f5800a69…)	real data
Theorem 4.2(ii) peak bracket; Remark 4.3; Theorem 4.4, 8.1 checks	`recompute_audit.py` → `R_audit_oct2026.json`	recomputed
Theorem 4.2 residues and confluent peak lag; Remark 5.6; Proposition 6.11(ii)	`hardening/p2_sec4_checks.py`, `p2_cheb_tradeoff.py`; unit tests `test_hardening_theory.py`	recomputed
Theorem 7.1(v), intersection row of Table 5	information-safety package, test T15	recomputed
Section 11.11	Part 7: `W6_bybit_test`; `P7_bybit.json`	real data
Table 22	Part 9: `U6_volbench_test`; `P9_volbench.json` (71983f1a…)	real data
Figure 7	`make_fig_forest.py`	real data